

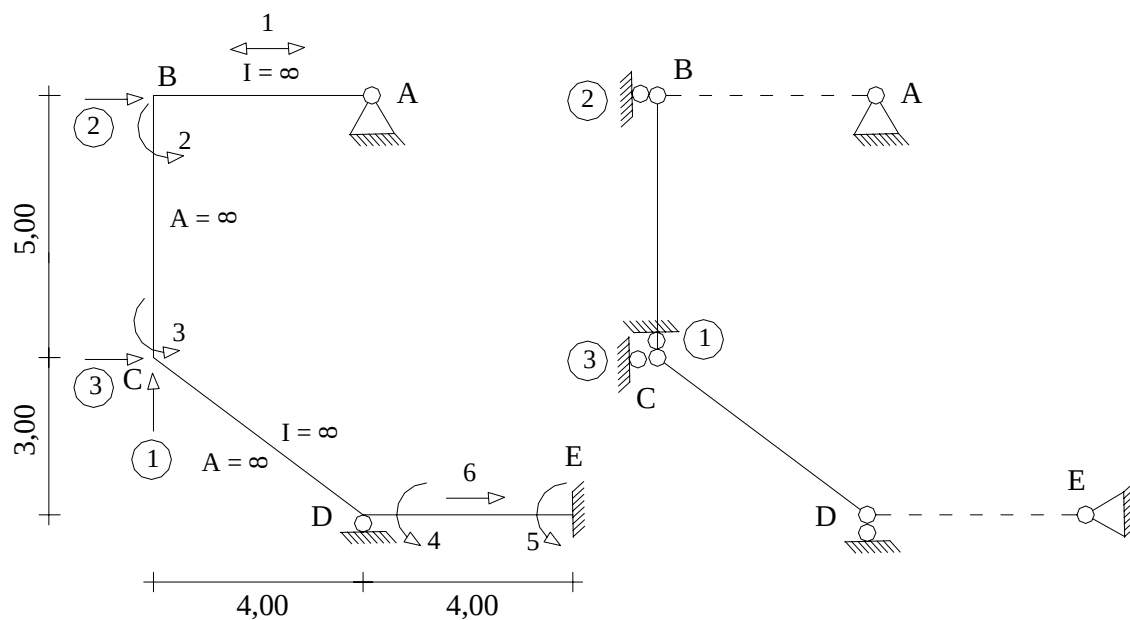
CALCULO DE LA MATRIZ DESPLAZAMIENTO-DEFORMACIÓN [A].

1.-

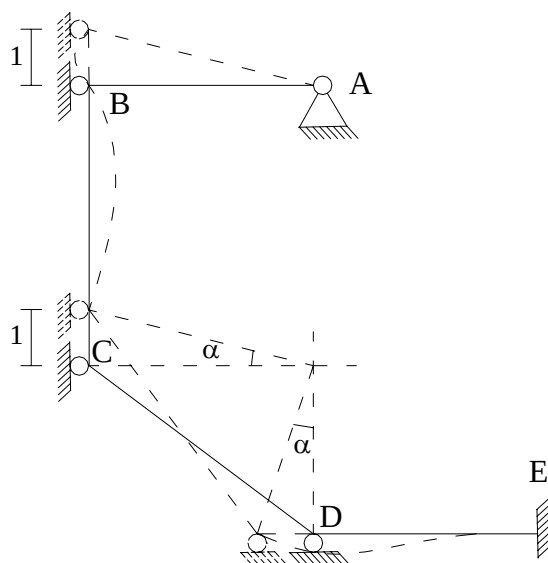
$$I_g^{S/R} = 9 \Rightarrow I_g = 3$$

$$R = 6$$

Imagen Cinemática.



Modo 1



$$\psi_{AB} = -1/4 = \theta_B$$

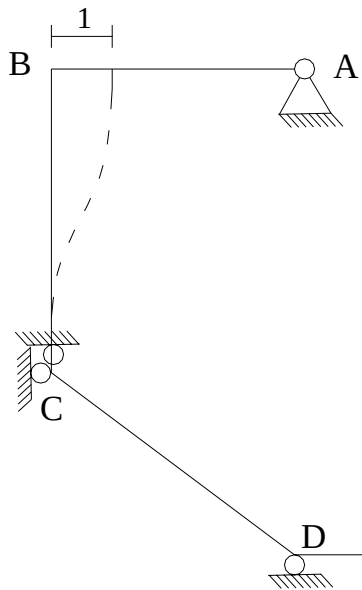
$$\alpha = 1/4 \Rightarrow \alpha = \Delta L / 3$$

$$\frac{1}{4} = \frac{\Delta L}{3} \Rightarrow \Delta L = \frac{3}{4}$$

$$\psi_{CD} = -1/4$$

$$d_1 = \begin{vmatrix} 0 \\ -1/4 - 0 \\ -1/4 - 0 \\ -1/4 - 0 \\ 0 - 0 \\ 3/4 \end{vmatrix} = \begin{vmatrix} 0 \\ -1/4 \\ -1/4 \\ -1/4 \\ 0 \\ 3/4 \end{vmatrix}$$

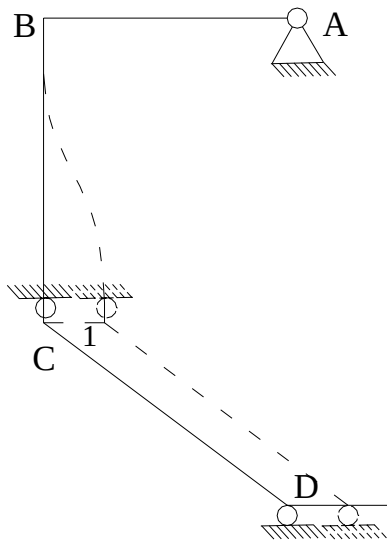
Modo 2



$$\psi_{BC} = 1/5$$

$$d_2 = \begin{bmatrix} -1 \\ 0 - (-1/5) \\ 0 - (-1/5) \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1/5 \\ 1/5 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Modo 3



$$\psi_{BC} = 1/5$$

$$d_3 = \begin{bmatrix} 0 \\ 0 - (1/5) \\ 0 - (1/5) \\ 0 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1/5 \\ -1/5 \\ 0 \\ 0 \\ -1 \end{bmatrix}$$

$$[A] = \begin{bmatrix} 0 & -1 & 0 \\ -1/4 & 1/5 & -1/5 \\ -1/4 & 1/5 & -1/5 \\ -1/4 & 0 & 0 \\ 0 & 0 & 0 \\ 3/4 & 0 & -1 \end{bmatrix}$$

2.- Hallar $[A]$

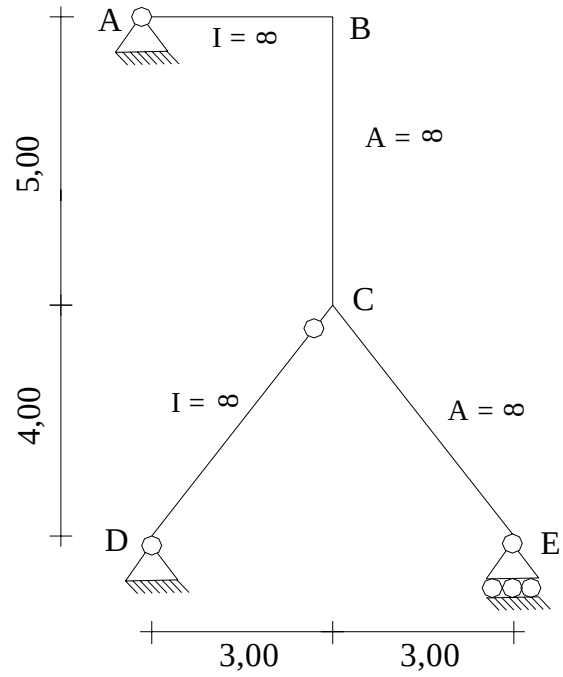
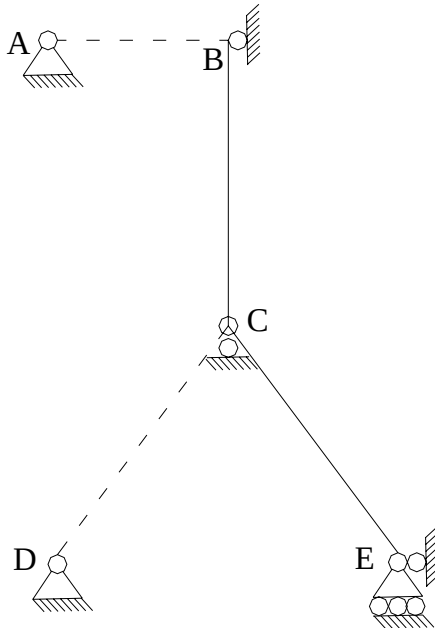
Definir $\begin{cases} Q-D \\ q-d \end{cases}$

$$I_g^{S/R} = 10$$

$$R = 6$$

$$I_g = 4$$

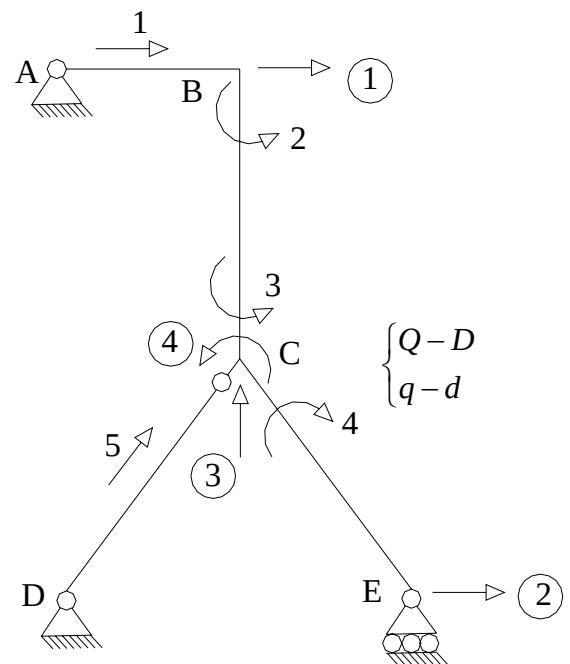
Imagen Cinemática



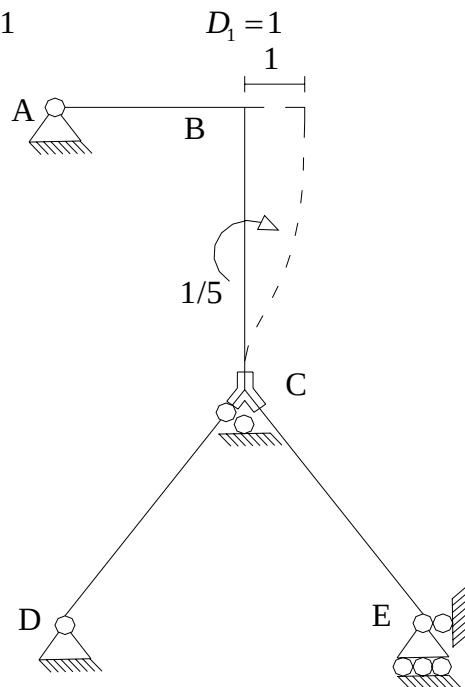
$$G.L. = G.D. = 3$$

$$\text{Rotaciones Independientes} = 1 : \theta_C$$

$$I_g = 4$$



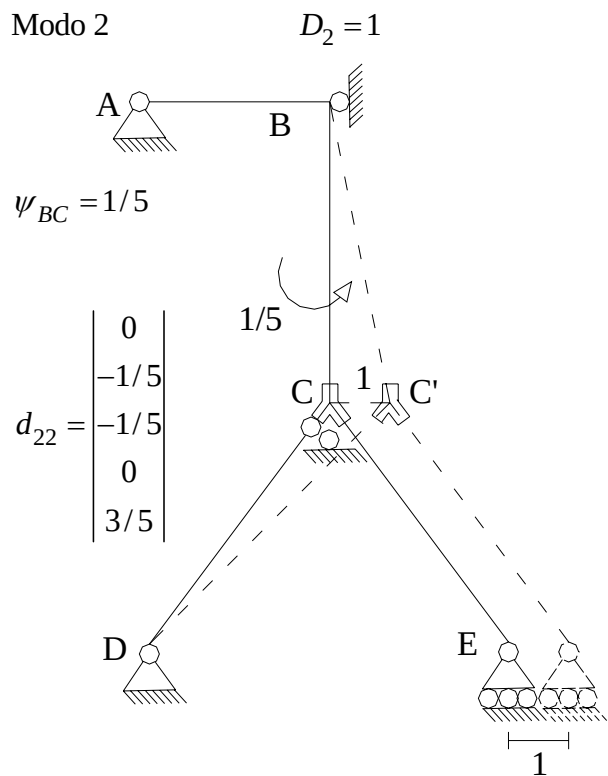
Modo 1



$$\psi_{BC} = -1/5$$

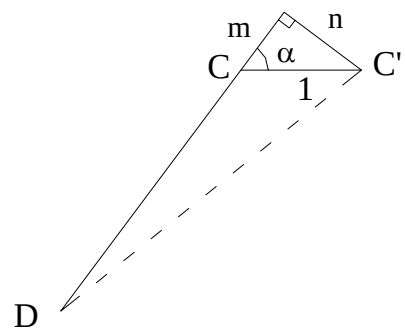
$$d_{21} = \begin{vmatrix} 1 \\ 1/5 \\ 1/5 \\ 0 \\ 0 \end{vmatrix}$$

Modo 2



$$\psi_{BC} = 1/5$$

$$d_{22} = \begin{vmatrix} 0 \\ -1/5 \\ -1/5 \\ 0 \\ 3/5 \end{vmatrix}$$



$$\cos \alpha = 3/5 = m$$

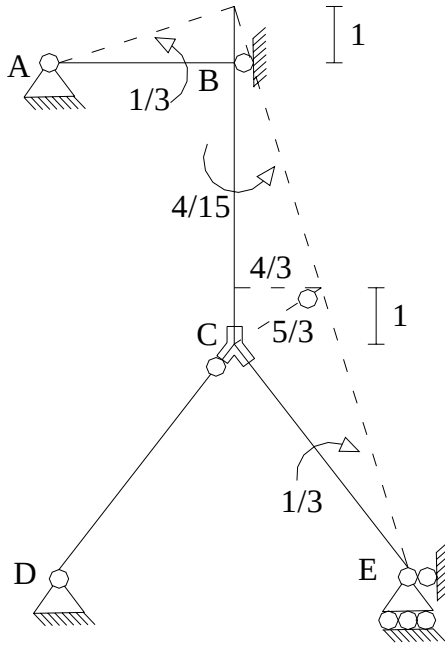
$$\sin \alpha = 4/5 = n$$

$$\psi_{DC} = -4/25$$

$$\Delta L_{DC} = 3/5$$

Modo 3

$$D_3 = 1$$

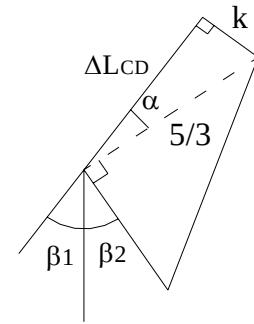
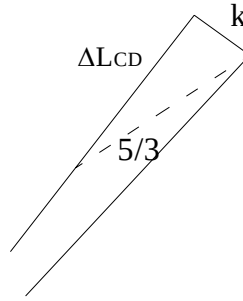


$$\theta_B = 1/3$$

$$\psi_{CE} = -1/3$$

$$\psi_{BC} = 4/15$$

$$d_{i3} = \begin{vmatrix} 0 \\ 1/3 - 4/15 \\ -4/15 \\ 1/3 \\ 1,60 \end{vmatrix}$$



$$\cos \alpha = \frac{\Delta L_{CD}}{5/3} \Rightarrow \Delta L_{CD} = 0,96(5/3) = 1,6$$

$$\beta_1 = \beta_2$$

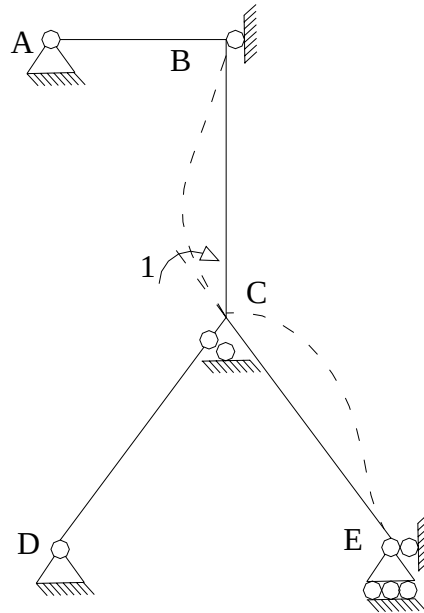
$$\cos \beta_1 = 3/5 \Rightarrow \cos \beta_1 = 0,6$$

$$\Rightarrow \beta_1 = 36,87$$

$$\Rightarrow \alpha = 180 - 90 - 2\beta_1$$

$$\Rightarrow \alpha = 16,26$$

Modo 4



$$d_{i4} = \begin{vmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 0 \end{vmatrix}$$

$$[A] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/5 & -1/5 & -1/15 & 0 \\ 1/5 & -1/5 & -4/5 & 1 \\ 0 & 0 & 1/3 & 1 \\ 0 & 3/5 & 1,60 & 0 \end{bmatrix}$$

3.-

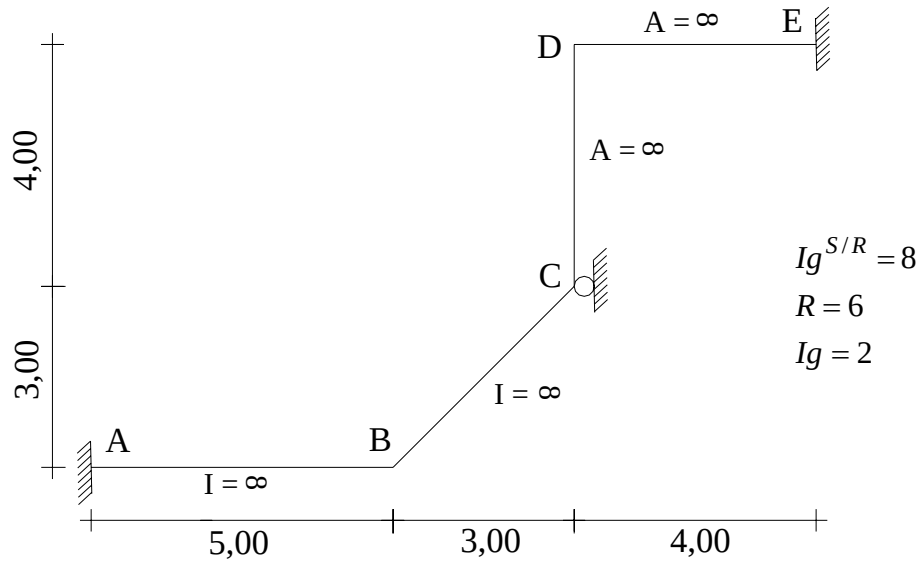
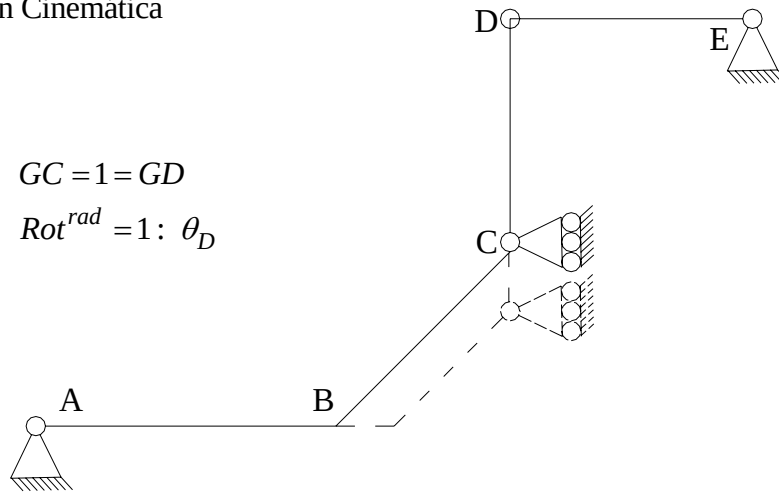
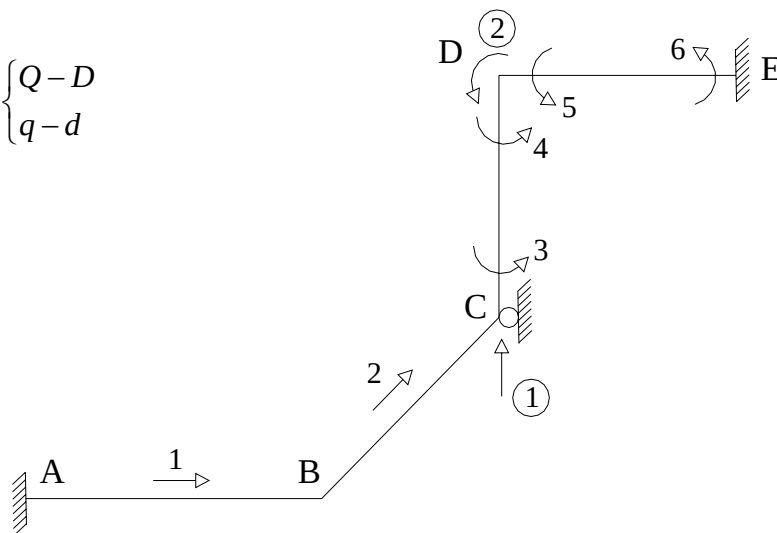


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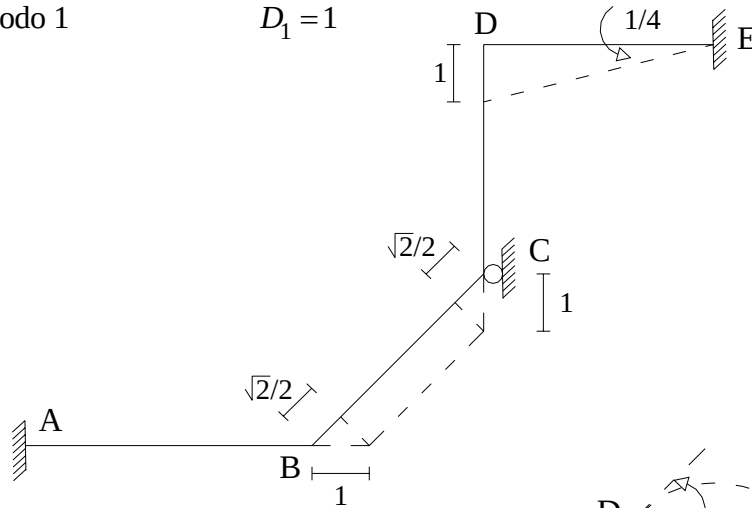


Sistema $\begin{cases} Q-D \\ q-d \end{cases}$



Modo 1

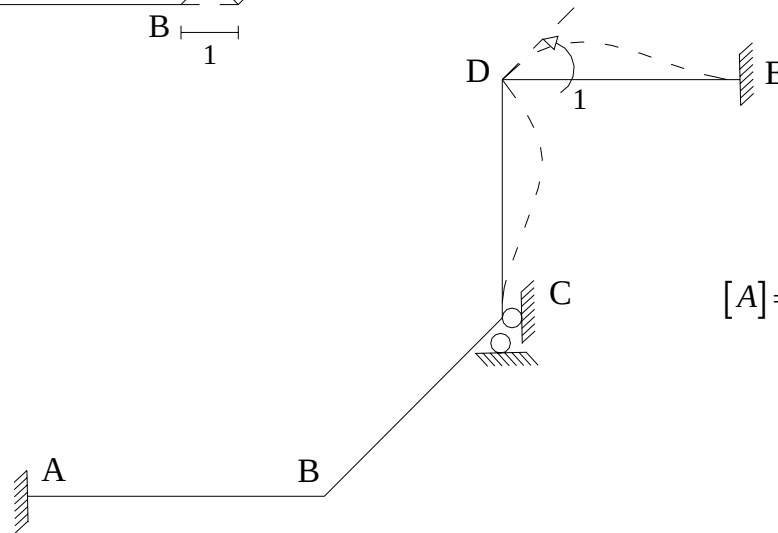
$$D_1 = 1$$



$$d_{i1} = \begin{bmatrix} 1 \\ -2 \frac{\sqrt{2}}{2} = -\sqrt{2} \\ 0 \\ 0 \\ -1/4 \\ -1/4 \end{bmatrix}$$

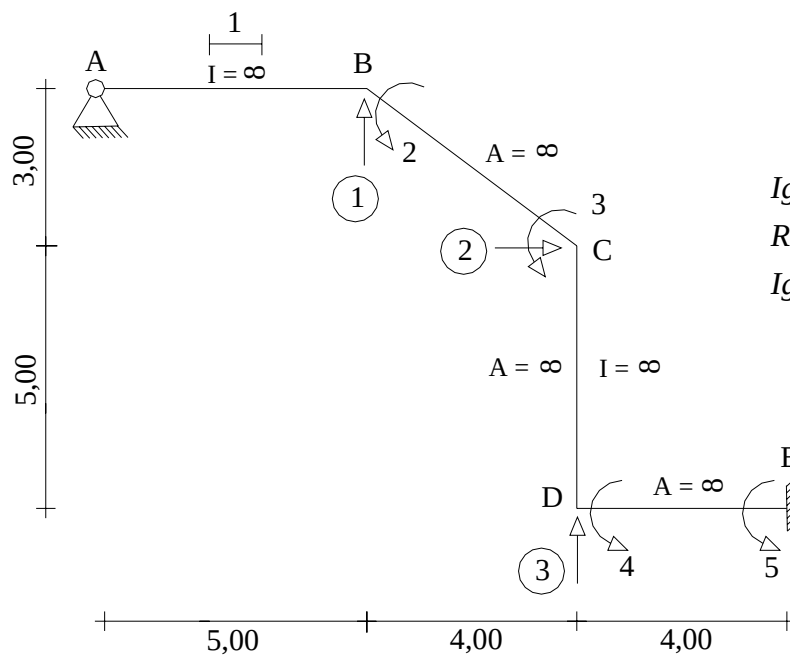
Modo 2

$$d_{2i} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$



$$[A] = \left[\begin{array}{ccc|ccc} 1 & & & 0 & & \\ -\sqrt{2} & & & 0 & & \\ 0 & & & 0 & & \\ 0 & & & 1 & & \\ -1/4 & & & 1 & & \\ -1/4 & & & 0 & & \end{array} \right]$$

4.-

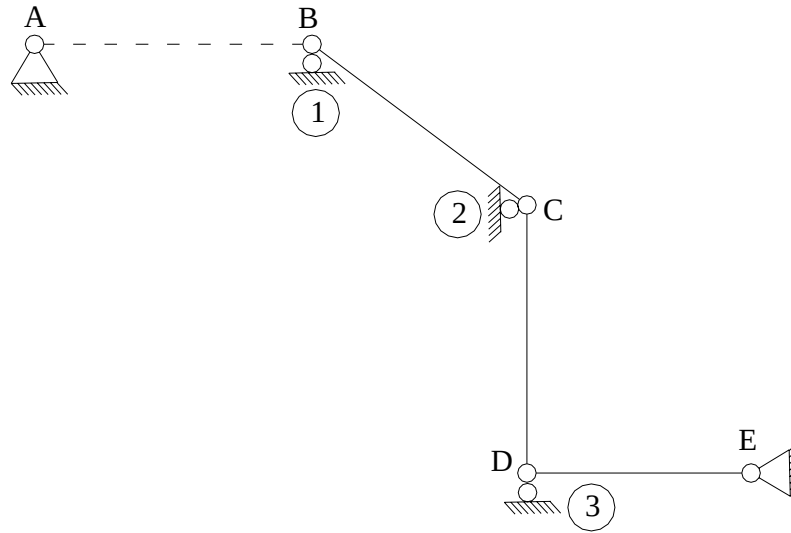


$$I_g^{S/R} = 10$$

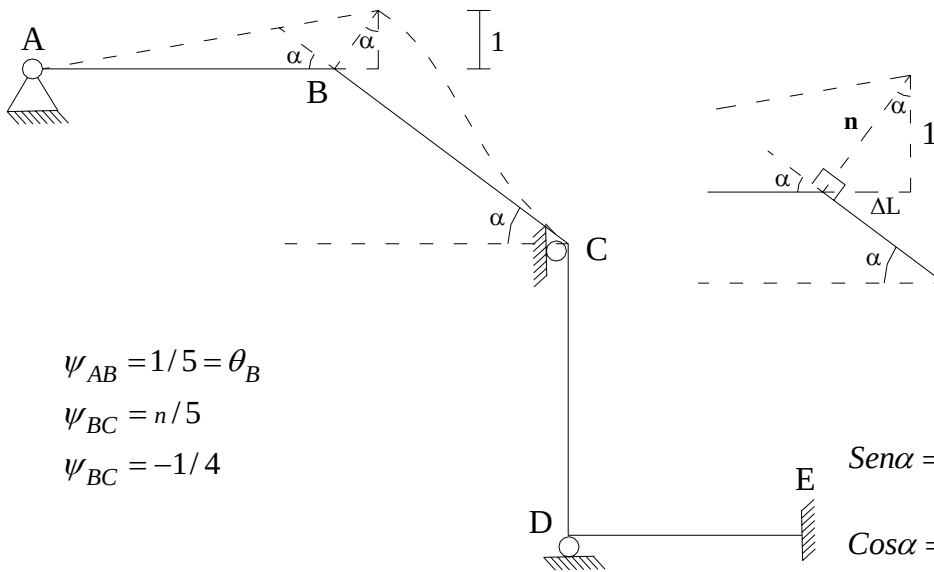
$$R = 7$$

$$I_g = 3$$

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Modo 1



$$\psi_{AB} = 1/5 = \theta_B$$

$$\psi_{BC} = n/5$$

$$\psi_{BC} = -1/4$$

$$d_{i1} = \begin{vmatrix} 3/4 \\ 1/5 - (-1/4) \\ 0 - (-1/4) \\ 0 \\ 0 \end{vmatrix} = \begin{vmatrix} 3/4 \\ 9/20 \\ 1/4 \\ 0 \\ 0 \end{vmatrix}$$

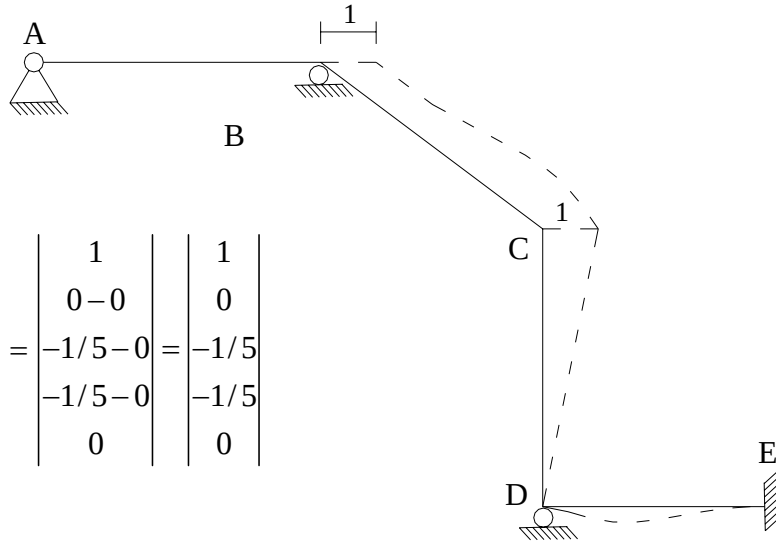
$$\text{Sen}\alpha = \frac{\Delta L}{n} \Rightarrow \text{Sen}\alpha = \frac{3}{5}$$

$$\text{Cos}\alpha = \frac{1}{n} \Rightarrow \text{Cos}\alpha = \frac{4}{5}$$

$$\frac{4}{5} = \frac{1}{n} \Rightarrow n = \frac{5}{4}$$

$$\frac{3}{5} = \frac{\Delta L}{5/4} \Rightarrow \Delta L = \frac{3}{4}$$

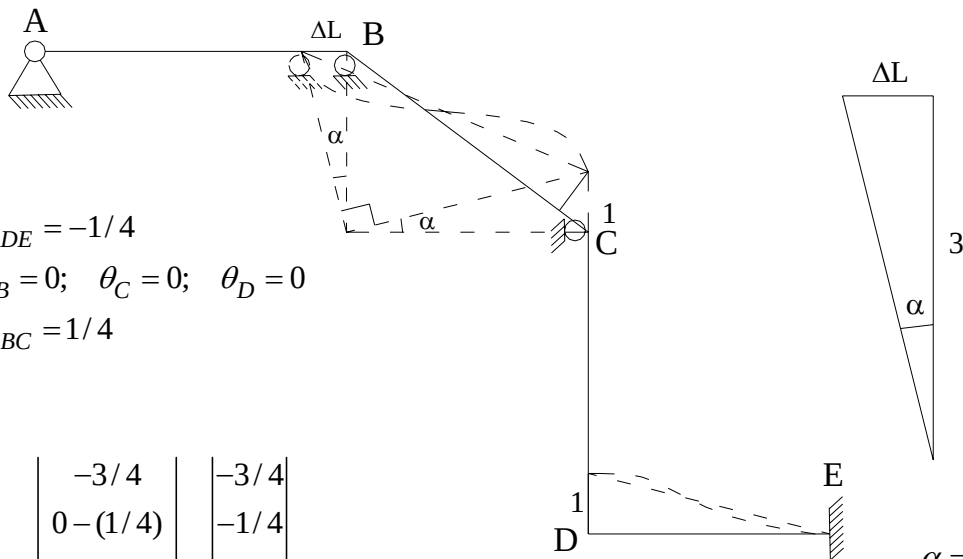
Modo 2



$$\psi_{DC} = -1/5$$

$$d_{i2} = \begin{bmatrix} 1 \\ 0-0 \\ -1/5-0 \\ -1/5-0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1/5 \\ -1/5 \\ 0 \end{bmatrix}$$

Modo 3



$$\psi_{DE} = -1/4$$

$$\theta_B = 0; \quad \theta_C = 0; \quad \theta_D = 0$$

$$\psi_{BC} = 1/4$$

$$d_{i3} = \begin{bmatrix} -3/4 \\ 0-(1/4) \\ 0-(1/4) \\ 0-(-1/4) \\ 0-(-1/4) \end{bmatrix} = \begin{bmatrix} -3/4 \\ -1/4 \\ -1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

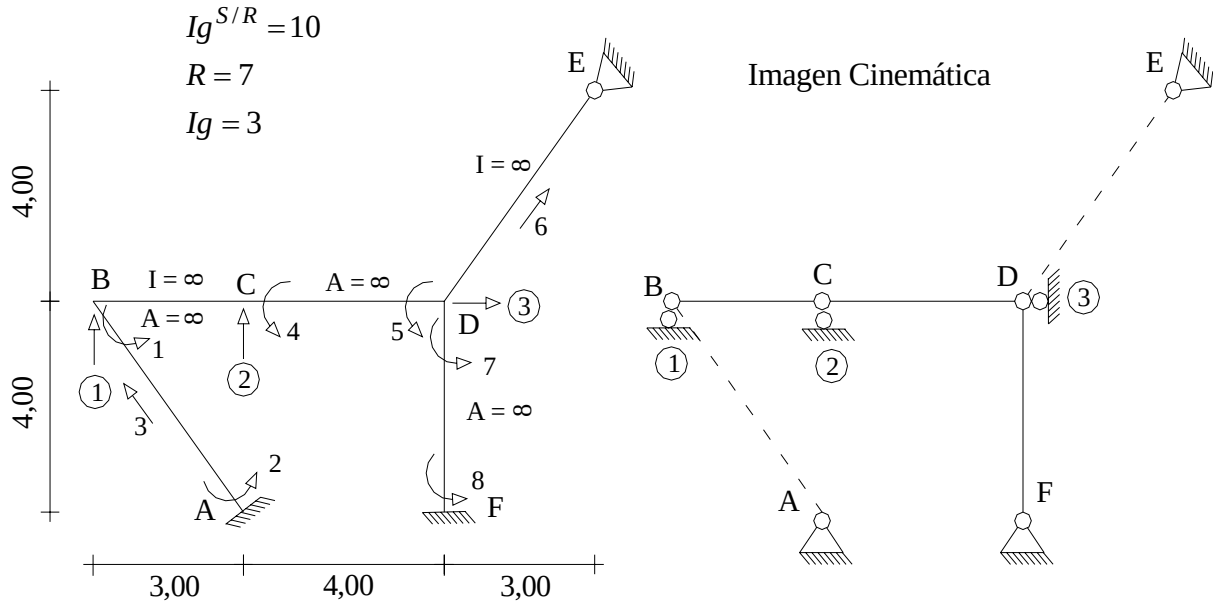
$$\alpha = 1/4$$

$$\alpha = \frac{\Delta L}{3}$$

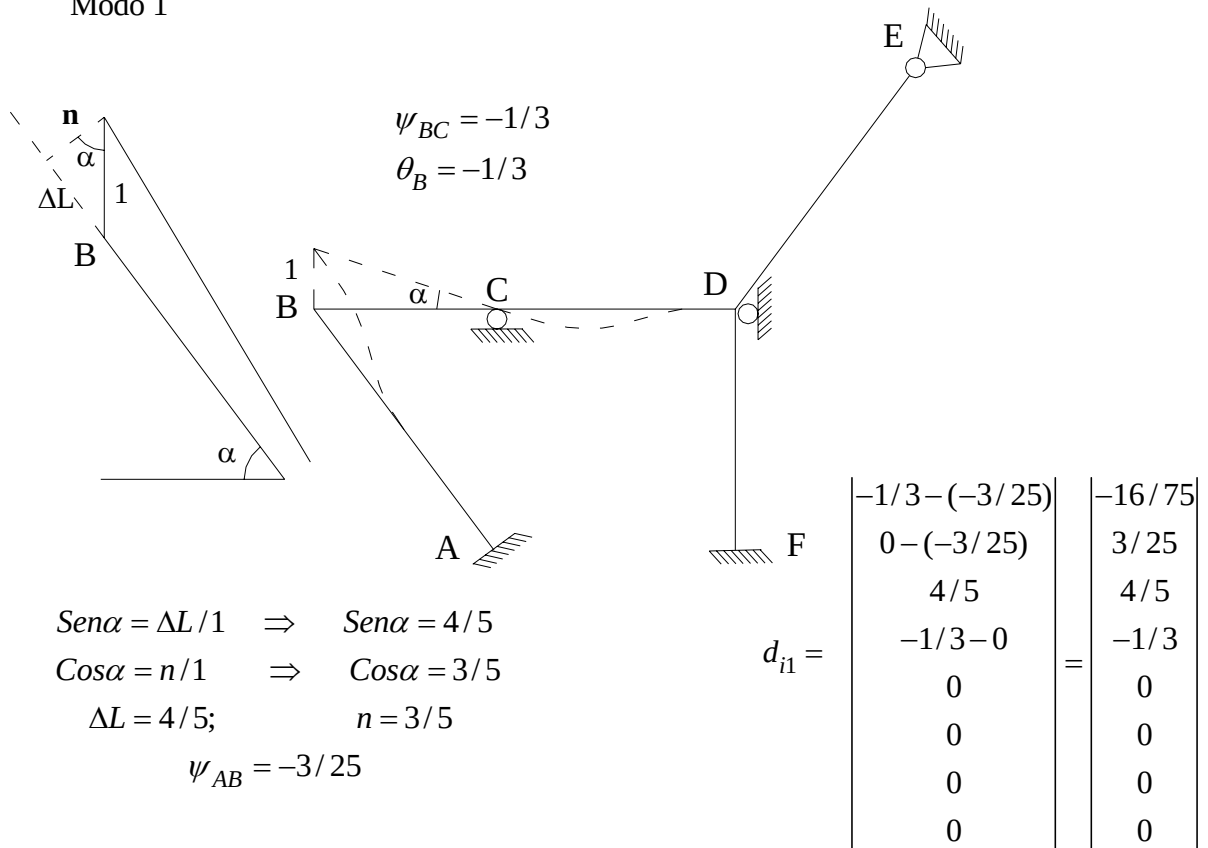
$$\Delta L = -3/4$$

$$[A] = \begin{bmatrix} 3/4 & 1 & -3/4 \\ 9/20 & 0 & -1/4 \\ 1/4 & -1/5 & -1/4 \\ 0 & -1/5 & 1/4 \\ 0 & 0 & 1/4 \end{bmatrix}$$

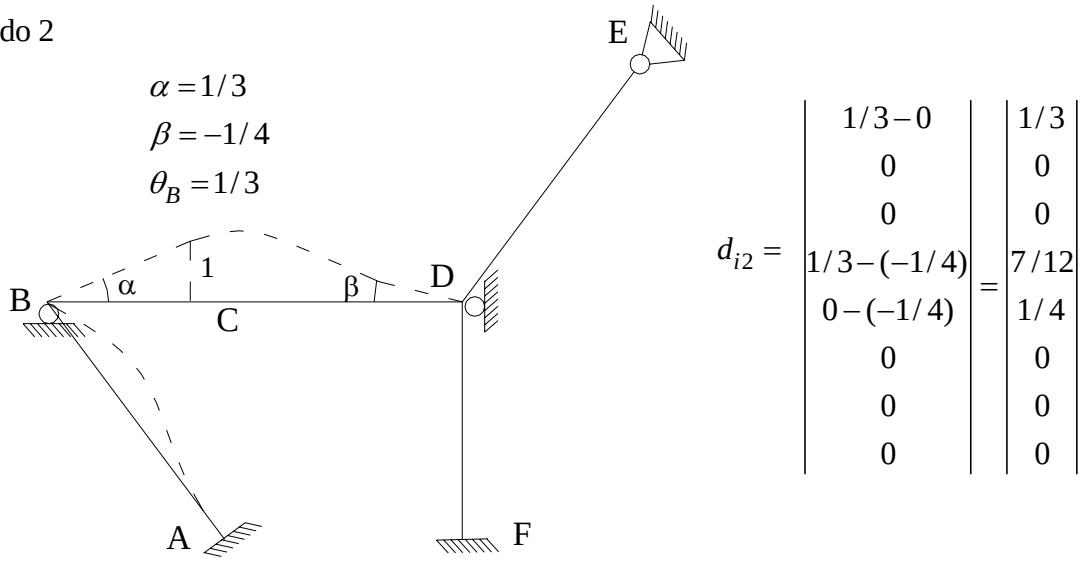
5.-



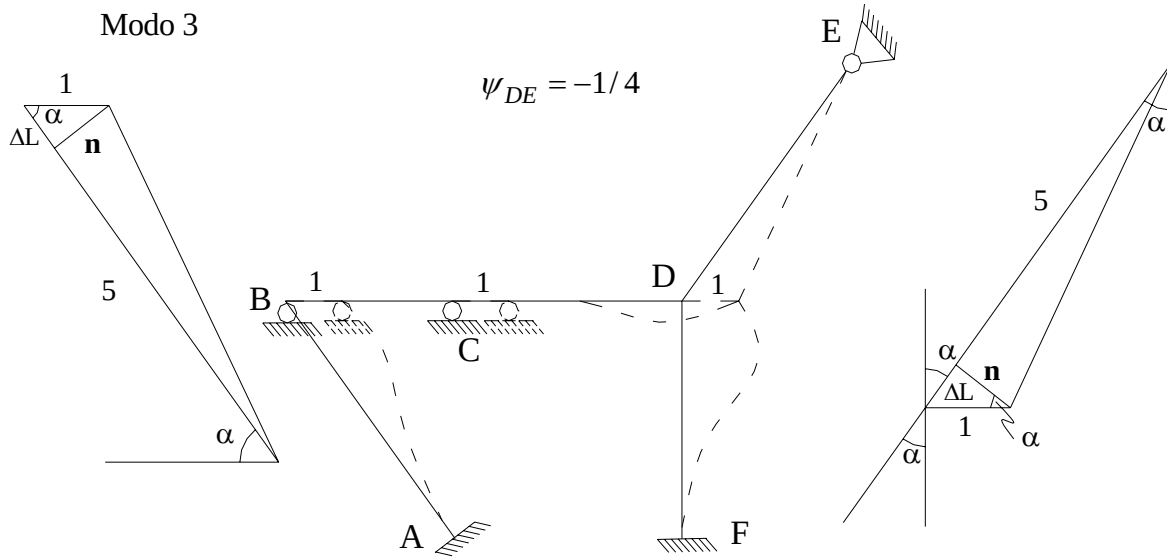
Modo 1



Modo 2



Modo 3



$$\cos \alpha = \frac{\Delta L}{1}; \quad \cos \alpha = \frac{3}{5}$$

$$\sin \alpha = \frac{n}{1}; \quad \sin \alpha = \frac{4}{5}$$

$$\Delta L = -3/5; \quad n = 4/5$$

$$\psi_A^B = -4/25; \quad \theta_B = 0$$

$$\sin \alpha = \frac{\Delta L}{1}; \quad \sin \alpha = \frac{3}{5}$$

$$\cos \alpha = \frac{n}{1}; \quad \cos \alpha = \frac{4}{5}$$

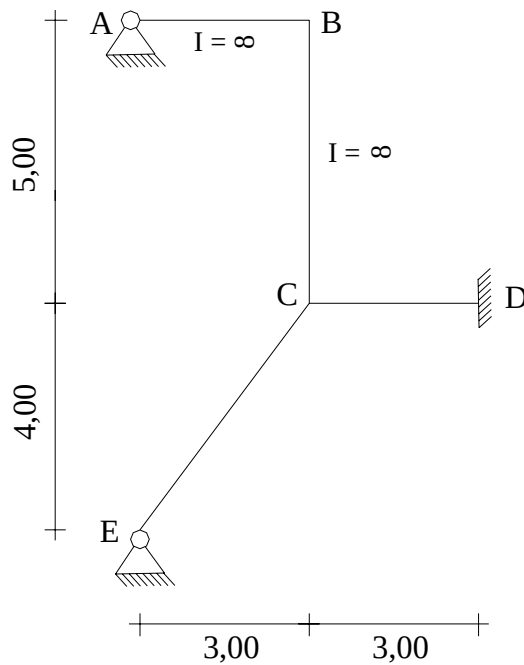
$$\Delta L = 3/5; \quad n = 4/5$$

$$\psi_D^E = 4/25; \quad \theta_D = 4/25$$

$$d_{i3} = \begin{vmatrix} 0 - (-4/25) \\ 0 - (-4/25) \\ -3/5 \\ 0 \\ 4/25 - 0 \\ -3/5 \\ 4/25 - (-1/4) \\ 0 - (-1/4) \end{vmatrix} = \begin{vmatrix} 4/25 \\ 4/25 \\ -3/5 \\ 0 \\ 4/25 \\ -3/5 \\ 41/100 \\ 1/4 \end{vmatrix}$$

$$[A] = \begin{bmatrix} -16/75 & 1/3 & 4/25 \\ 3/25 & 0 & 4/25 \\ 4/5 & 0 & -3/5 \\ -1/3 & 7/12 & 0 \\ 0 & 1/4 & 4/25 \\ 0 & 0 & -3/5 \\ 0 & 0 & 41/100 \\ 0 & 0 & 1/4 \end{bmatrix}$$

6.-



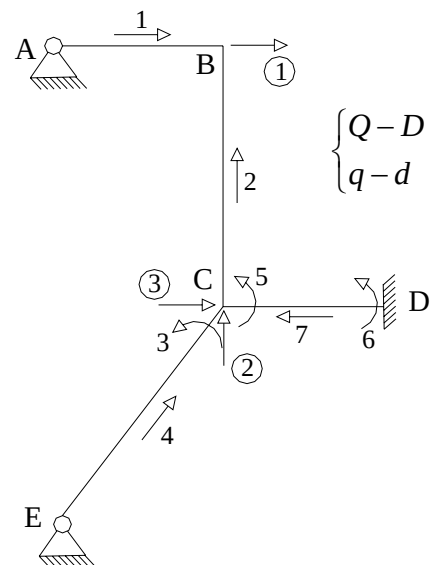
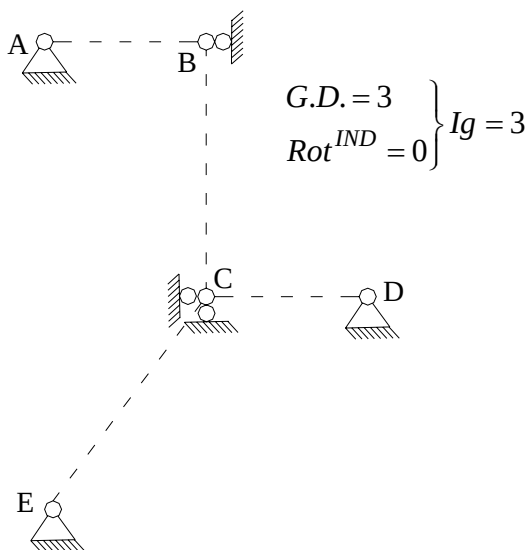
$$I_g = ?; \quad b) I_g = G.D \quad 1 \quad Rot^{IND}$$

$$a) I_g = I_g^{S/R} - R$$

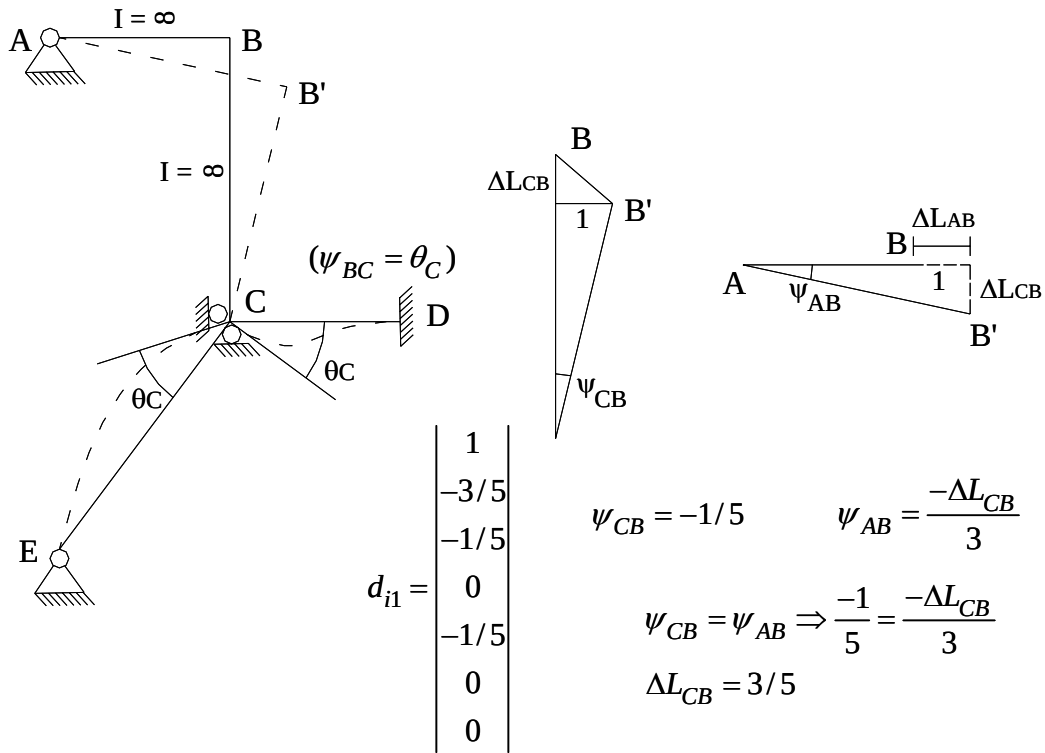
$$I_g^{S/R} = 7; \quad R = 4$$

$$I_g = 3$$

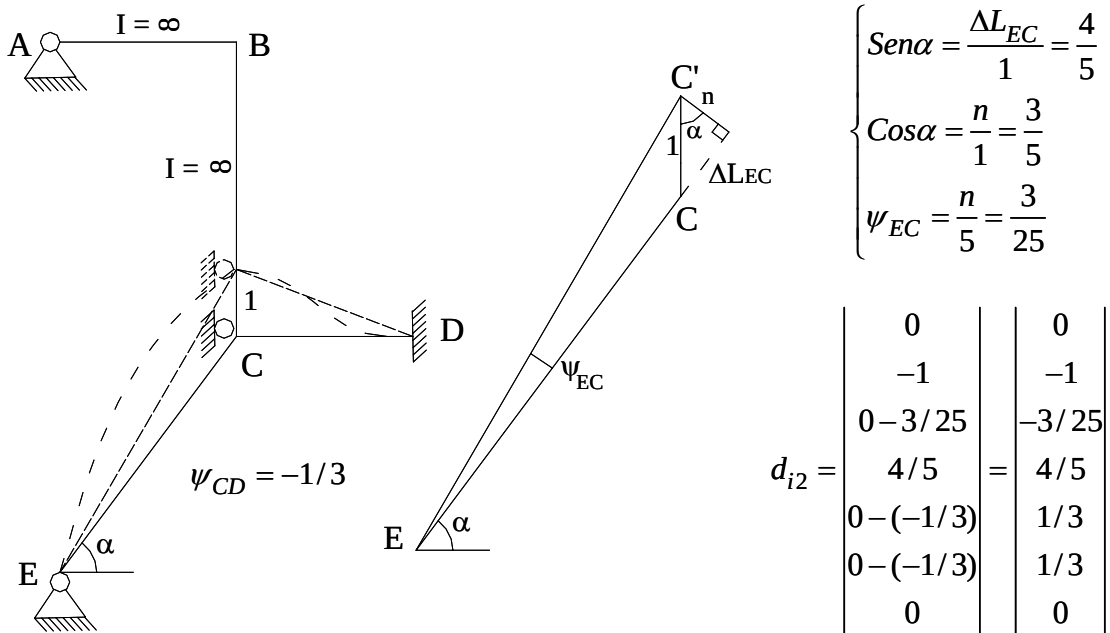
Imagen Cinemática



Modo 1 $D_1 = 1 // D_2 = D_3 = 0$



Modo 2 $D_2 = 1 // D_1 = D_3 = 0$



Modo 3 $D_3 = 1 // D_1 = D_2 = 0$

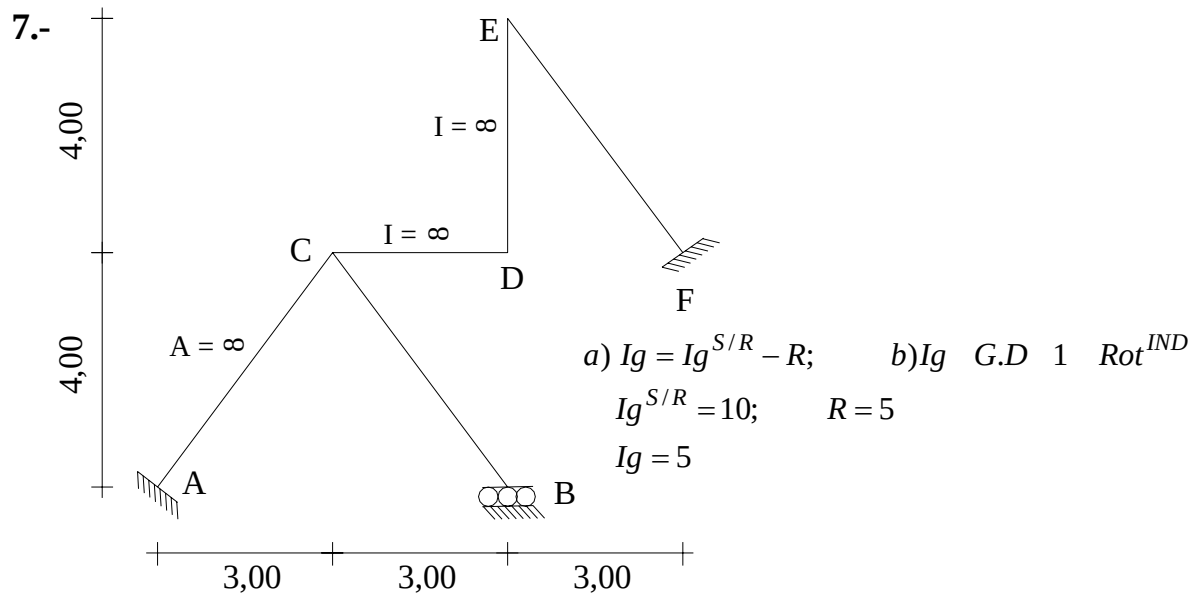
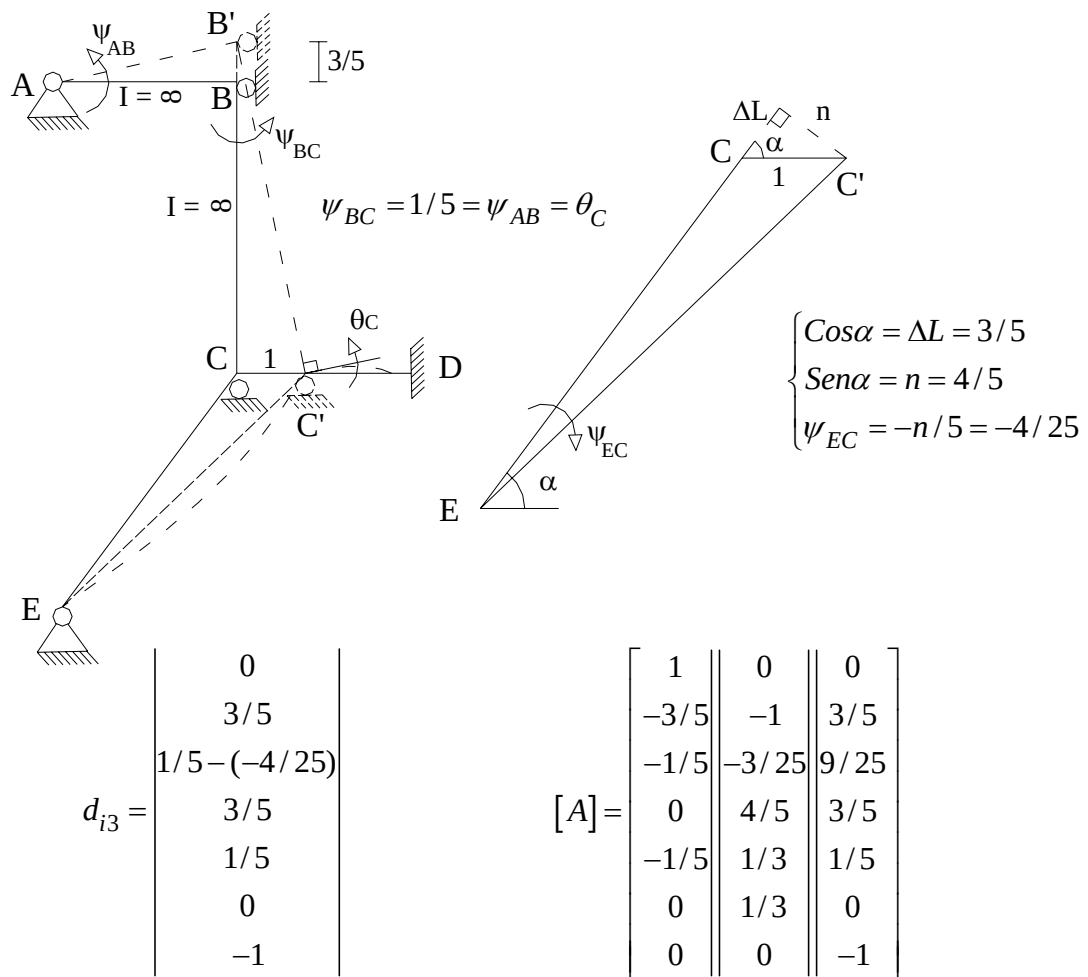
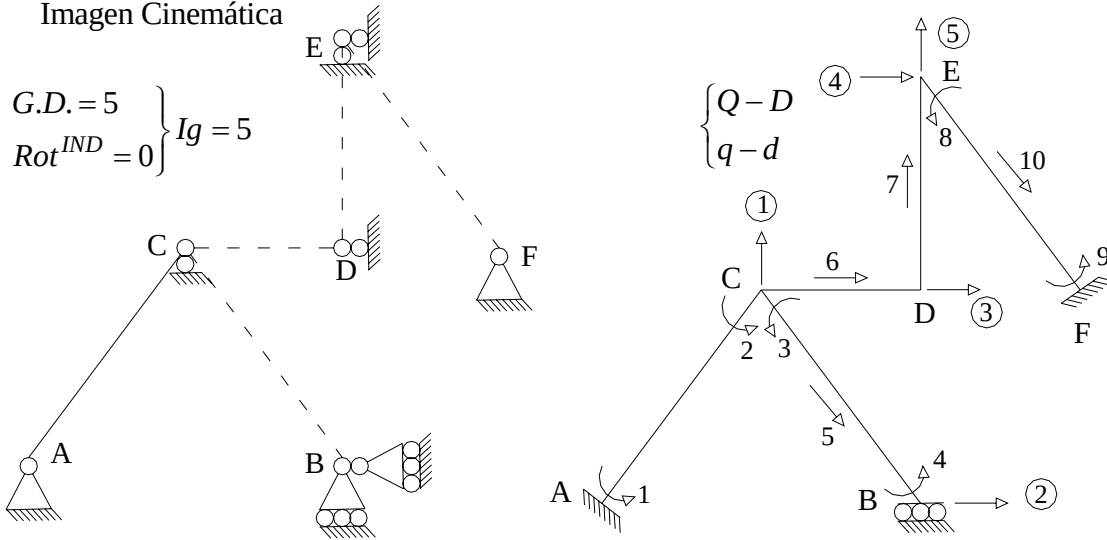
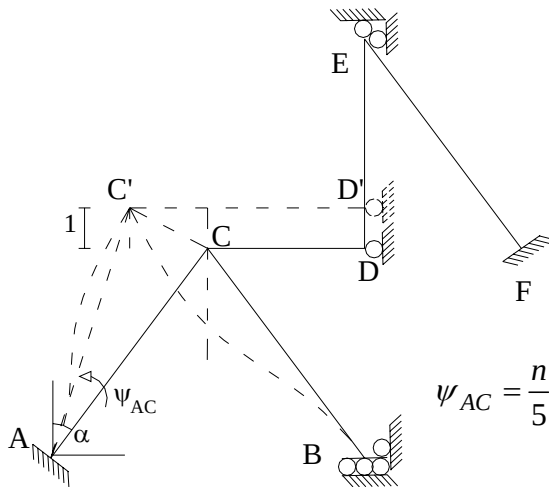


Imagen Cinemática

$$\left. \begin{array}{l} G.D. = 5 \\ Rot^{IND} = 0 \end{array} \right\} Ig = 5$$



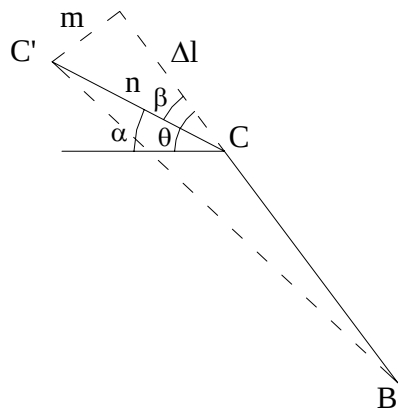
Modo 1 $D_1 = 1 // D_2 = D_3 = D_4 = D_5 = 0$



$$\psi_{AC} = \frac{n}{5} = \frac{1}{3}$$

$$Tg\alpha = \frac{1}{\Delta L} = \frac{3}{4} \Rightarrow \Delta L = \frac{4}{3}$$

$$Sen\alpha = \frac{1}{n} = \frac{3}{5} \Rightarrow n = \frac{5}{3}$$



$$Tg\theta = 4/3$$

$$\theta - \alpha = \beta$$

$$\beta = arcTg\theta - arcTg\alpha$$

$$\beta = artTg(4/3) - arcTg(3/4)$$

$$\beta = 16,26$$

$$Cos\beta = \Delta L / n$$

$$\Delta L = (5/3)Cos(16,26)$$

$$\Delta L = 1,6$$

$$d_{i1} = \begin{vmatrix} 0 & -1/3 \\ 0 & -1/3 \\ 0 & -0,092 \\ 0 & -0,092 \\ 1,6 \\ 4/3 \\ -1 \\ 0 \\ 0 \\ 0 \end{vmatrix} = \begin{vmatrix} -1/3 \\ -1/3 \\ -0,092 \\ -0,092 \\ 1,6 \\ 4/3 \\ -1 \\ 0 \\ 0 \\ 0 \end{vmatrix}$$

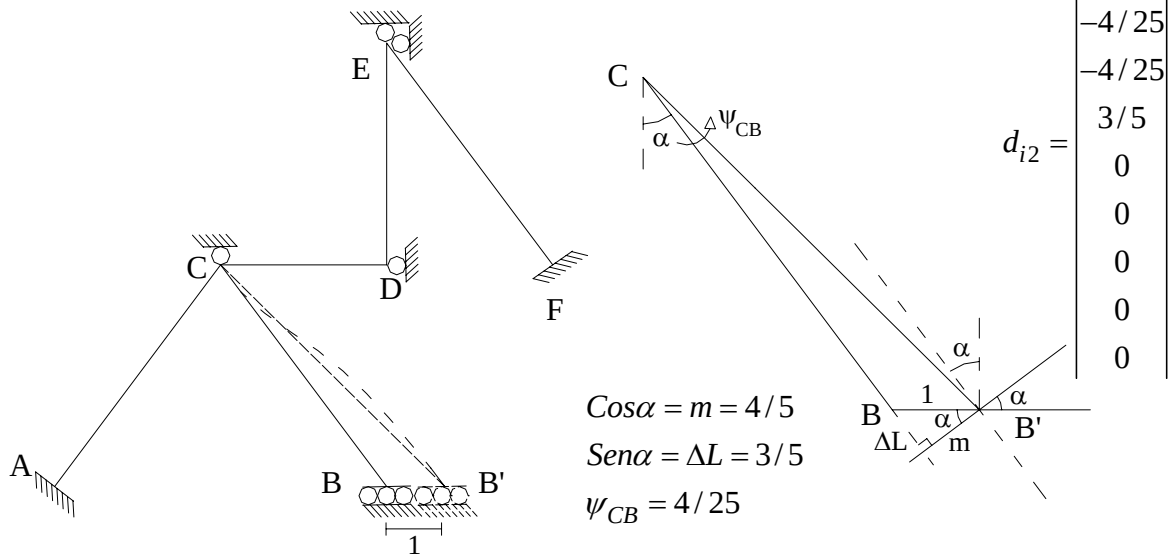
$$\text{Sen} \beta = m / n$$

$$m = \text{Sen}(16,26)(5/3)$$

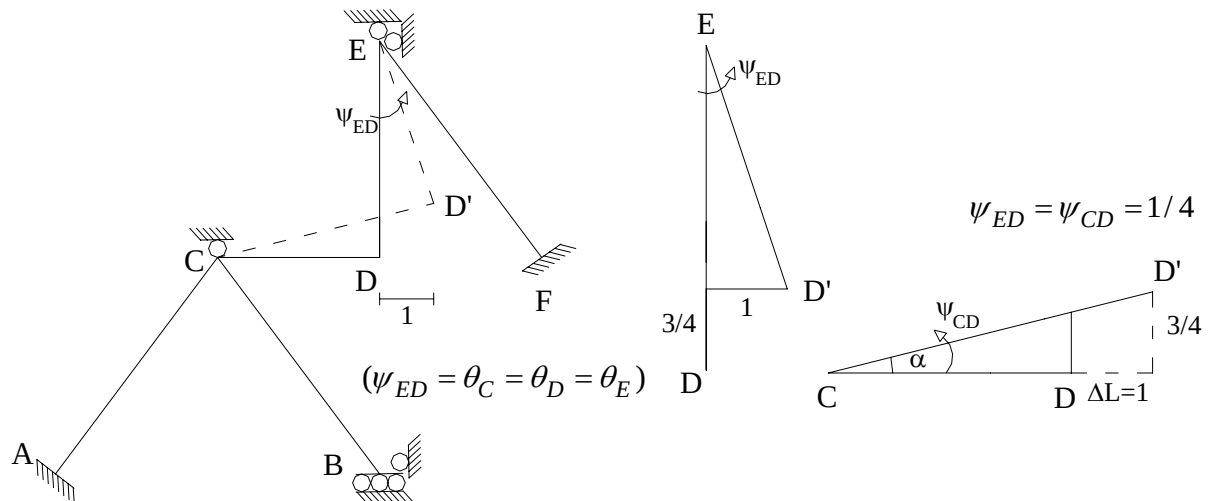
$$m = 0,46$$

$$\psi_{BC} = \frac{m}{5} = \frac{0,46}{5} = 0,092$$

Modo 2 $D_2 = 1 // D_1 = D_3 = D_4 = D_5 = 0$



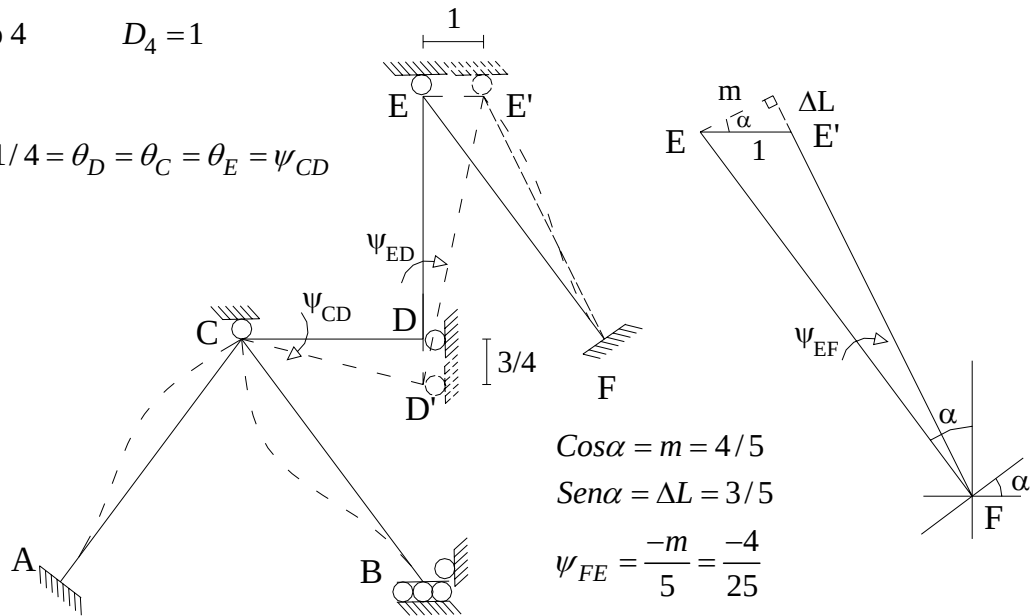
Modo 3 $D_3 = 1 // D_1 = D_2 = D_4 = D_5 = 0$



$$d_{i3} = \begin{vmatrix} 0 \\ 1/4 \\ 1/4 \\ 0 \\ 0 \\ 1 \\ -3/4 \\ 1/4 \\ 0 \\ 0 \end{vmatrix}$$

Modo 4 $D_4 = 1$

$$\psi_{ED} = -1/4 = \theta_D = \theta_C = \theta_E = \psi_{CD}$$



$$\cos \alpha = m = 4/5$$

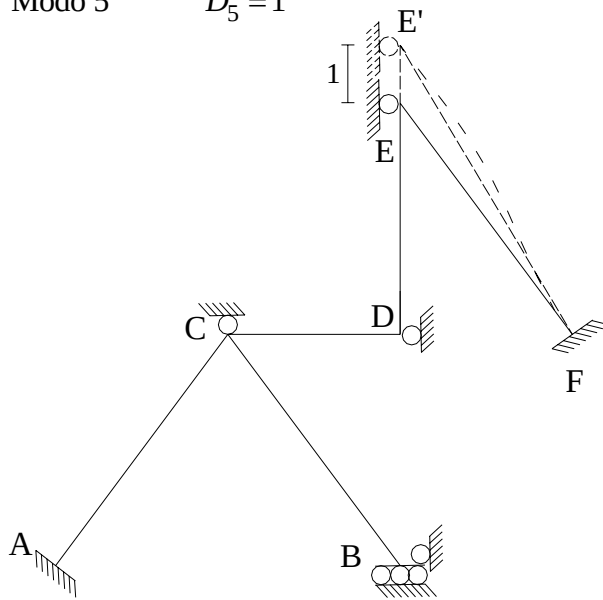
$$\sin \alpha = \Delta L = 3/5$$

$$\psi_{FE} = \frac{-m}{5} = \frac{-4}{25}$$

$$d_{i4} = \begin{vmatrix} 0 \\ -1/4 \\ -1/4 \\ 0 \\ 0 \\ 0 \\ 3/4 \\ -1/4 + 4/25 \\ 0 + 4/25 \\ -3/5 \end{vmatrix} = \begin{vmatrix} 0 \\ -1/4 \\ -1/4 \\ 0 \\ 0 \\ 0 \\ 3/4 \\ -9/100 \\ 4/25 \\ -3/5 \end{vmatrix}$$

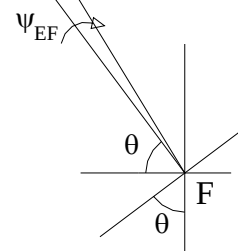
Modo 5

$$D_5 = 1$$

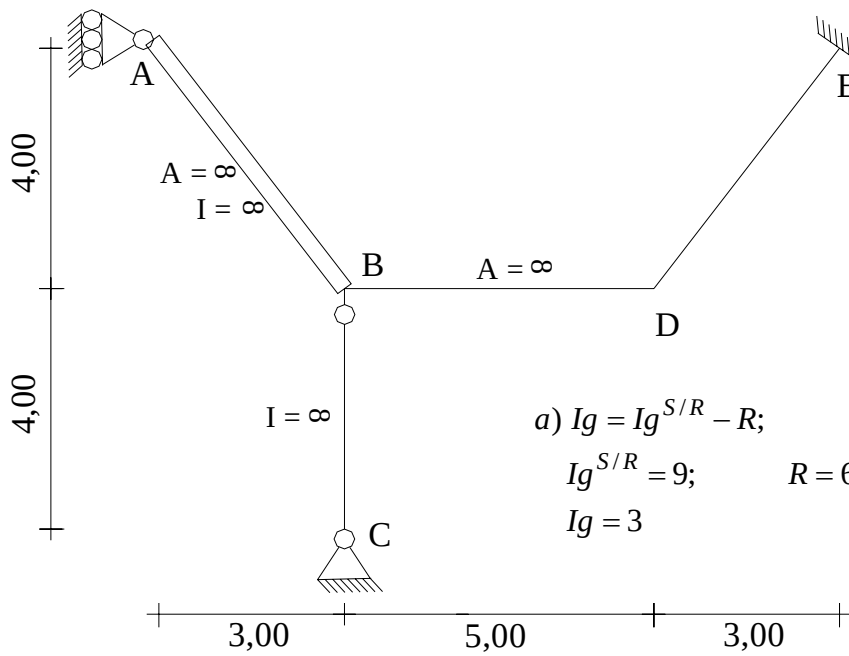


$$\begin{aligned} \cos \theta &= m = 3/5 \\ \sin \theta &= \Delta L = 4/5 \\ \psi_{EF} &= \frac{-m}{5} = \frac{-3}{25} \end{aligned}$$

$$d_{i5} = \begin{vmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 3/25 \\ 3/25 \\ 4/5 \end{vmatrix}$$

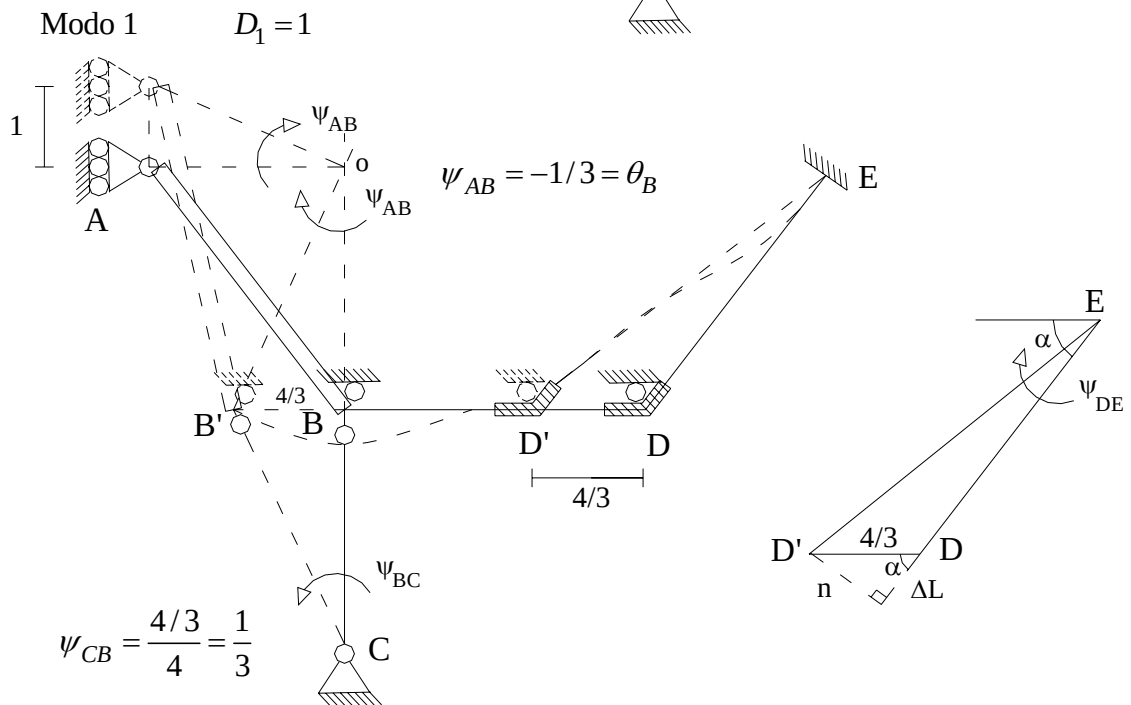
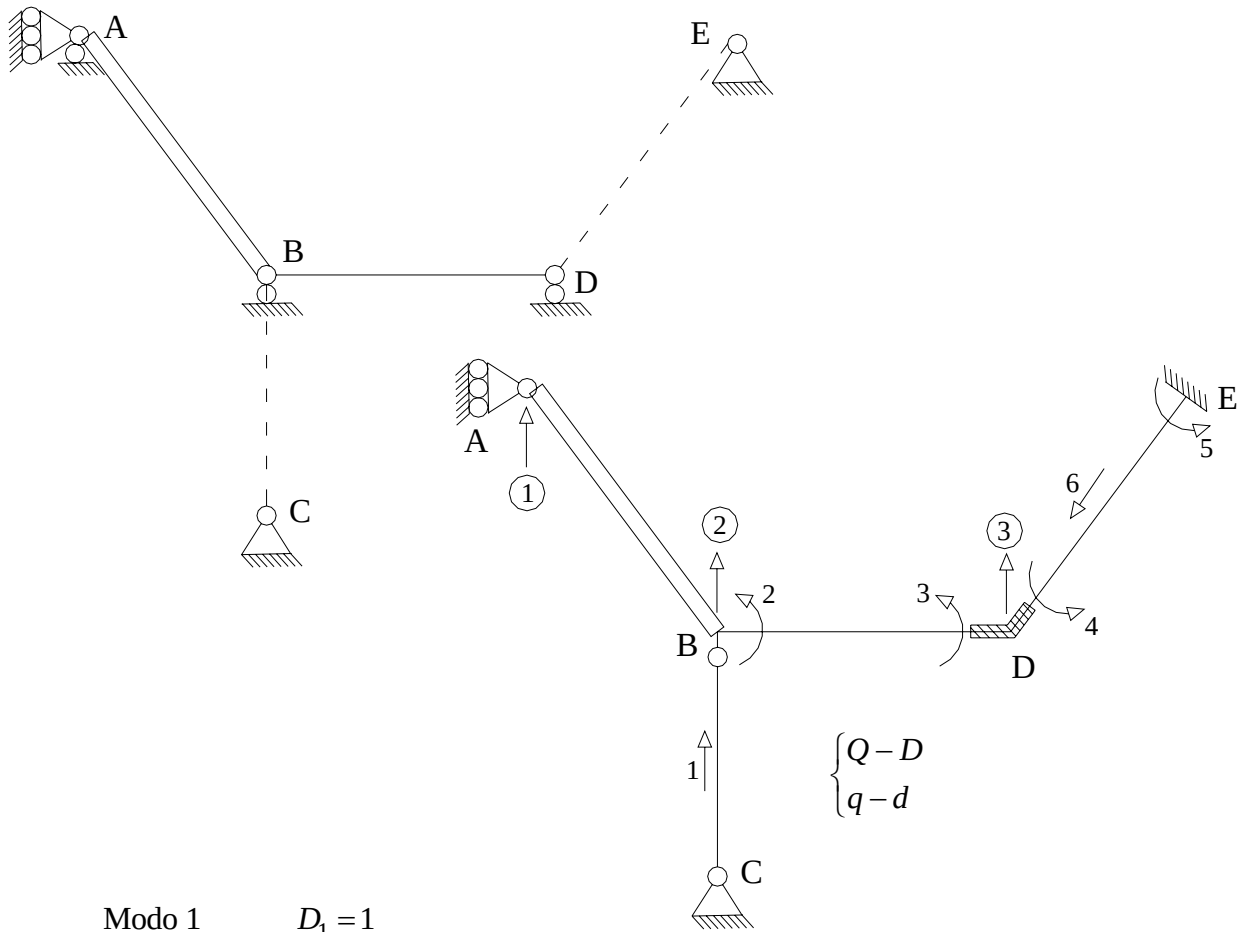


8.-



$$\begin{aligned} a) \quad I_g &= I_g^{S/R} - R; & b) \quad I_g &= G.D. + Rot^{IND} \\ I_g^{S/R} &= 9; & R &= 6 \\ I_g &= 3 \end{aligned}$$

Imagen Cinemática

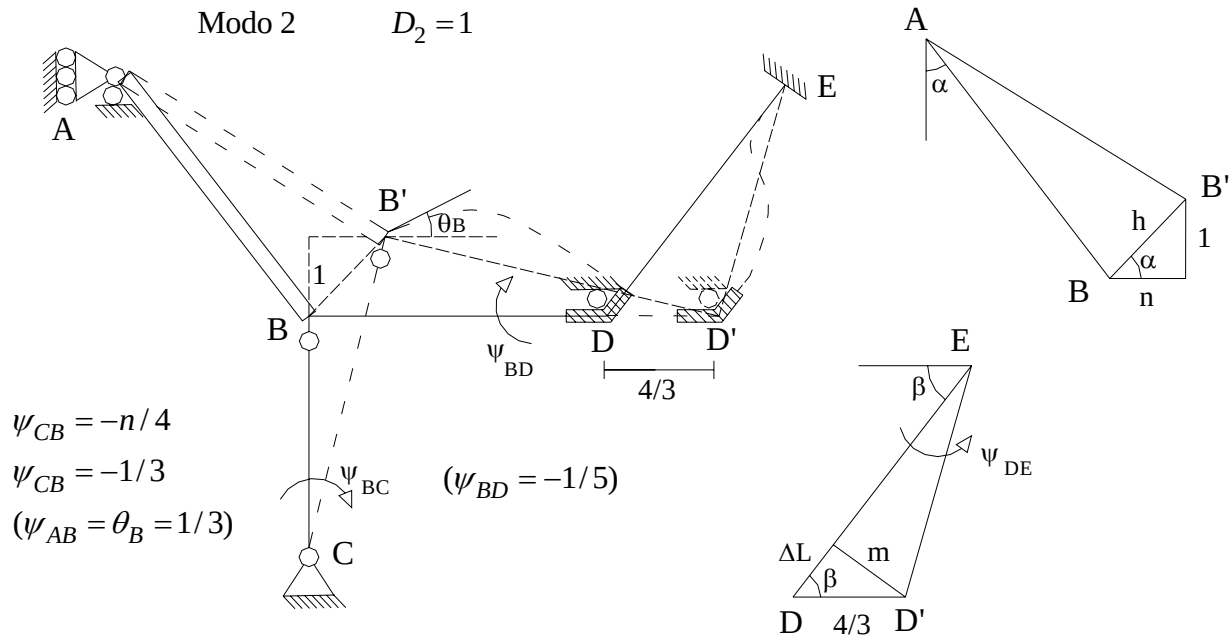


$$d_{i1} = \begin{vmatrix} 0 \\ -1/3 \\ 0 \\ 0+16/75 \\ 0+16/75 \\ 4/5 \end{vmatrix}$$

$$\cos \alpha = \frac{\Delta L}{4/3} = \frac{3}{5} \Rightarrow \Delta L = \frac{3}{5} \times \frac{4}{3} = \frac{4}{5}$$

$$\sin \alpha = \frac{n}{4/3} = \frac{4}{5} \Rightarrow n = \frac{4}{5} \times \frac{4}{3} = \frac{16}{15}$$

$$\psi_{ED} = \frac{-16}{75}$$



$$\tan \alpha = \frac{1}{n} = \frac{3}{4} \Rightarrow n = \frac{3}{4}$$

$$\sin \alpha = \frac{1}{h} = \frac{3}{5} \Rightarrow h = \frac{5}{3}$$

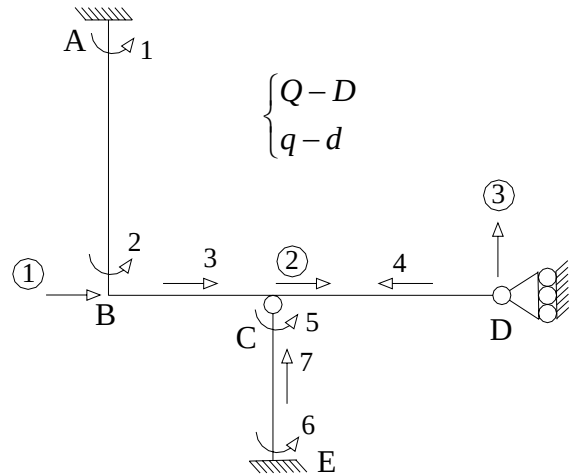
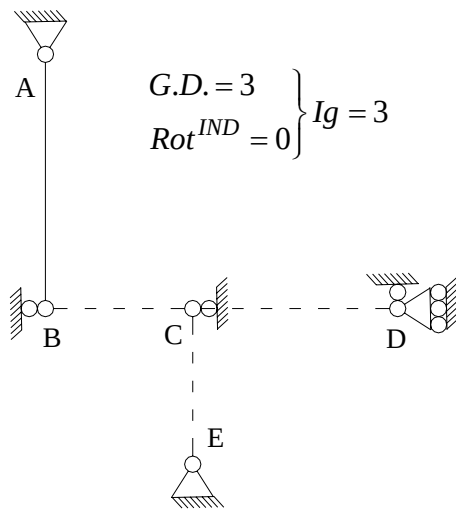
$$\cos \beta = \frac{\Delta L}{4/3} = \frac{3}{5} \Rightarrow \Delta L = \frac{-4}{5}$$

$$\sin \beta = \frac{m}{4/3} = \frac{4}{5} \Rightarrow m = \frac{16}{15}$$

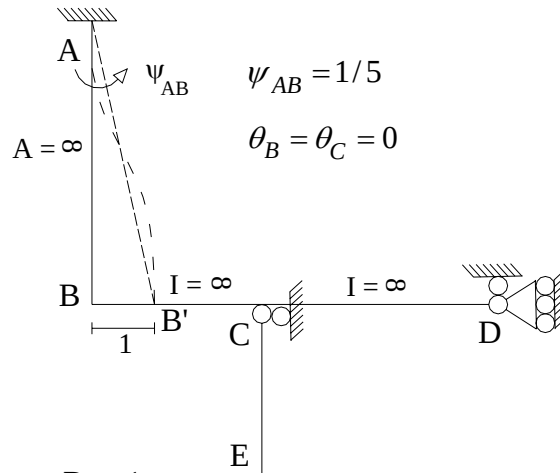
$$\psi_{ED} = 16/75$$

$$d_{i2} = \begin{vmatrix} 1 \\ 1/3+1/5 \\ 0+1/5 \\ 0-16/75 \\ 0-16/75 \\ -4/5 \end{vmatrix} = \begin{vmatrix} 1 \\ 8/15 \\ 1/5 \\ -16/75 \\ -16/75 \\ -4/5 \end{vmatrix}$$

Imagen Cinemática



Modo 1 $D_1 = 1$



$$d_{i1} = \begin{vmatrix} -1/5 \\ -1/5 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{vmatrix}$$

Modo 2 $D_2 = 1$

$$d_{i2} = \begin{vmatrix} 0 \\ 0 \\ 1 \\ -1 \\ 1/3 \\ 1/3 \\ 0 \end{vmatrix}$$

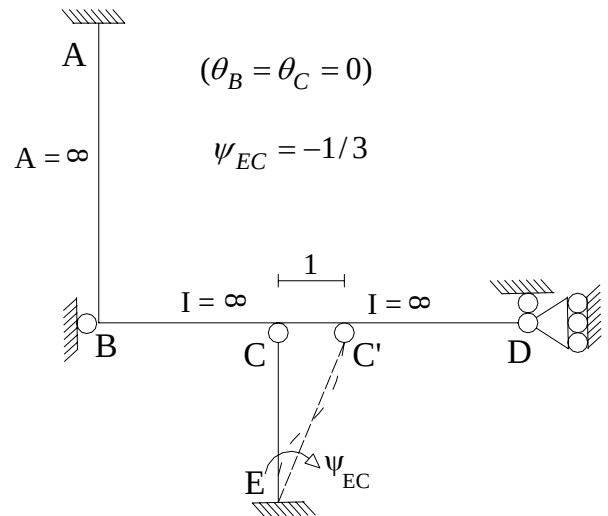
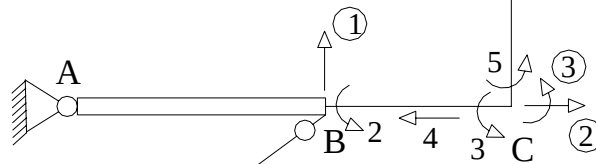
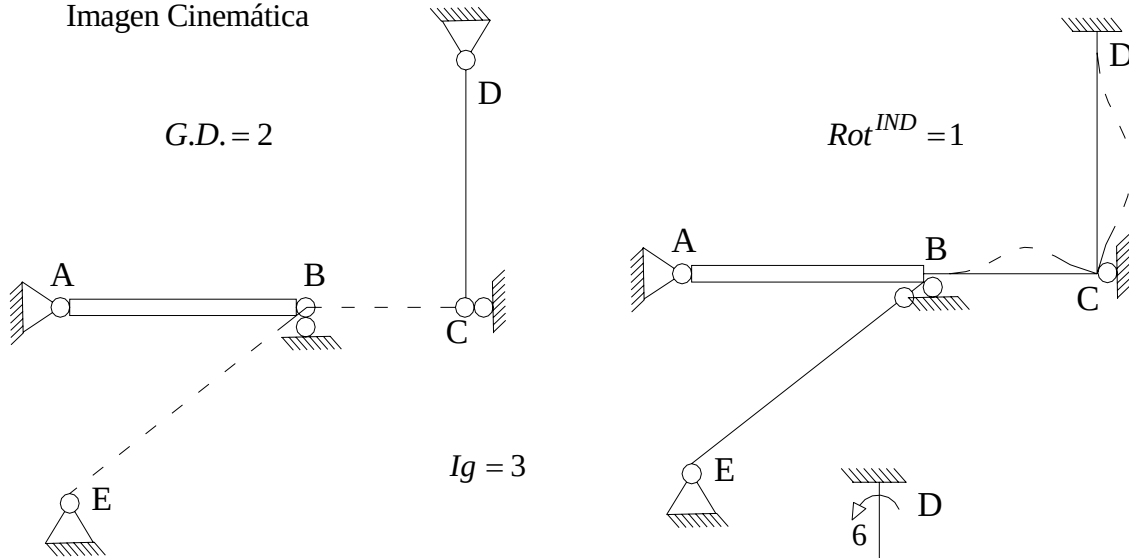


Imagen Cinemática

$$G.D. = 2$$

$$Rot^{IND} = 1$$

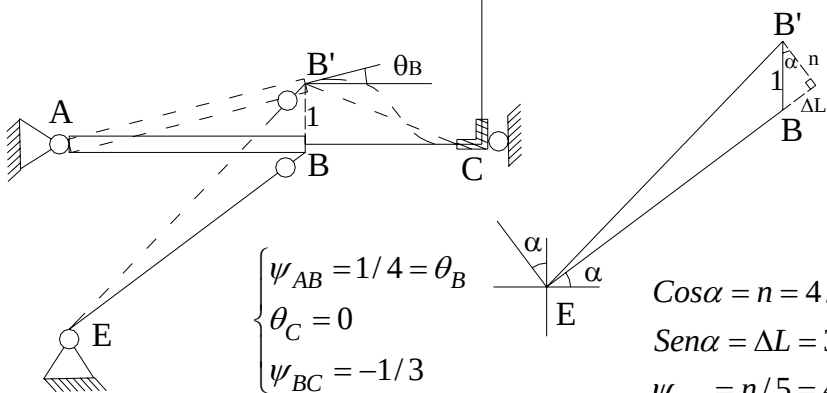
$$I_g = 3$$



$$\begin{cases} Q-D \\ q-d \end{cases}$$

Modo 1

$$D_1 = 1$$

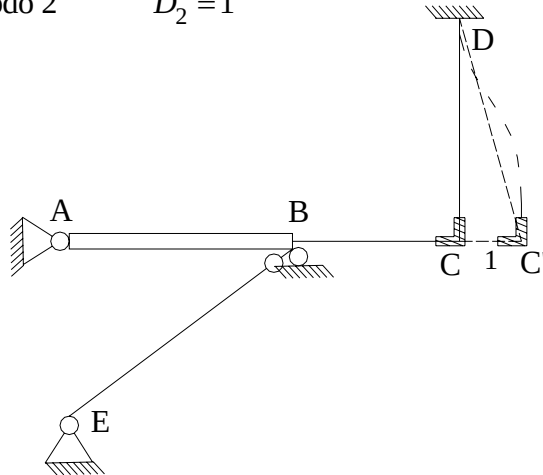


$$\begin{cases} \psi_{AB} = 1/4 = \theta_B \\ \theta_C = 0 \\ \psi_{BC} = -1/3 \end{cases}$$

$$\begin{aligned} \cos \alpha &= n = 4/5 \\ \sin \alpha &= \Delta L = 3/5 \\ \psi_{EB} &= n/5 = 4/25 \end{aligned}$$

$$d_{i1} = \begin{vmatrix} 3/5 \\ 1/4 + 1/3 \\ 0 + 1/3 \\ 0 \\ 0 \\ 0 \end{vmatrix}$$

Modo 2 $D_2 = 1$

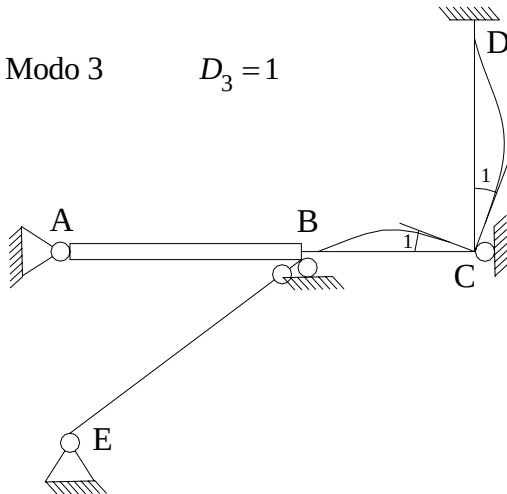


$$(\psi_{DC} = 1/4)$$

$$(\theta_C = \theta_D = 0)$$

$$d_{i2} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ -1/4 \\ -1/4 \end{bmatrix}$$

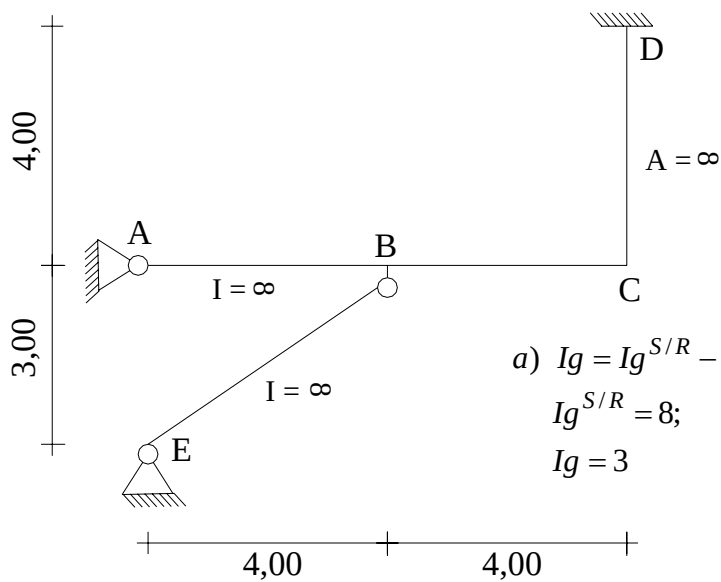
Modo 3 $D_3 = 1$



$$d_{i3} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

$$[A] = \begin{bmatrix} 3/5 & 0 & 0 \\ 7/12 & 0 & 0 \\ 1/3 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & -1/4 & -1 \\ 0 & -1/4 & 0 \end{bmatrix}$$

11.-

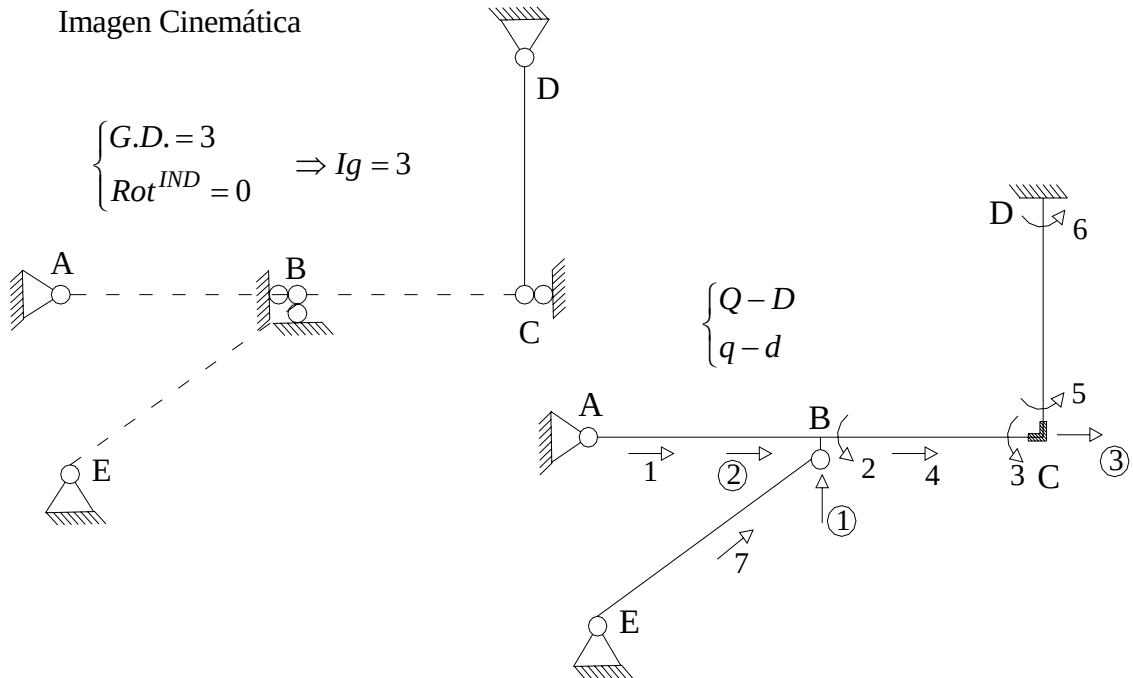


$$a) I_g = I_g^{S/R} - R; \quad b) I_g = G.D. + Rot^{IND}$$

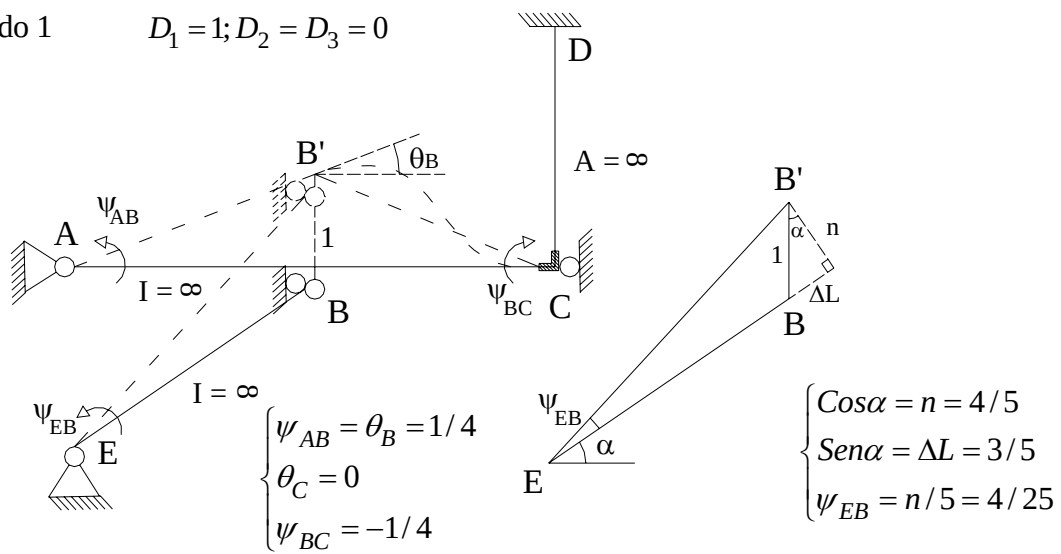
$$I_g^{S/R} = 8; \quad R = 5$$

$$I_g = 3$$

Imagen Cinemática

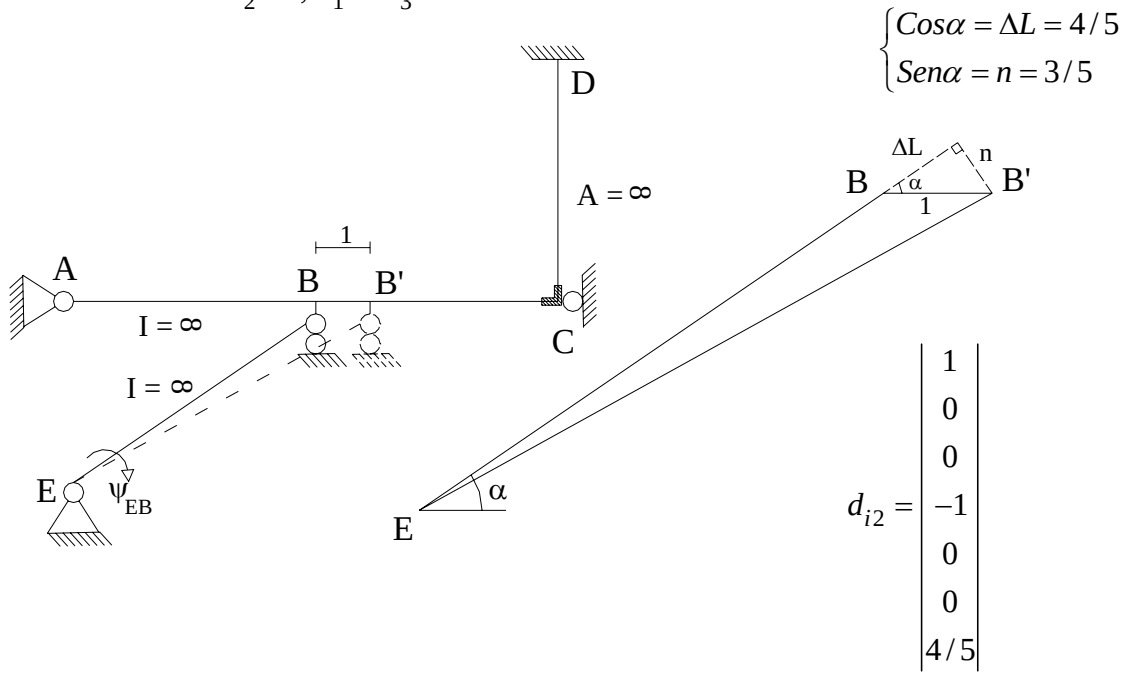


Modo 1 $D_1 = 1; D_2 = D_3 = 0$

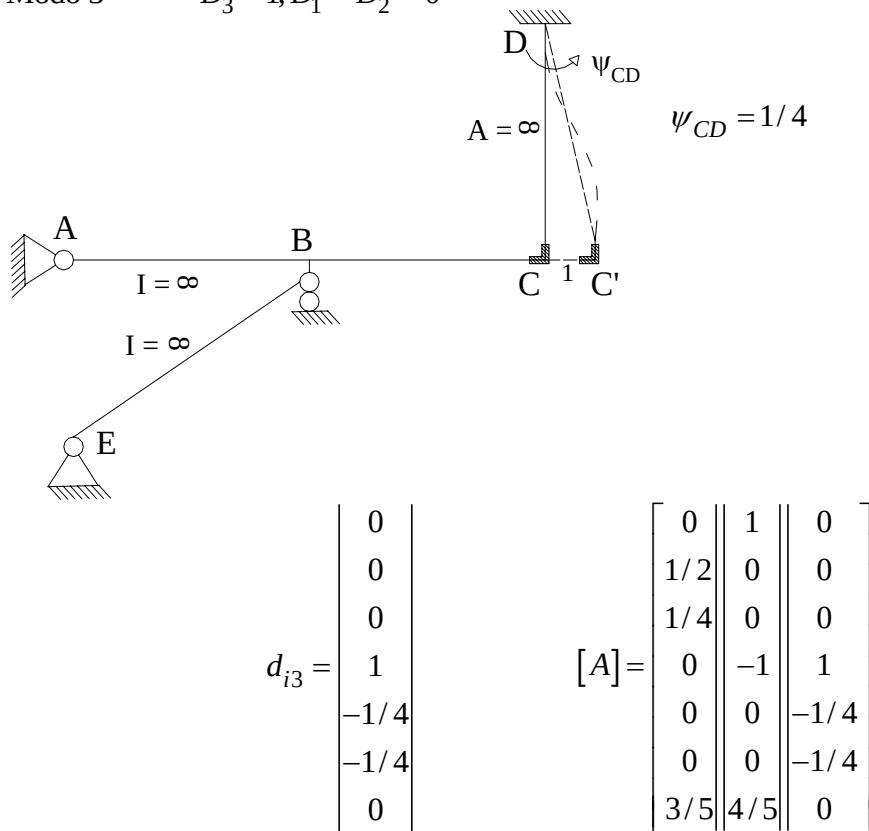


$$d_{i1} = \begin{vmatrix} 0 \\ 1/4 - (-1/4) \\ 0 - (-1/4) \\ 0 \\ 0 \\ 0 \\ 3/5 \end{vmatrix} = \begin{vmatrix} 0 \\ 1/2 \\ 1/4 \\ 0 \\ 0 \\ 0 \\ 3/5 \end{vmatrix}$$

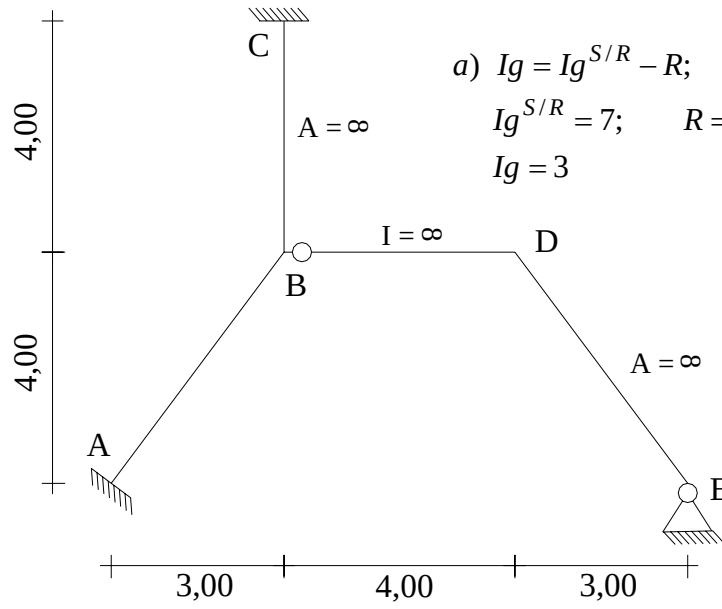
Modo 2 $D_2 = 1; D_1 = D_3 = 0$



Modo 3 $D_3 = 1; D_1 = D_2 = 0$



12.-

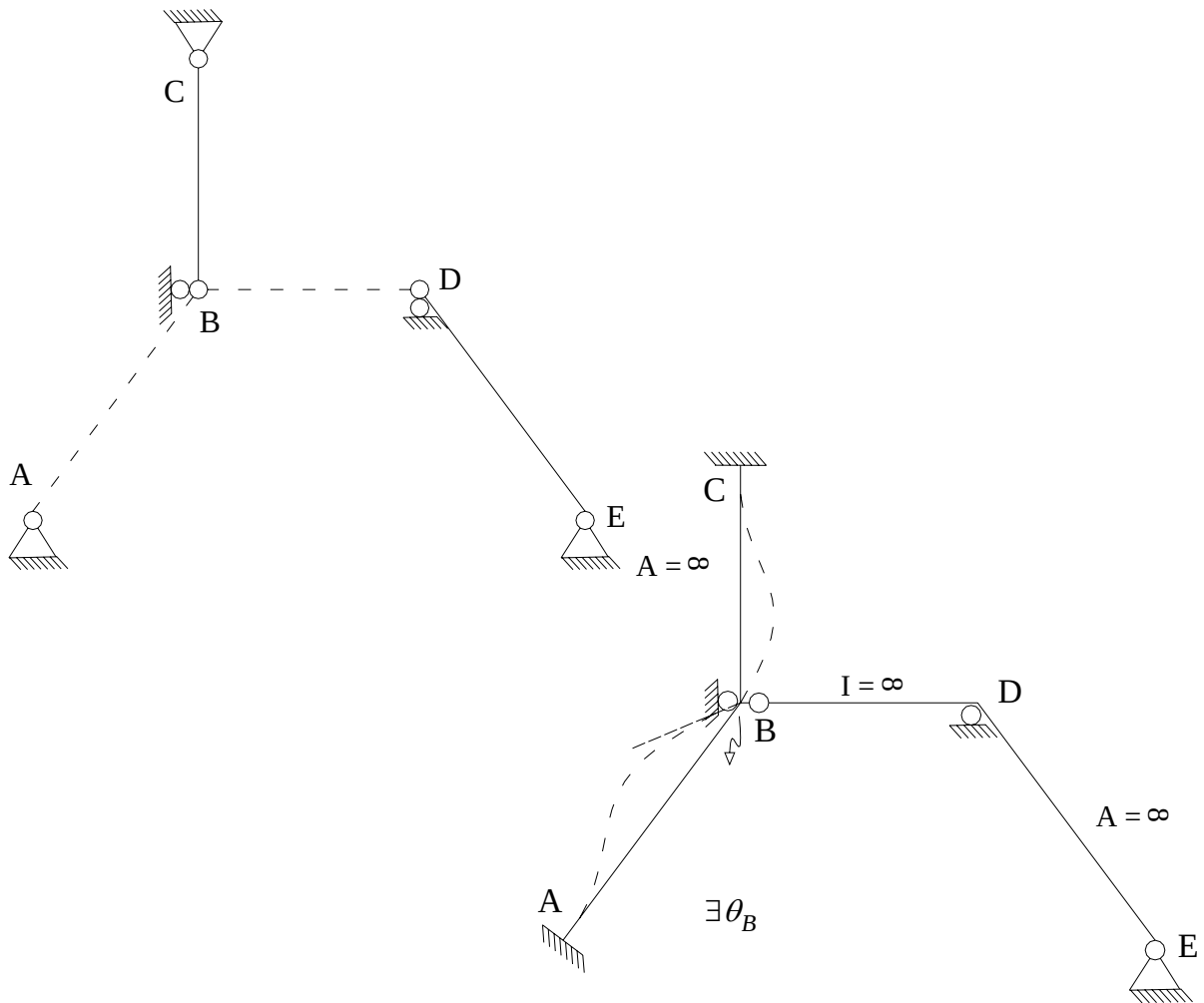


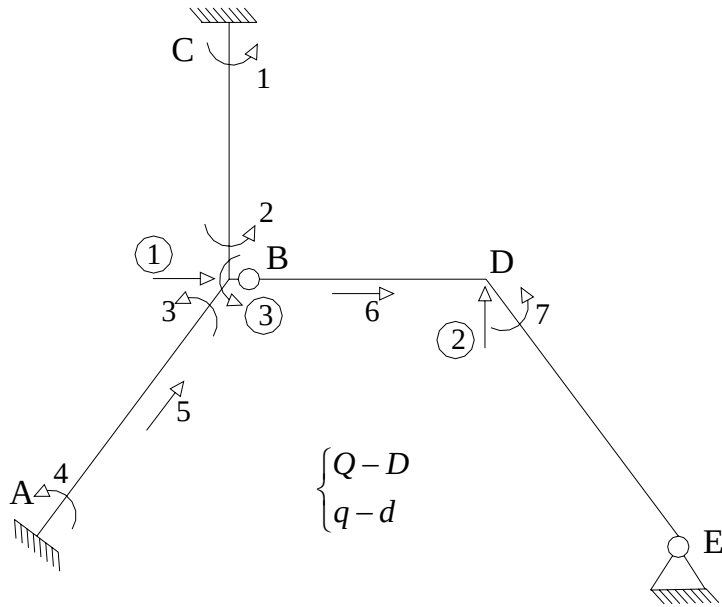
$$a) I_g = I_g^{S/R} - R; \quad b) I_g = G.D. + Rot^{IND}$$

$$I_g^{S/R} = 7; \quad R = 4$$

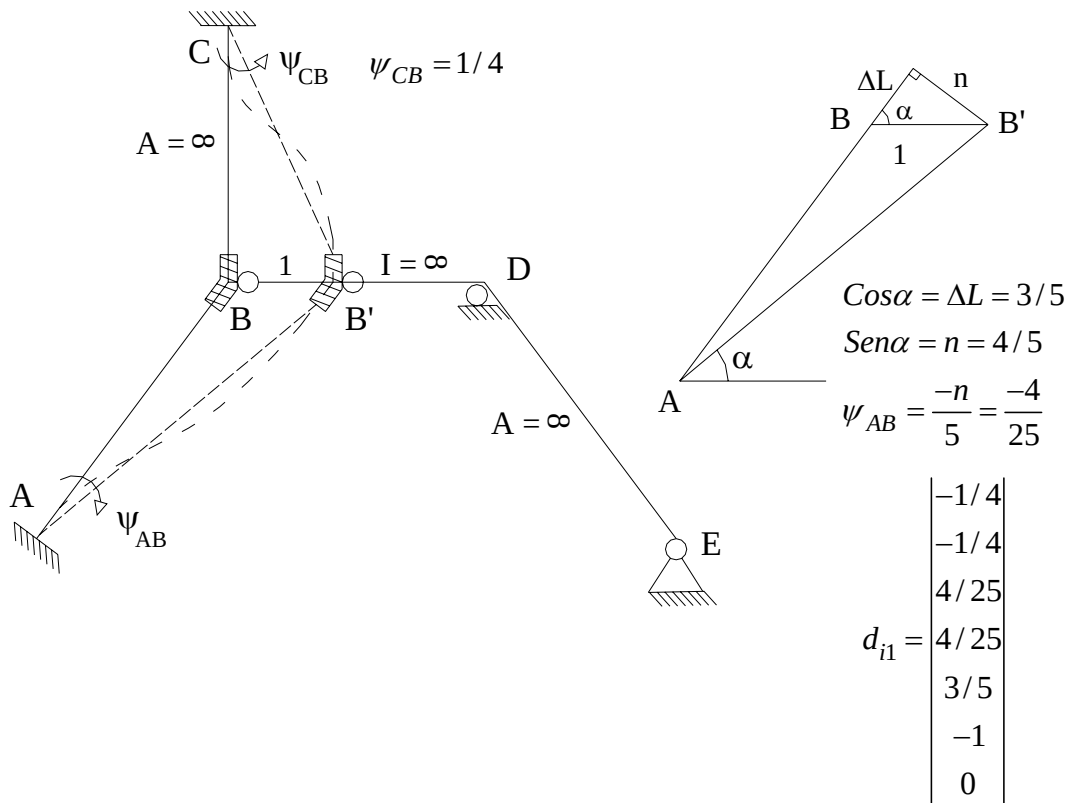
$$I_g = 3$$

Imagen Cinemática

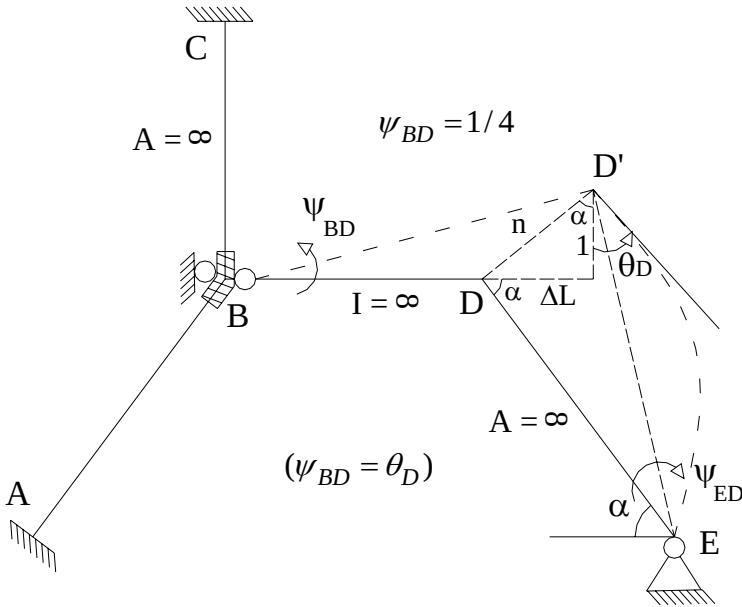




Modo 1 $D_1 = 1; D_2 = D_3 = 0$



Modo 2 $D_2 = 1; D_1 = D_3 = 0$



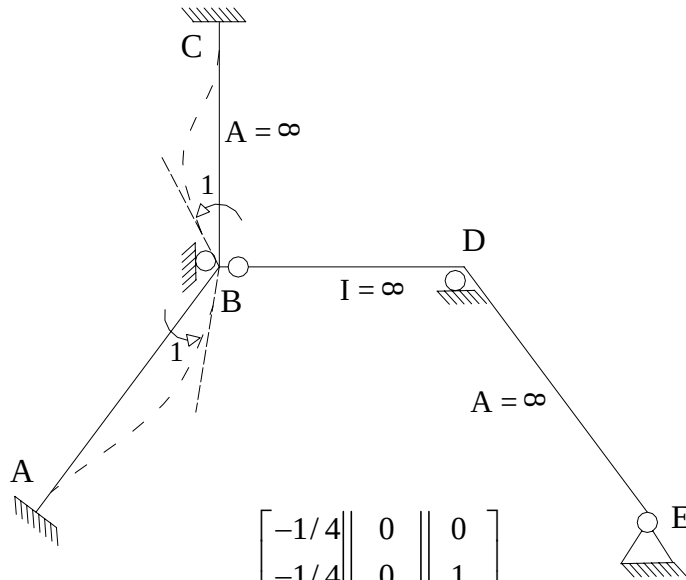
$$\operatorname{Tg} \alpha = \frac{\Delta L}{1} = \frac{4}{3} \Rightarrow \Delta L = \frac{4}{3}$$

$$\operatorname{Cos} \alpha = \frac{1}{n} = \frac{3}{5} \Rightarrow n = \frac{5}{3}$$

$$\psi_{ED} = \frac{-n}{5} = \frac{-1}{3}$$

$$d_{i2} = \begin{vmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 4/3 \\ 1/4 - (-1/3) \end{vmatrix}$$

Modo 3 $D_3 = 1; D_1 = D_2 = 0$



$$d_{i3} = \begin{vmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{vmatrix}$$

$$[A] = \begin{bmatrix} -1/4 & 0 & 0 \\ -1/4 & 0 & 1 \\ 4/25 & 0 & 1 \\ 4/25 & 0 & 0 \\ 3/5 & 0 & 0 \\ -1 & 4/3 & 0 \\ 0 & 7/12 & 0 \end{bmatrix}$$

Diagram of a frame structure for a structural analysis problem. The structure consists of a horizontal beam BCD and a vertical column DF. A diagonal member DE is also present. The beam BCD has a total length of 7.00 m, with a hinge at C (3.00 m from B). The column DF is 4.00 m high. The diagonal member DE is 5.00 m long. The structure is supported by a pin support at A (3.00 m from B) and a roller support at F. The frame has a constant flexural rigidity EI . The problem asks for the influence of a unit load moving on the beam BCD on the reaction at F and the bending moment at D.

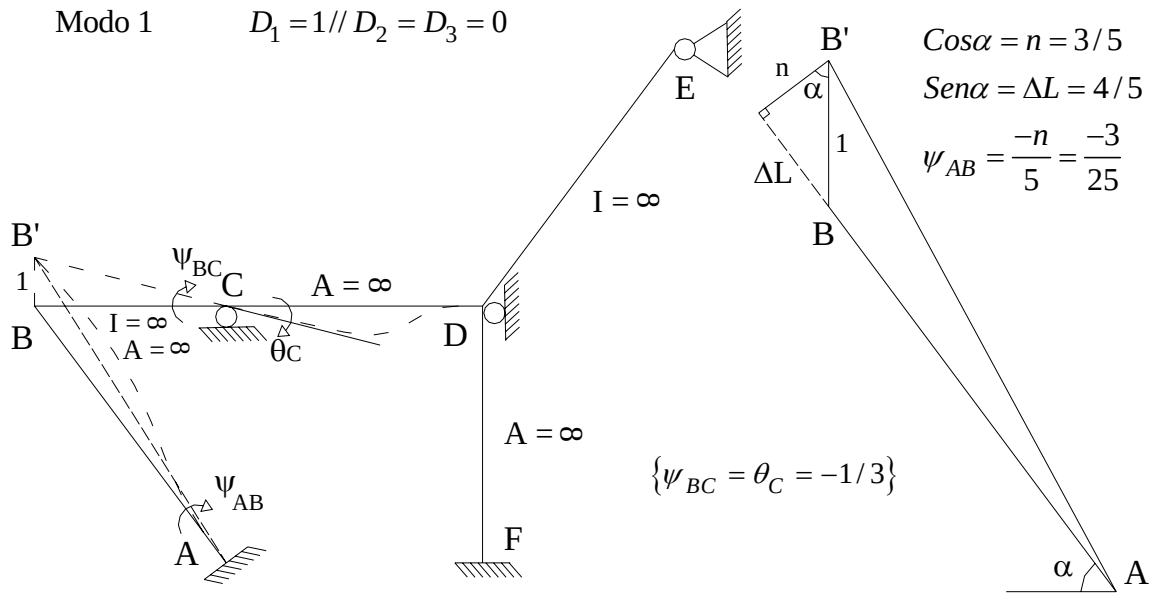
$$b) Ig = G.D. + Rot^{IND}$$
$$I_g = 3$$

Imagen Cinemática

$G.D. = 3$
 $Rot^{IND} = 0$ } $Ig = 3$

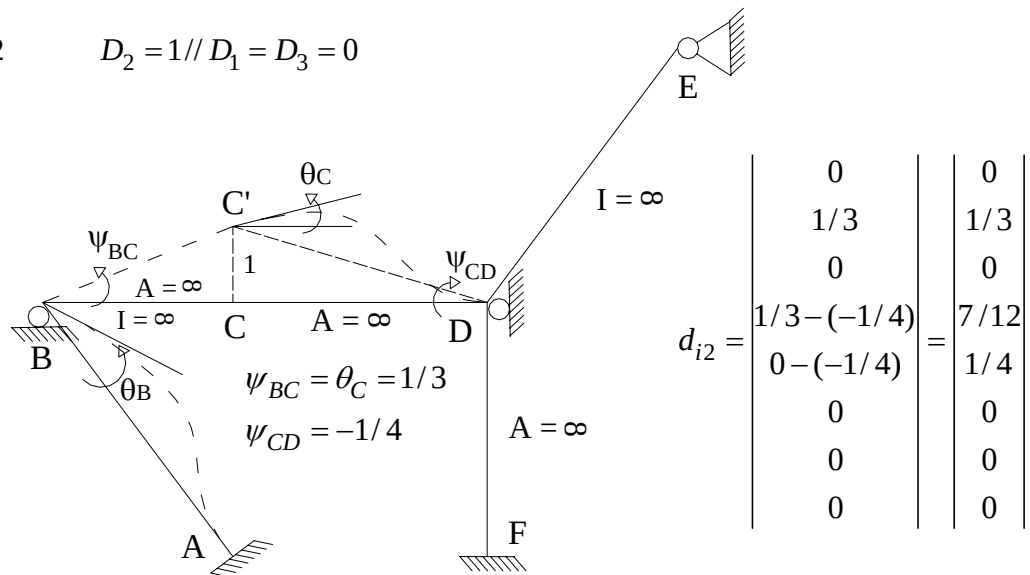
$\begin{cases} Q-D \\ q-d \end{cases}$

Modo 1 $D_1 = 1 // D_2 = D_3 = 0$



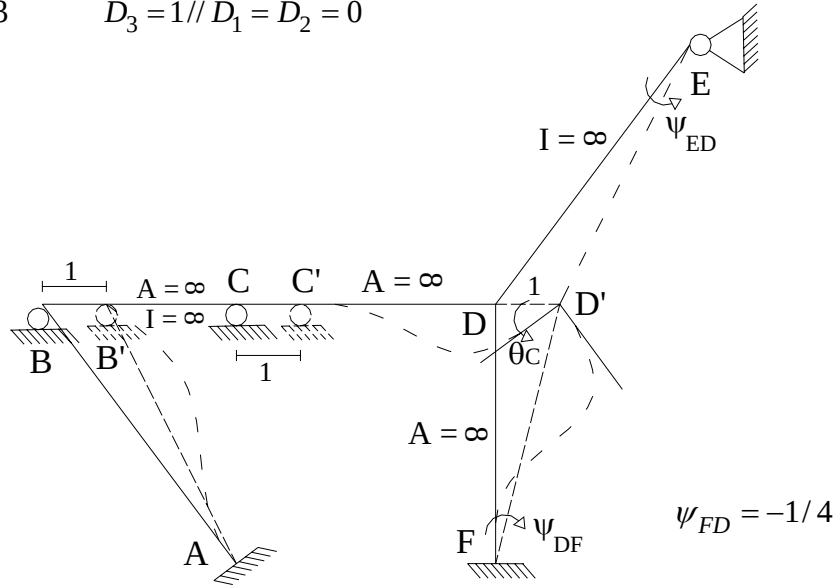
$$d_{i1} = \begin{vmatrix} 0 - (-3/15) \\ -1/3 - (-3/25) \\ 4/5 \\ -1/3 \\ 0 \\ 0 \\ 0 \\ 0 \end{vmatrix} = \begin{vmatrix} 3/25 \\ -16/75 \\ 4/5 \\ -1/3 \\ 0 \\ 0 \\ 0 \\ 0 \end{vmatrix}$$

Modo 2 $D_2 = 1 // D_1 = D_3 = 0$



$$d_{i2} = \begin{vmatrix} 0 \\ 1/3 \\ 0 \\ 1/3 - (-1/4) \\ 0 - (-1/4) \\ 0 \\ 0 \\ 0 \end{vmatrix} = \begin{vmatrix} 0 \\ 1/3 \\ 0 \\ 7/12 \\ 1/4 \\ 0 \\ 0 \\ 0 \end{vmatrix}$$

Modo 3 $D_3 = 1 // D_1 = D_2 = 0$



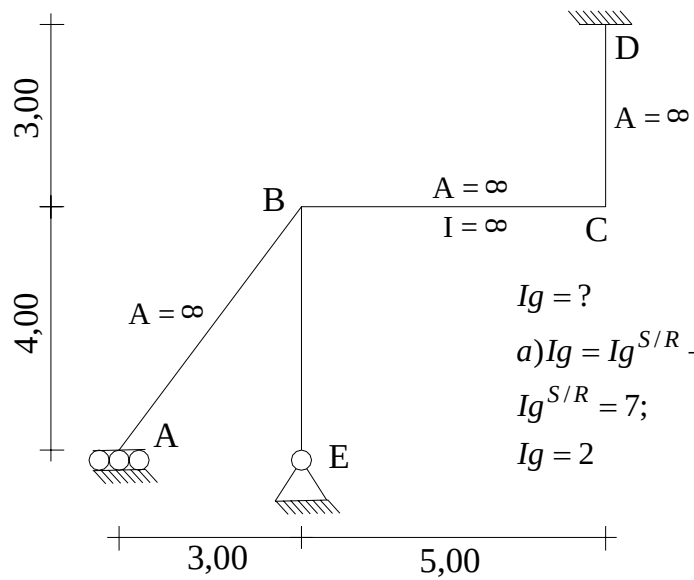
$\cos \alpha = \Delta L = -3/5$
 $\sin \alpha = n = 4/5$
 $\psi_{AB} = \frac{-n}{5} = \frac{-4}{25}$

$\cos \alpha = \Delta L = -3/5$
 $\sin \alpha = n = 4/5$
 $\psi_{ED} = \frac{n}{5} = \frac{4}{25} = \theta_D$

$$d_{i3} = \begin{vmatrix} 4/25 & 4/25 \\ 4/25 & 4/25 \\ -3/25 & -3/5 \\ 0 & 0 \\ 4/25 & 4/25 \\ -3/5 & -3/5 \\ 4/25 + 1/4 & 41/100 \\ 0 + 1/4 & 1/4 \end{vmatrix} = \begin{vmatrix} 4/25 & 4/25 \\ 4/25 & 4/25 \\ -3/25 & -3/5 \\ 0 & 0 \\ 4/25 & 4/25 \\ -3/5 & -3/5 \\ 41/100 & 41/100 \\ 1/4 & 1/4 \end{vmatrix}$$

$$[A] = \begin{bmatrix} 3/25 & 0 & 4/25 \\ -16/75 & 1/3 & 4/25 \\ 4/5 & 0 & -3/5 \\ -1/3 & 7/12 & 0 \\ 0 & 1/4 & 4/25 \\ 0 & 0 & -3/5 \\ 0 & 0 & 41/100 \\ 0 & 0 & 1/4 \end{bmatrix}$$

14.-



$$I_g = ?$$

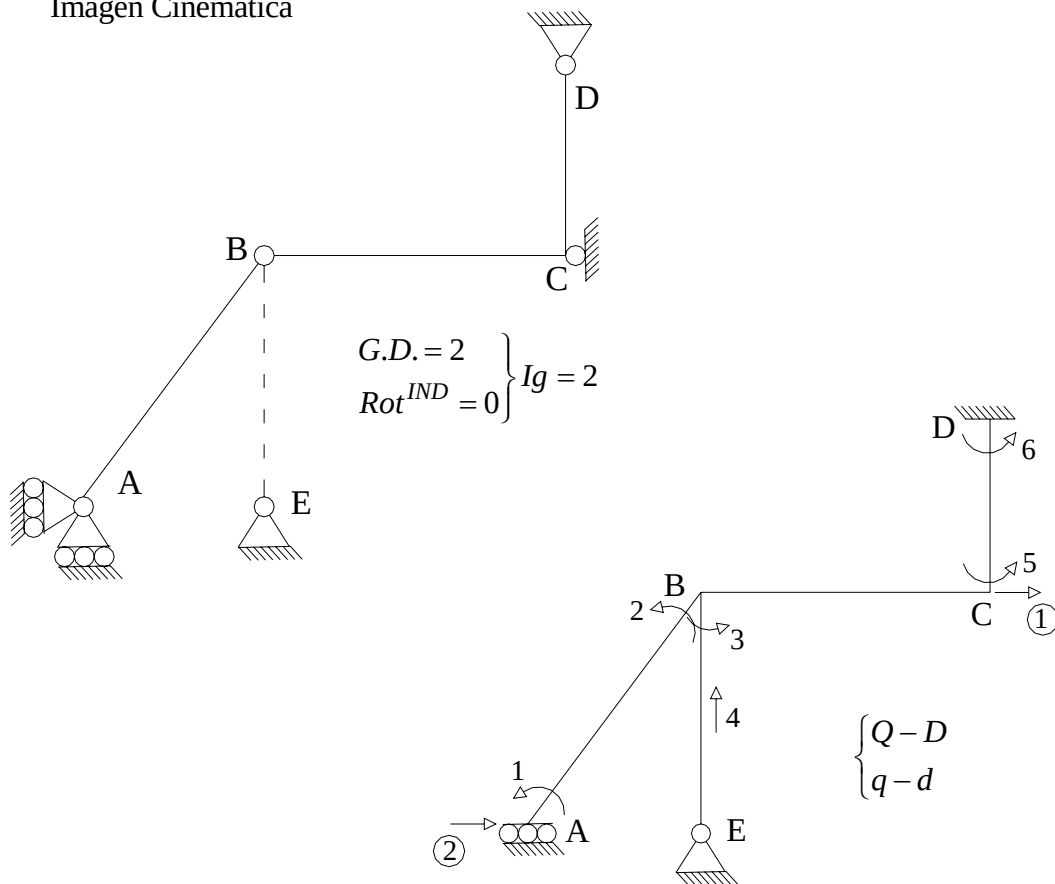
$$a) I_g = I_g^{S/R} - R;$$

$$I_g^{S/R} = 7; \quad R = 5$$

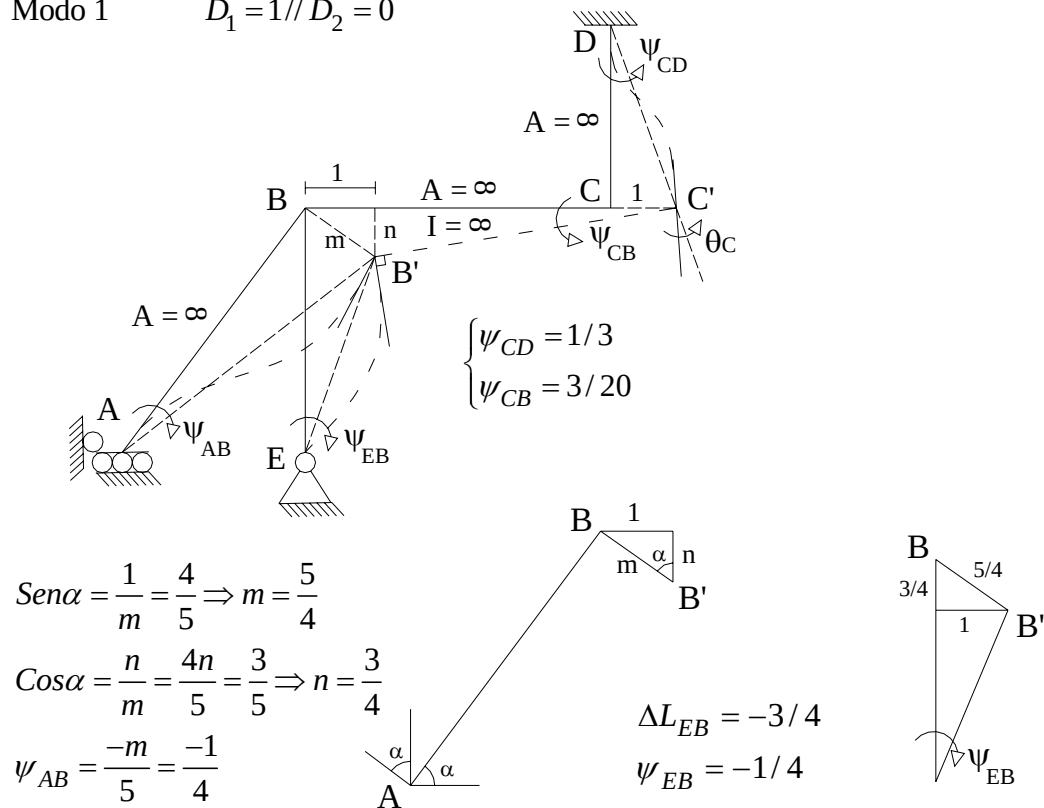
$$I_g = 2$$

$$b) I_g = G.D. + Rot^{IND}$$

Imagen Cinemática

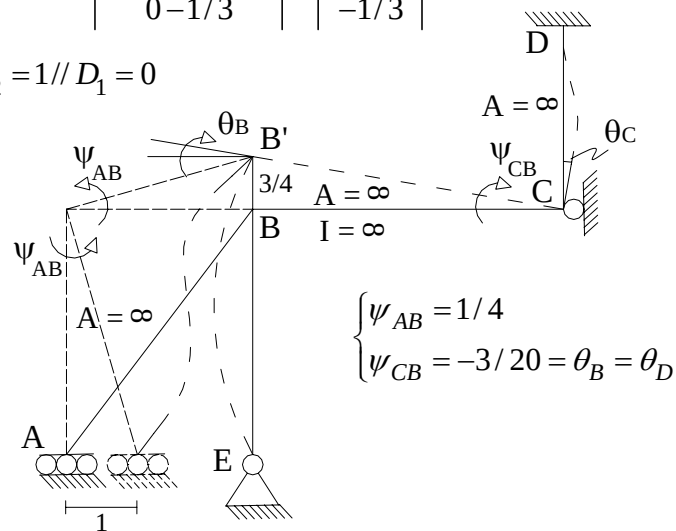


Modo 1 $D_1 = 1 // D_2 = 0$



$$d_{i1} = \begin{vmatrix} 0 - (-1/4) \\ 3/20 - (-1/4) \\ 3/20 - (-1/4) \\ -3/4 \\ 3/20 - 1/3 \\ 0 - 1/3 \end{vmatrix} = \begin{vmatrix} 1/4 \\ 2/5 \\ 2/5 \\ -3/4 \\ -11/60 \\ -1/3 \end{vmatrix}$$

Modo 2 $D_2 = 1 // D_1 = 0$



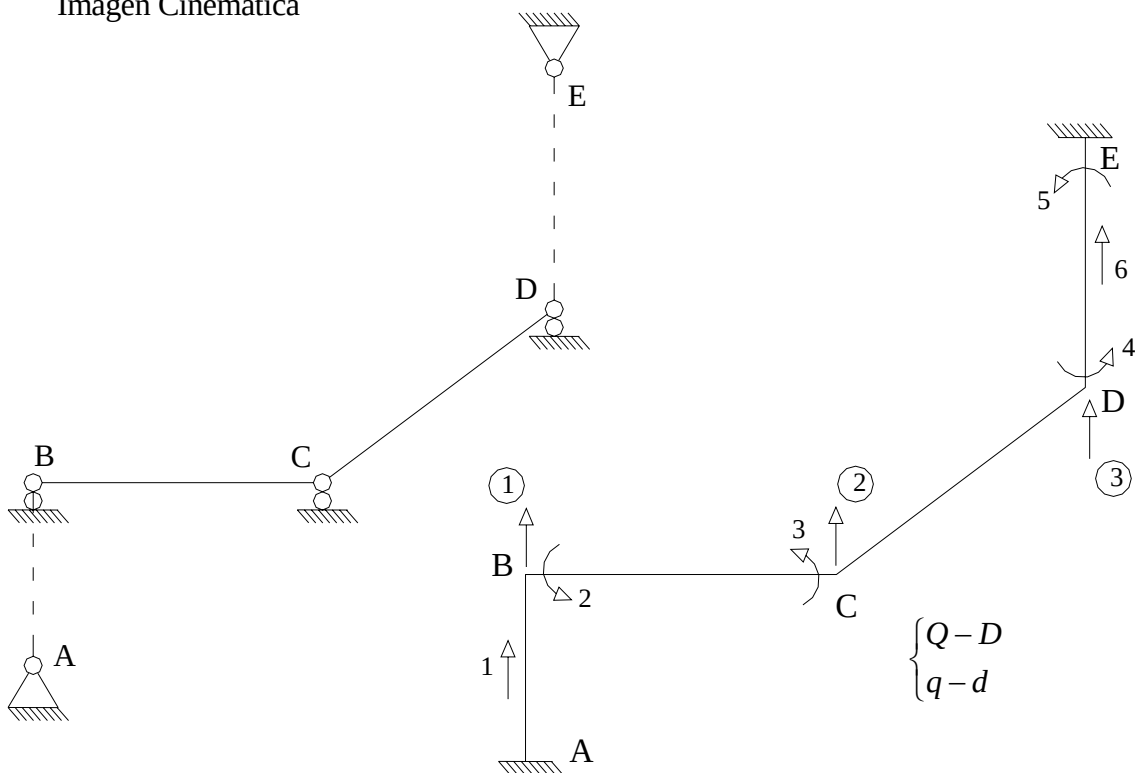
15.-

$$[A] = \begin{bmatrix} 1/4 & -1/4 \\ 2/5 & -2/5 \\ 2/5 & -3/20 \\ -3/4 & 3/4 \\ -11/60 & -3/20 \\ -1/3 & 0 \end{bmatrix}$$

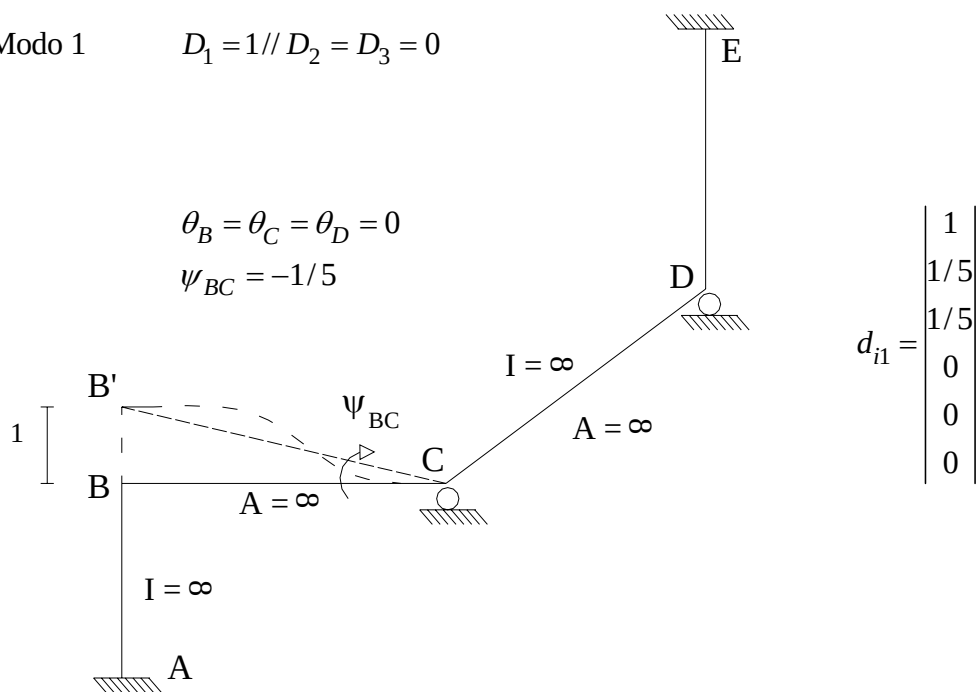


$$Ig = 3$$

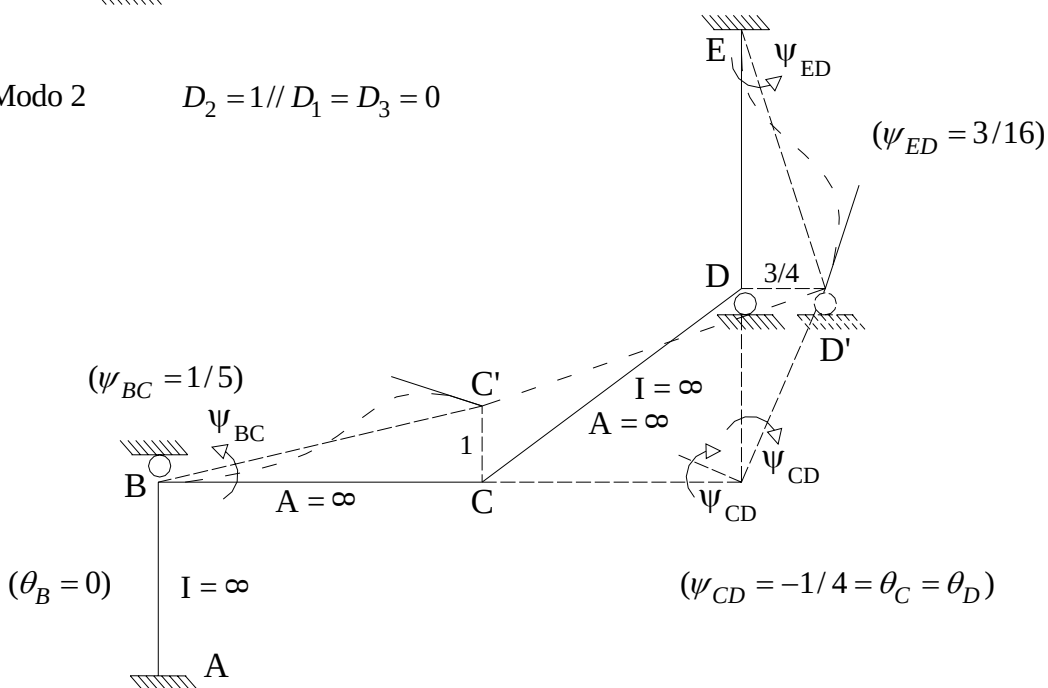
$$R = 6$$



Modo 1 $D_1 = 1 // D_2 = D_3 = 0$



Modo 2 $D_2 = 1 // D_1 = D_3 = 0$



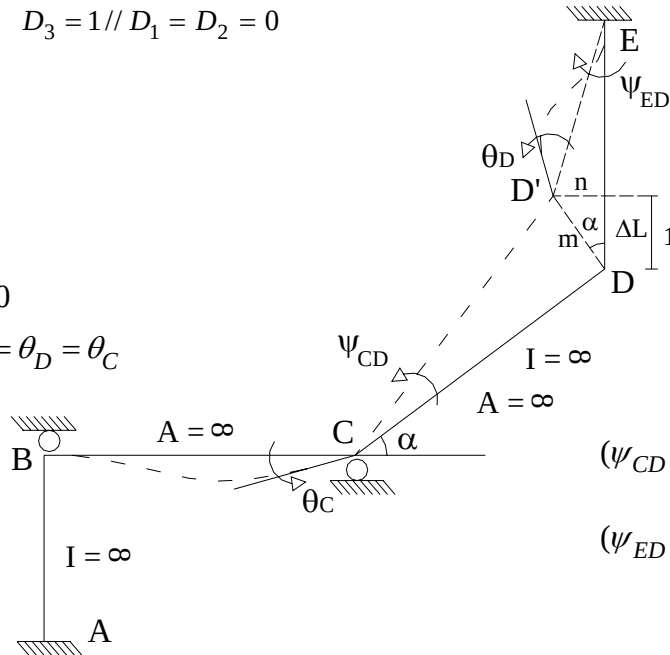
$$d_{i2} = \begin{vmatrix} 0 \\ 0 - 1/5 \\ -1/4 - 1/5 \\ -1/4 - 3/16 \\ 0 - 3/16 \\ 0 \end{vmatrix} = \begin{vmatrix} 0 \\ -1/5 \\ -9/20 \\ -7/16 \\ -3/16 \\ 0 \end{vmatrix}$$

Modo 3

$$D_3 = 1 // D_1 = D_2 = 0$$

$$\theta_B = 0$$

$$\psi_{CD} = \theta_D = \theta_C$$



$$\Delta L_{ED} = 1$$

$$\cos \alpha = \frac{\Delta L}{m} = \frac{4}{5}$$

$$m = \frac{5\Delta L}{4} = \frac{5}{4}$$

$$\sin \alpha = \frac{n}{m} = \frac{3}{5}$$

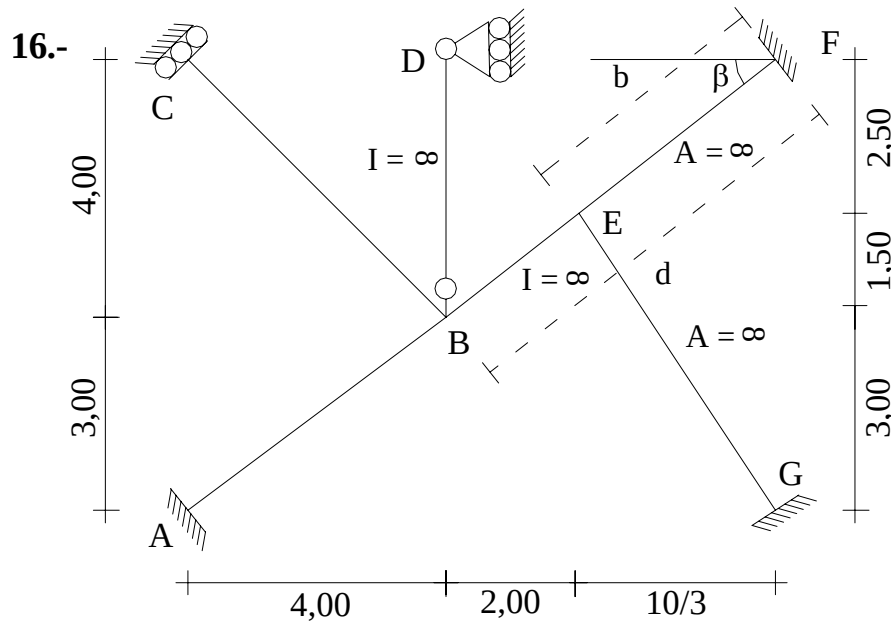
$$n = \frac{3m}{5} = \frac{3}{4}$$

$$(\psi_{CD} = \frac{m}{5} = \frac{1}{4})$$

$$(\psi_{ED} = \frac{-n}{4} = \frac{-3}{16})$$

$$d_{i3} = \begin{vmatrix} 0 & 0 \\ 0 & 0 \\ 1/4 & 1/4 \\ 1/4 - (-3/16) & 7/16 \\ 0 - (-3/16) & 3/16 \\ -1 & -1 \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ 0 & 0 \\ 1/4 & 1/4 \\ 7/16 & 7/16 \\ 3/16 & 3/16 \\ -1 & -1 \end{vmatrix}$$

$$[A] = \begin{bmatrix} 0 & 0 & 0 \\ 1/5 & -1/5 & 0 \\ 1/5 & -9/20 & 1/4 \\ 0 & -7/16 & 7/16 \\ 0 & -3/16 & 3/16 \\ 0 & 0 & -1 \end{bmatrix}$$



$$\sin \beta = \frac{3}{5} = \frac{4}{d}$$

$$d = 20/3$$

$$\sin \beta = \frac{3}{5} = \frac{2.5}{b}$$

$$b = 12.5/3$$

$$d_{BF} = 7.5/3 = 2.5$$

$$I_g = ?$$

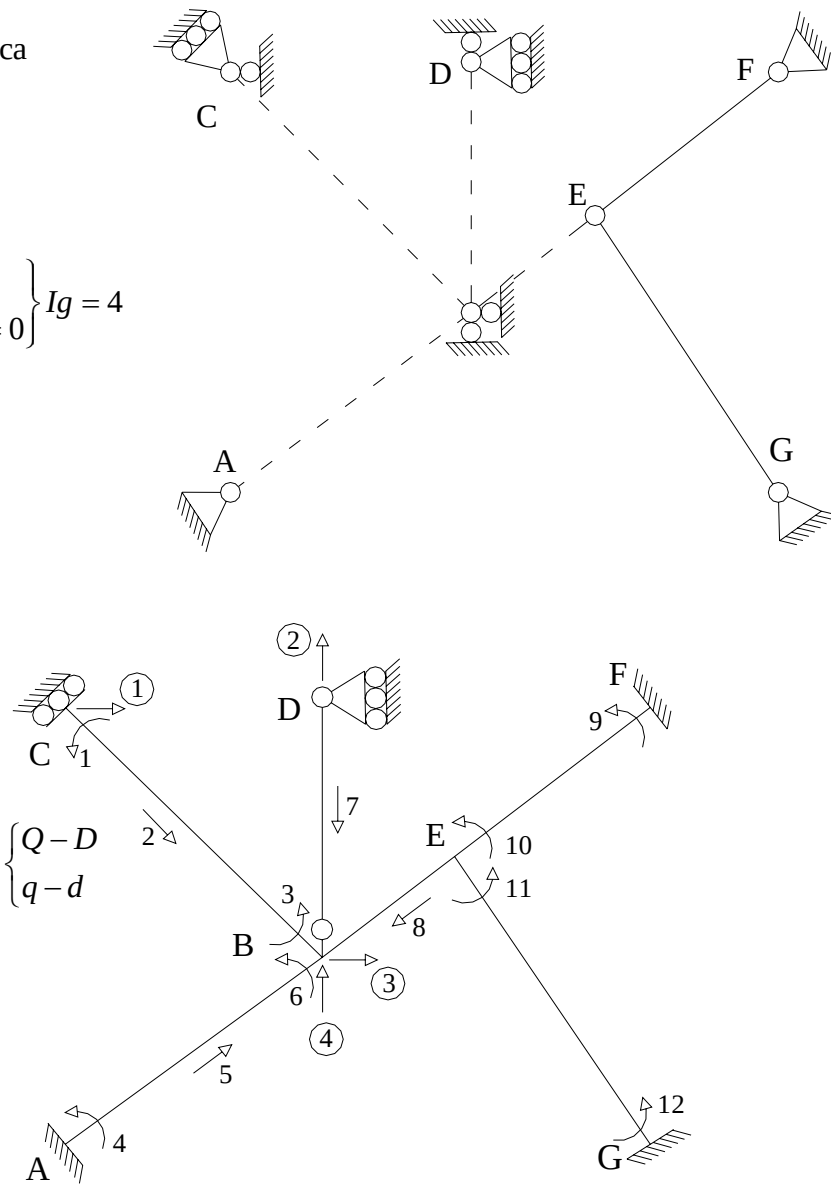
$$a) I_g = I_g^{S/R} - R; \quad b) I_g = G.D. + Rot^{IND}$$

$$I_g^{S/R} = 10; \quad R = 6$$

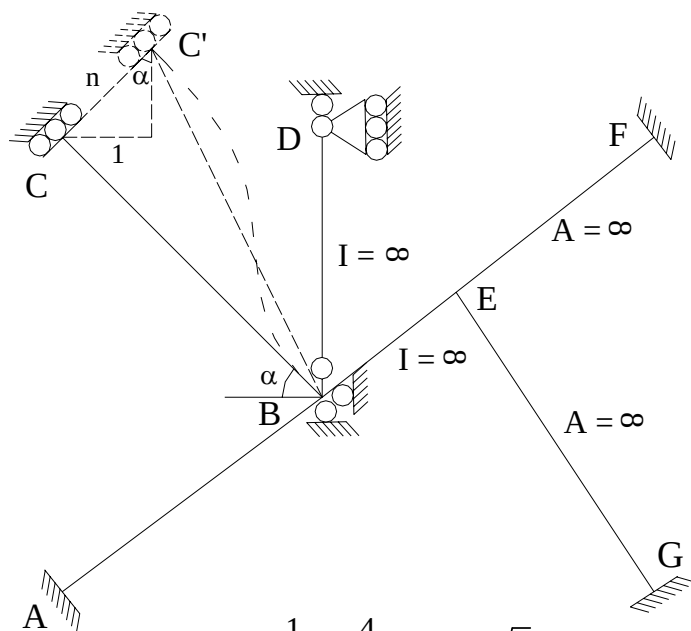
$$I_g = 6$$

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$$\left. \begin{array}{l} G.D. = 4 \\ Rot^{IND} = 0 \end{array} \right\} I_g = 4$$



Modo 1 $D_1 = 1 // D_2 = D_3 = D_4 = 0$

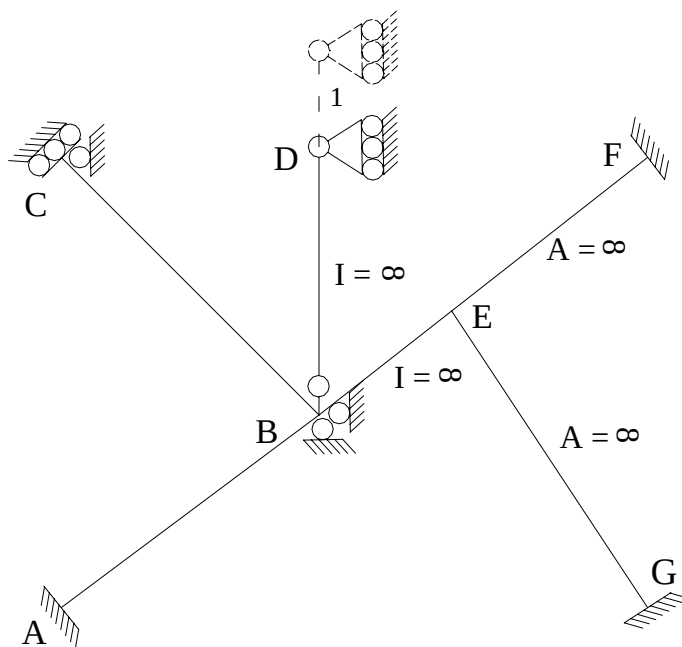


$$\text{Sen} \alpha = \frac{1}{n} = \frac{4}{4\sqrt{2}} \Rightarrow n = \sqrt{2}$$

$$\psi_{CB} = \frac{-n}{4\sqrt{2}} = \frac{-\sqrt{2}}{4\sqrt{2}} = \frac{-1}{4}$$

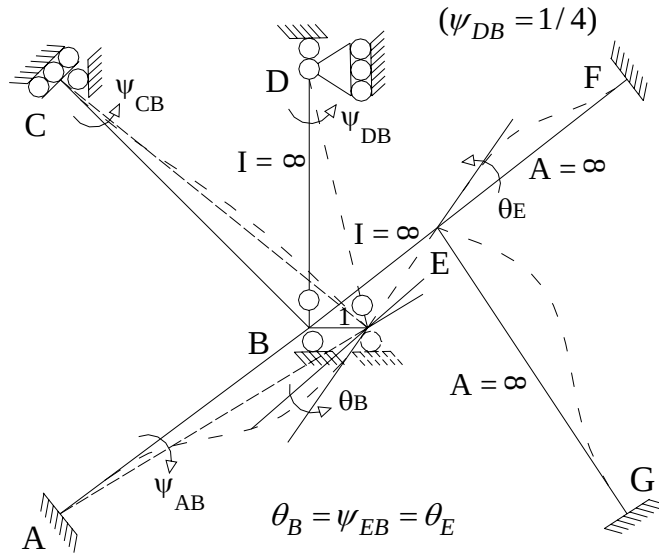
$$d_{i1} = \begin{bmatrix} 1/4 \\ 0 \\ 1/4 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Modo 2 $D_2 = 1 // D_1 = D_3 = D_4 = 0$



$$d_{i2} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Modo 3 $D_3 = 1 // D_1 = D_2 = D_4 = 0$



$$\cos \alpha = \Delta L = \frac{\sqrt{2}}{2}$$

$$\sin \alpha = n = \frac{\sqrt{2}}{2}$$

$$\psi_{CB} = \frac{n}{4\sqrt{2}} = \frac{\sqrt{2}/2}{4\sqrt{2}} = \frac{1}{8}$$

$$\cos \beta = \Delta L = 4/5$$

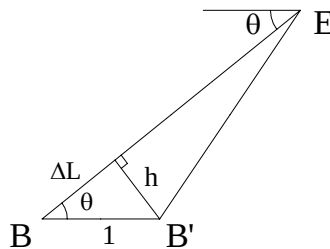
$$\sin \beta = m = 3/5$$

$$\psi_{AB} = \frac{-m}{5} = \frac{-3}{25}$$

$$\cos \theta = \Delta L = -4/5$$

$$\sin \theta = h = 3/5$$

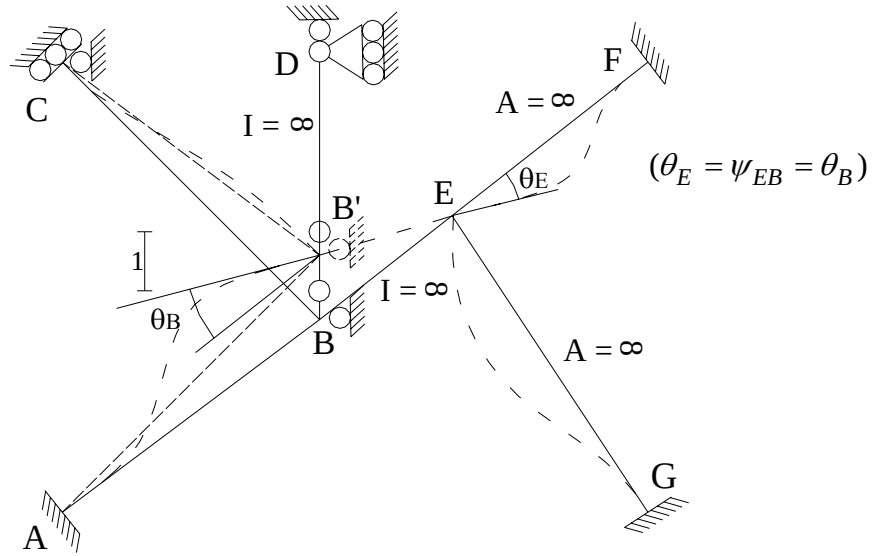
$$\psi_{EB} = \frac{h}{2.5} = \frac{h}{5/2} = \frac{3/5}{5/2} = \frac{6}{25}$$



$$d_{i3} = \begin{vmatrix} 0 & -1/8 \\ 1/\sqrt{2} & 1/\sqrt{2} \\ 6/25 & -1/8 \\ 0 & -(-3/25) \\ 4/5 & 4/5 \\ 6/25 & -(-3/25) \\ 0 & 0 \\ -4/5 & -4/5 \\ 0 & 0 \\ 6/25 & 6/25 \\ 6/25 & 6/25 \\ 0 & 0 \end{vmatrix} = \begin{vmatrix} -1/8 \\ 1/\sqrt{2} \\ 23/200 \\ 3/25 \\ 4/5 \\ 9/25 \\ 0 \\ -4/5 \\ 0 \\ 6/25 \\ 6/25 \\ 0 \end{vmatrix}$$

Modo 4

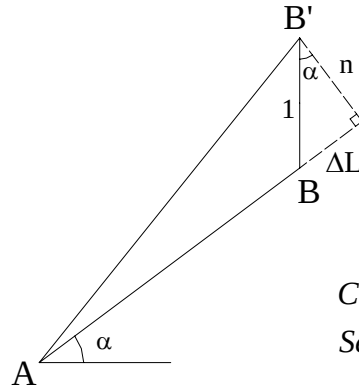
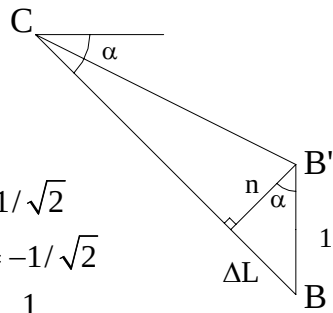
$$D_4 = 1 // D_1 = D_2 = D_3 = 0$$



$$\cos \alpha = m = 1/\sqrt{2}$$

$$\sin \alpha = \Delta L = -1/\sqrt{2}$$

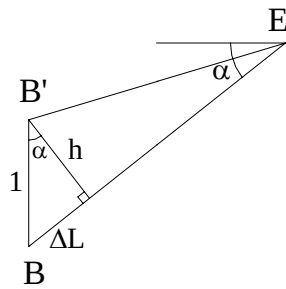
$$\psi_{CB} = \frac{m}{4\sqrt{2}} = \frac{1}{8}$$



$$\cos \alpha = n = 4/5$$

$$\sin \alpha = \Delta L = 3/5$$

$$\psi_{AB} = \frac{n}{5} = \frac{4}{25}$$



$$\cos \alpha = h = 4/5$$

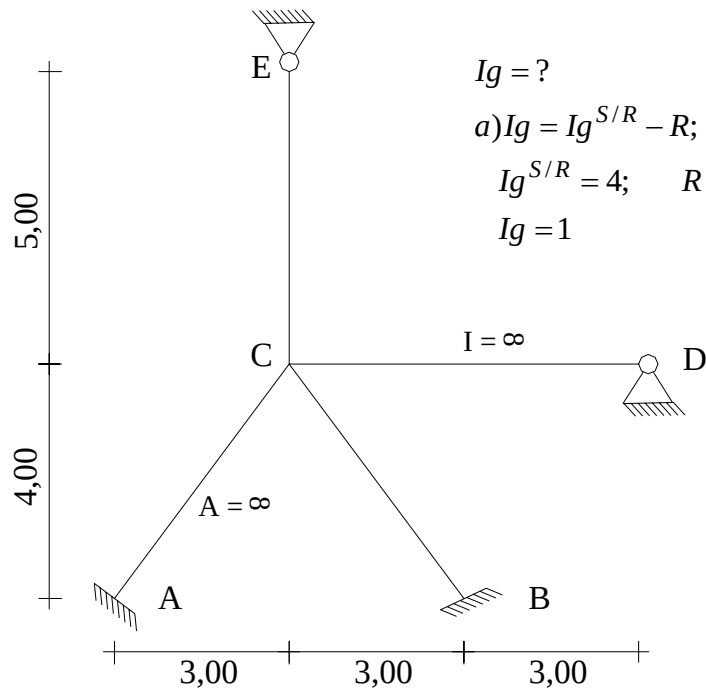
$$\sin \alpha = \Delta L = -3/5$$

$$\psi_{EB} = \frac{-h}{5/2} = \frac{-4/5}{5/2} = \frac{-8}{25}$$

$$d_{i4} = \begin{bmatrix} 0 - 1/8 \\ -1/\sqrt{2} \\ -8/25 - 1/8 \\ 0 - 4/25 \\ 3/5 \\ -8/25 - 4/25 \\ -1 \\ -3/5 \\ 0 \\ -8/25 \\ -8/25 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/8 \\ -1/\sqrt{2} \\ -89/200 \\ -4/25 \\ 3/5 \\ -12/25 \\ -1 \\ -3/5 \\ 0 \\ -8/25 \\ -8/25 \\ 0 \end{bmatrix}$$

$$[A] = \begin{bmatrix} 1/4 & 0 & -1/8 & -1/8 \\ 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1/4 & 0 & 23/200 & -89/200 \\ 0 & 0 & 3/25 & -4/25 \\ 0 & 0 & 4/5 & 3/5 \\ 0 & 0 & 9/25 & -12/25 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -4/5 & -3/5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 6/25 & -8/25 \\ 0 & 0 & 6/25 & -8/25 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

17.-



$$I_g = ?$$

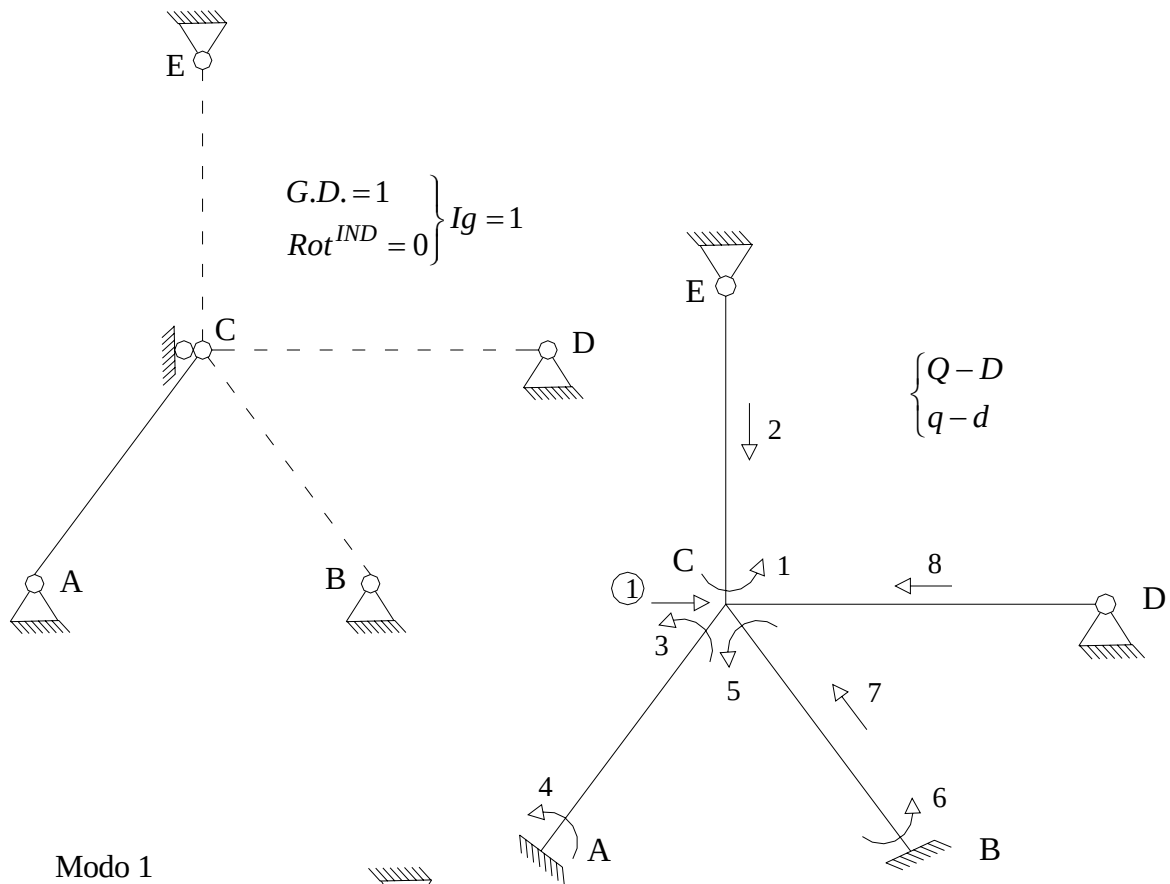
$$a) I_g = I_g^{S/R} - R;$$

$$I_g^{S/R} = 4; \quad R = 3$$

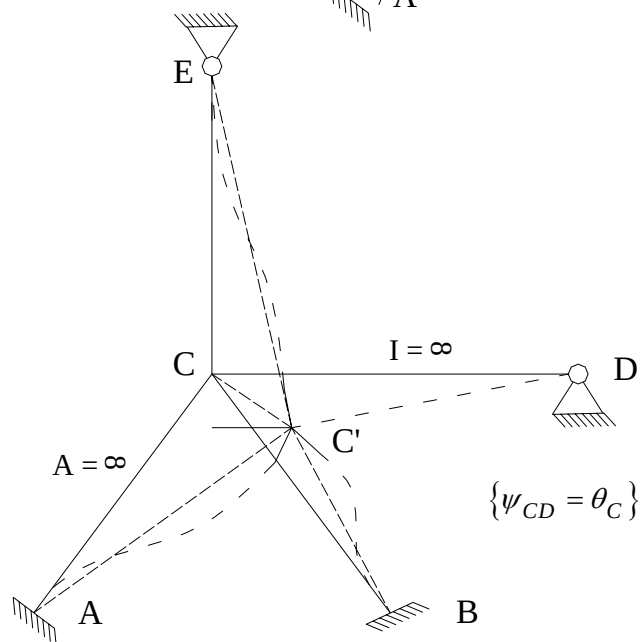
$$I_g = 1$$

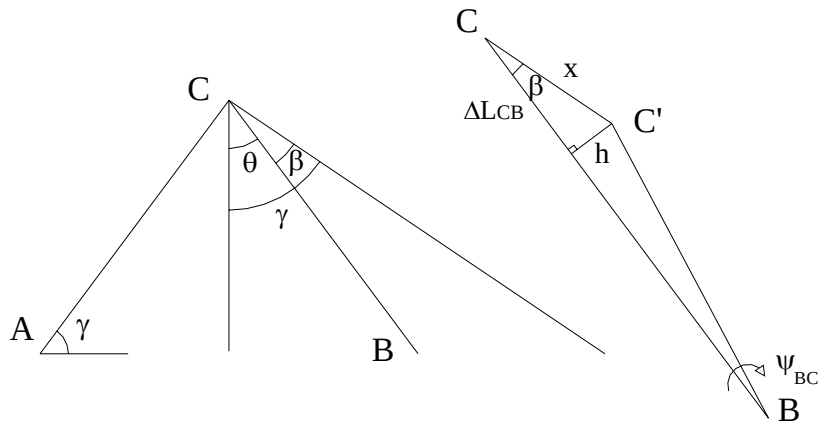
$$b) I_g = G.D. + Rot^{IND}$$

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Modo 1





$$\theta + \beta = \varphi \Rightarrow \beta = \varphi - \theta$$

$$\text{Tg} \varphi = 4/3 \Rightarrow \varphi = 53,13$$

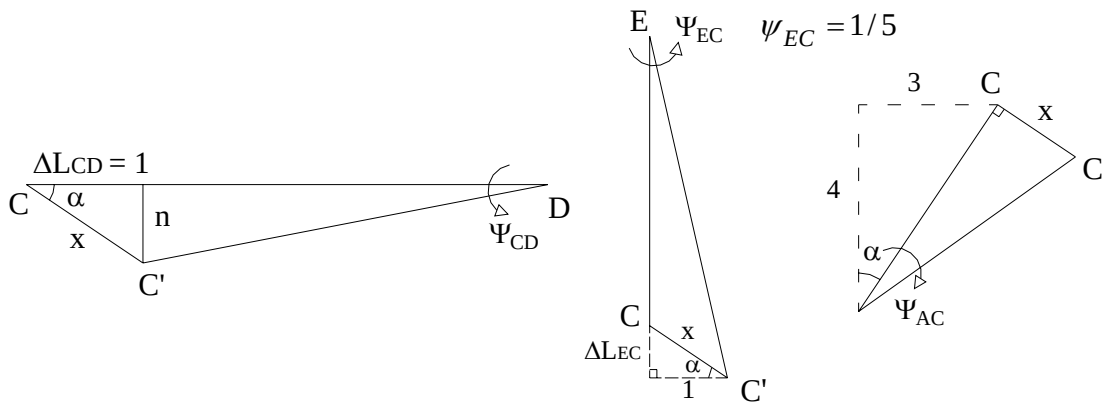
$$\text{Tg} \theta = 3/4 \Rightarrow \theta = 36,86$$

$$\beta = 16,26$$

$$\cos \beta = \frac{\Delta L}{x} = 0,86 \Rightarrow \Delta L = -1,2 = 6/5$$

$$\sin \beta = \frac{h}{x} = 0,779 \Rightarrow h = 0,345$$

$$\psi_{BC} = \frac{-h}{5} = -6,99 \times 10^{-2}$$



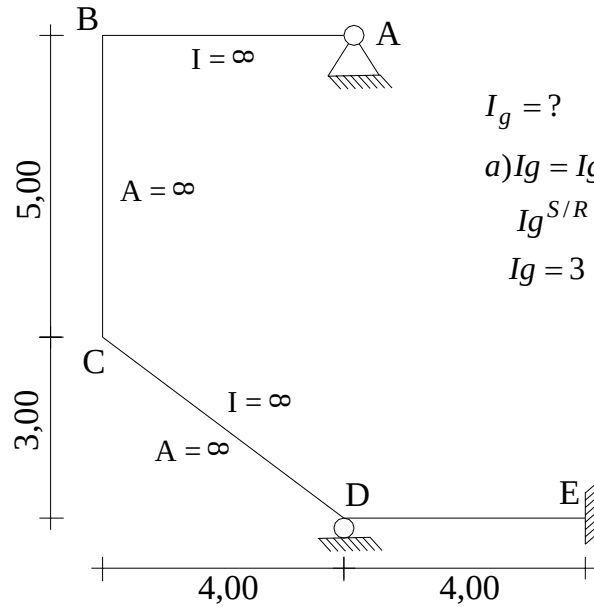
$$\text{Tg} \alpha = n = \frac{3}{4} \Rightarrow \psi_{DC} = \frac{n}{6} = \frac{3}{24} = \frac{1}{8}$$

$$\cos \alpha = \frac{1}{x} = \frac{4}{5} \Rightarrow x = \frac{5}{4} \Rightarrow \psi_{AC} = \frac{-x}{5} = \frac{-1}{4}$$

$$\sin \alpha = \frac{\Delta L_{EC}}{x} = \frac{3}{5} \Rightarrow \Delta L_{EC} = \frac{3x}{5} = \frac{3}{4}$$

$$d_{i1} = \begin{vmatrix} 1/8 - 1/5 \\ 3/4 \\ 1/8 - (-1/4) \\ 0 - (-1/4) \\ 1/8 - (-6,99 \times 10^{-2}) \\ 0 - (-6,99 \times 10^{-2}) \\ -6/5 \\ -1 \end{vmatrix} = \begin{vmatrix} -3/40 \\ 3/4 \\ 3/8 \\ 1/4 \\ 0,194 \\ 0,0699 \\ -6/5 \\ -1 \end{vmatrix}$$

18.-



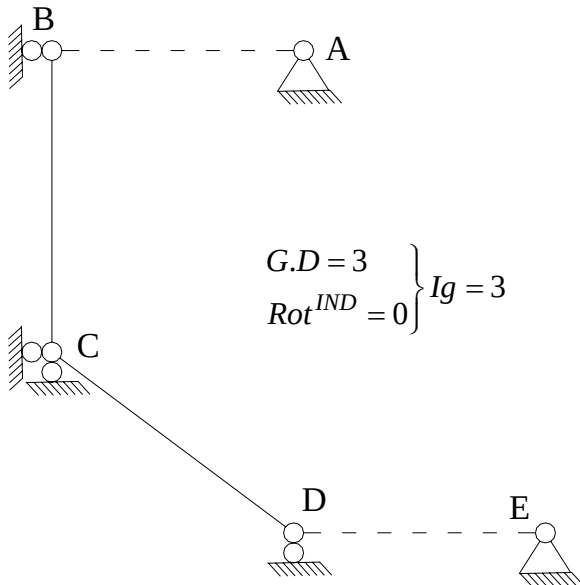
$$I_g = ?$$

$$a) I_g = I_g^{S/R} - R; \quad I_g = G.D. + Rot^{IND}$$

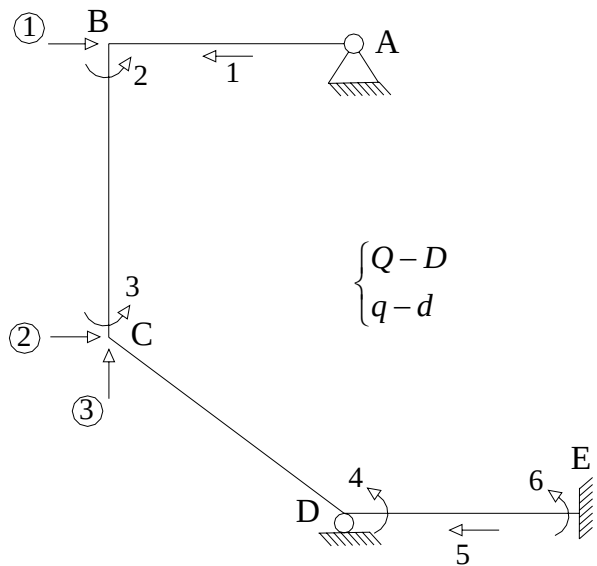
$$I_g^{S/R} = 9; \quad R = 6$$

$$I_g = 3$$

Imagen Cinemática



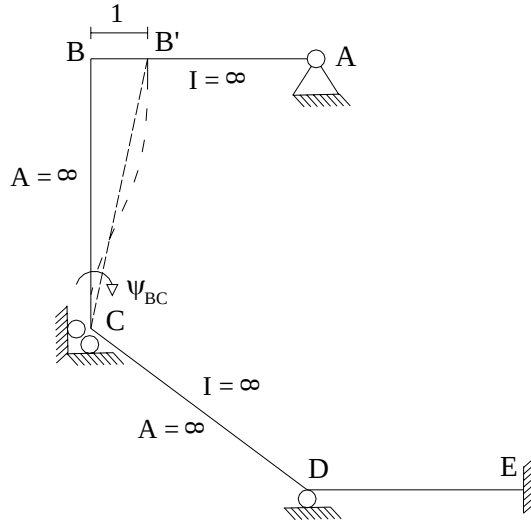
$$\left. \begin{array}{l} G.D = 3 \\ Rot^{IND} = 0 \end{array} \right\} I_g = 3$$



Modo 1

$$D_1 = 1 // D_2 = D_3 = 0$$

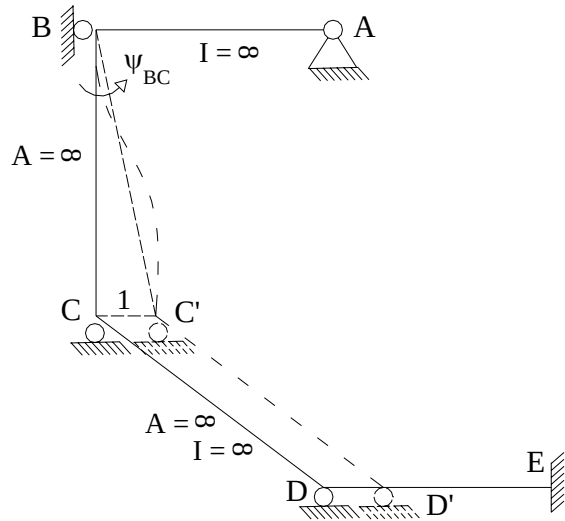
$$d_{i1} = \begin{vmatrix} -1 \\ 1/5 \\ 1/5 \\ 0 \\ 0 \\ 0 \end{vmatrix}$$



Modo 2

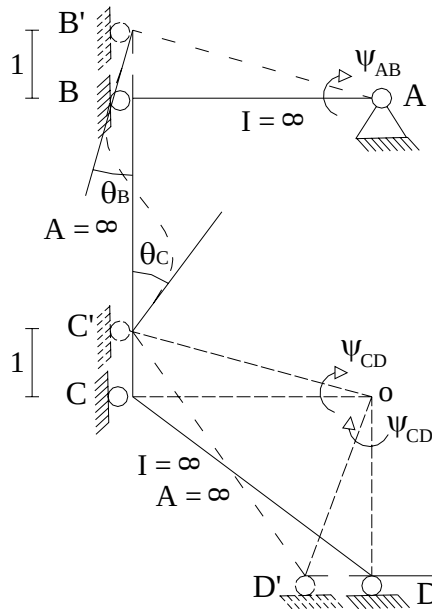
$$D_2 = 1 // D_1 = D_3 = 0$$

$$d_{i2} = \begin{vmatrix} 0 \\ -1/5 \\ -1/5 \\ 0 \\ -1 \\ 0 \end{vmatrix}$$



Modo 3

$$D_3 = 1 // D_1 = D_2 = 0$$



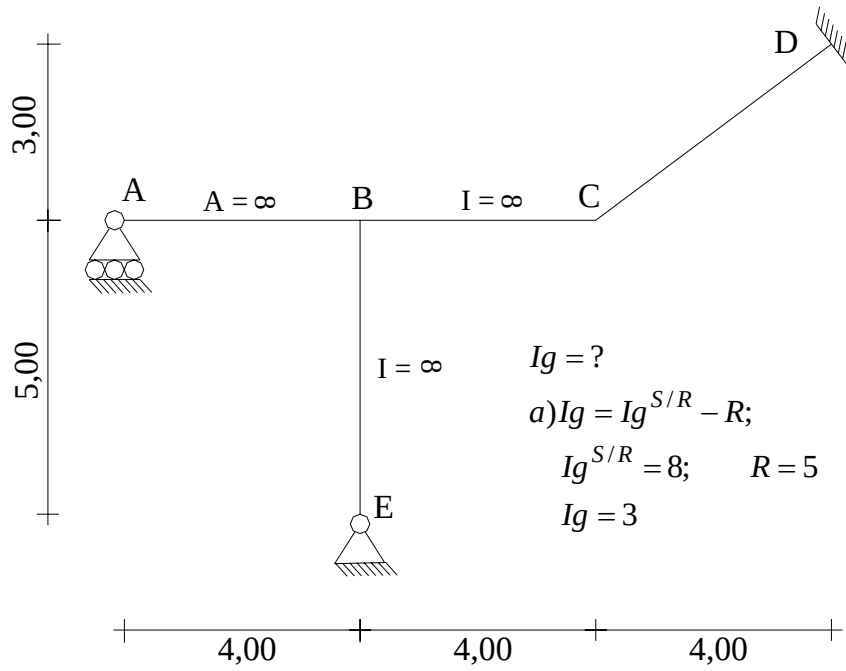
$$d_{i3} = \begin{vmatrix} 0 \\ -1/4 \\ -1/4 \\ -1/4 \\ 3/4 \\ 0 \end{vmatrix}$$

$$\psi_{AB} = \theta_B = -1/4$$

$$\theta_C = \theta_D = \psi_{CD} = -1/4$$

$$[A] = \left[\begin{array}{c|c|c} -1 & 0 & 0 \\ 1/5 & -1/5 & -1/4 \\ 1/5 & -1/5 & -1/4 \\ 0 & 0 & -1/4 \\ 0 & -1 & 3/4 \\ 0 & 0 & 0 \end{array} \right]$$

19.-



$$I_g = ?$$

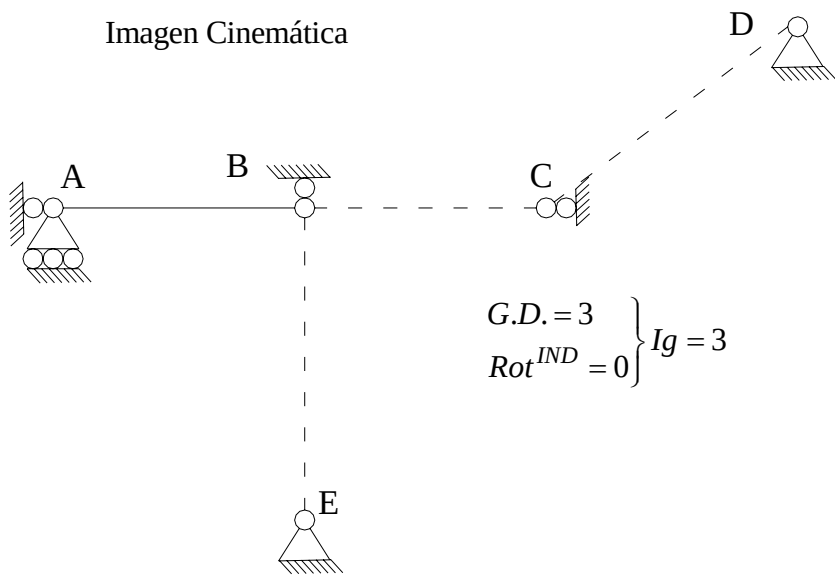
$$a) I_g = I_g^{S/R} - R;$$

$$b) I_g = G.D. + Rot^{IND}$$

$$I_g^{S/R} = 8; \quad R = 5$$

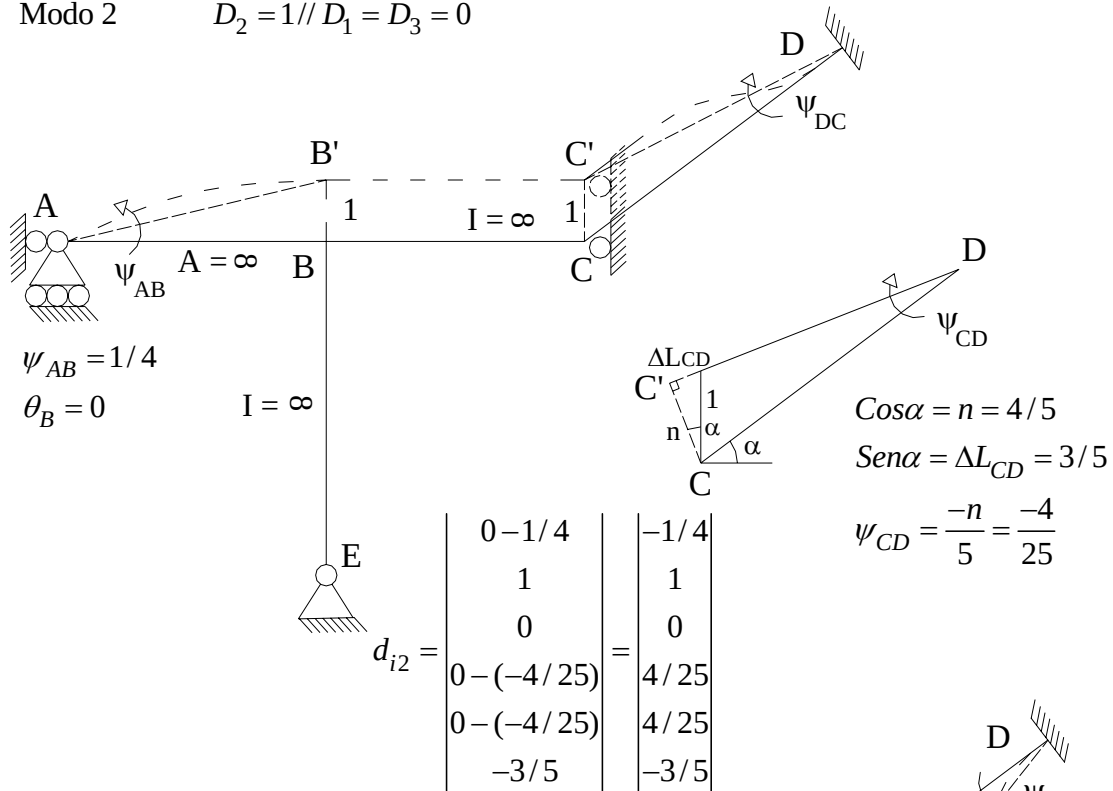
$$I_g = 3$$

Imagen Cinemática

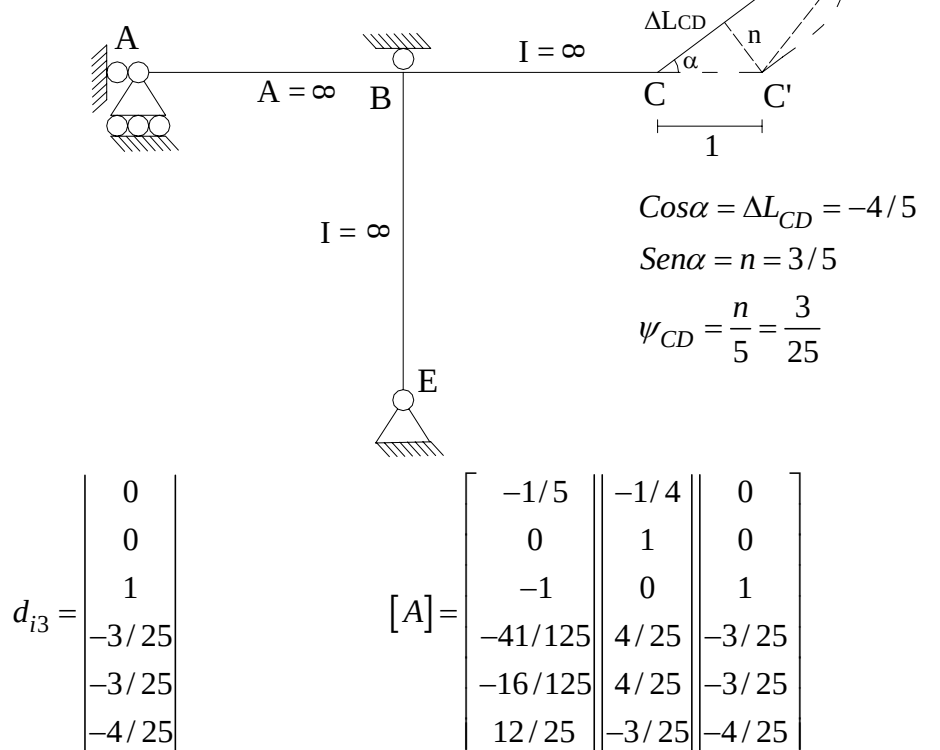


$$\left. \begin{array}{l} G.D. = 3 \\ Rot^{IND} = 0 \end{array} \right\} I_g = 3$$

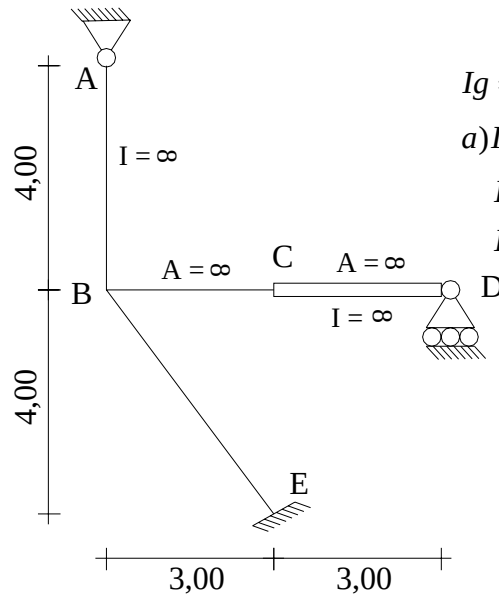
Modo 2 $D_2 = 1 // D_1 = D_3 = 0$



Modo 3 $D_3 = 1 // D_1 = D_2 = 0$



20.-



$$I_g = ?$$

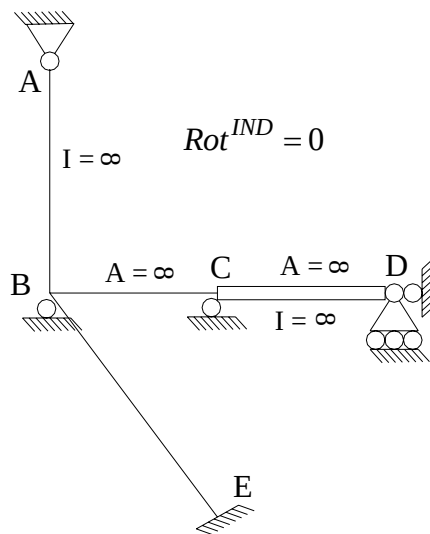
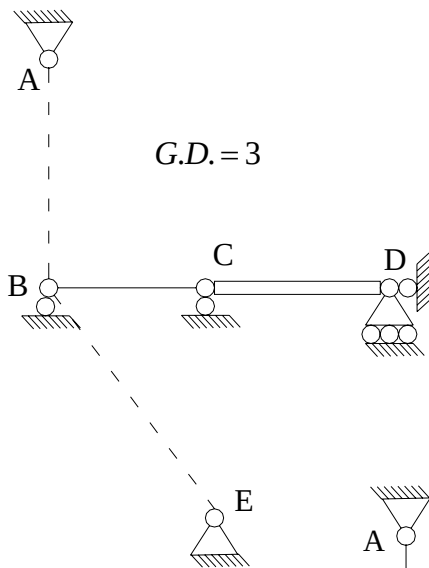
$$a) I_g = I_g^{S/R} - R;$$

$$I_g^{S/R} = 9; \quad R = 6$$

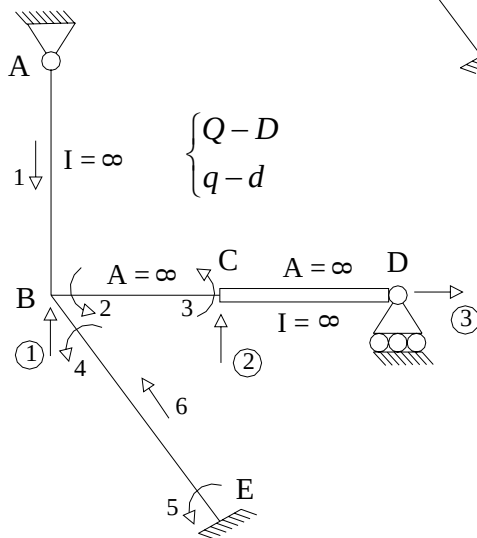
$$I_g = 3$$

$$b) I_g = G.D. + Rot^{IND}$$

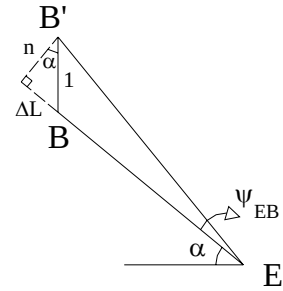
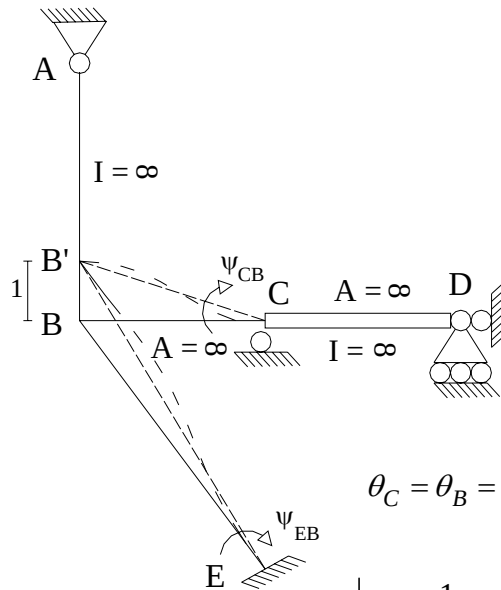
Imagen Cinemática



$$I_g = 3$$



Modo 1 $D_1 = 1 // D_2 = D_3 = 0$



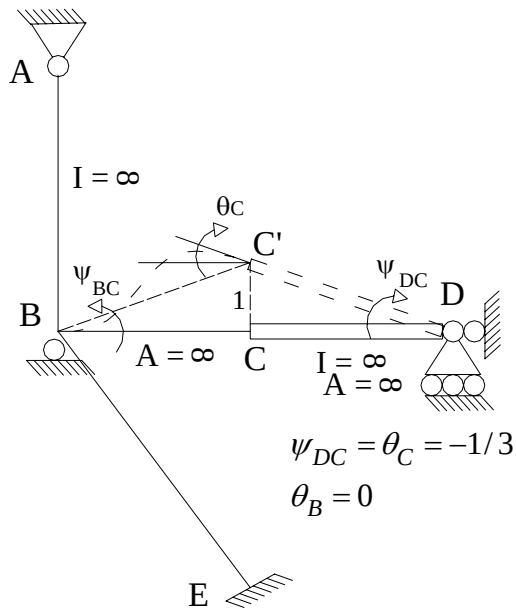
$$\cos \alpha = n = 3/5$$

$$\sin \alpha = \Delta L = 4/5$$

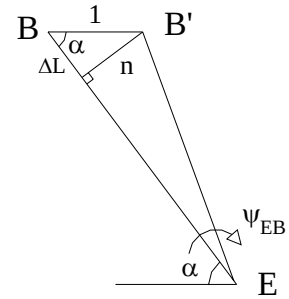
$$\psi_{EB} = \frac{-n}{5} = \frac{-3}{25}$$

$$d_{i1} = \begin{vmatrix} -1 \\ 0 - (-1/3) \\ 0 - (-1/3) \\ 0 - (-3/25) \\ 0 - (-3/25) \\ 4/5 \end{vmatrix} = \begin{vmatrix} -1 \\ 1/3 \\ 1/3 \\ 3/25 \\ 2/35 \\ 4/5 \end{vmatrix}$$

Modo 2 $D_2 = 1 // D_1 = D_3 = 0$



$$d_{i2} = \begin{vmatrix} 0 \\ 0 - 1/3 \\ -1/3 - 1/3 \\ 0 \\ 0 \\ 0 \end{vmatrix} = \begin{vmatrix} 0 \\ -1/3 \\ -2/3 \\ 0 \\ 0 \\ 0 \end{vmatrix}$$

$$D_3 = 1 // D_1 = D_2 = 0$$


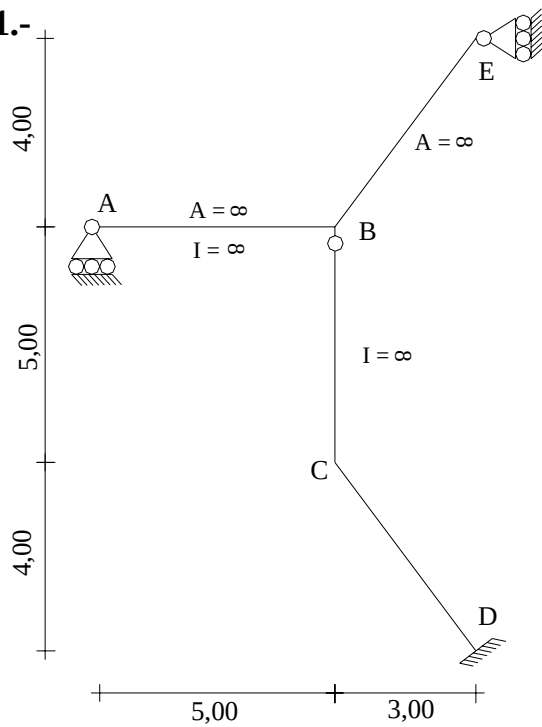
$$\text{Sen}\alpha = n = 4/5$$

$$\psi_{AB} = \theta_B = 1/4$$

$$\theta_C = 0$$

$$[A] = \begin{bmatrix} -1 & 0 & 0 \\ 1/3 & -1/3 & 1/4 \\ 1/3 & -2/3 & 0 \\ 3/25 & 0 & 41/100 \\ 3/25 & 0 & 4/25 \\ 4/5 & 0 & -3/5 \end{bmatrix}$$

21.-



$$I_g = ?$$

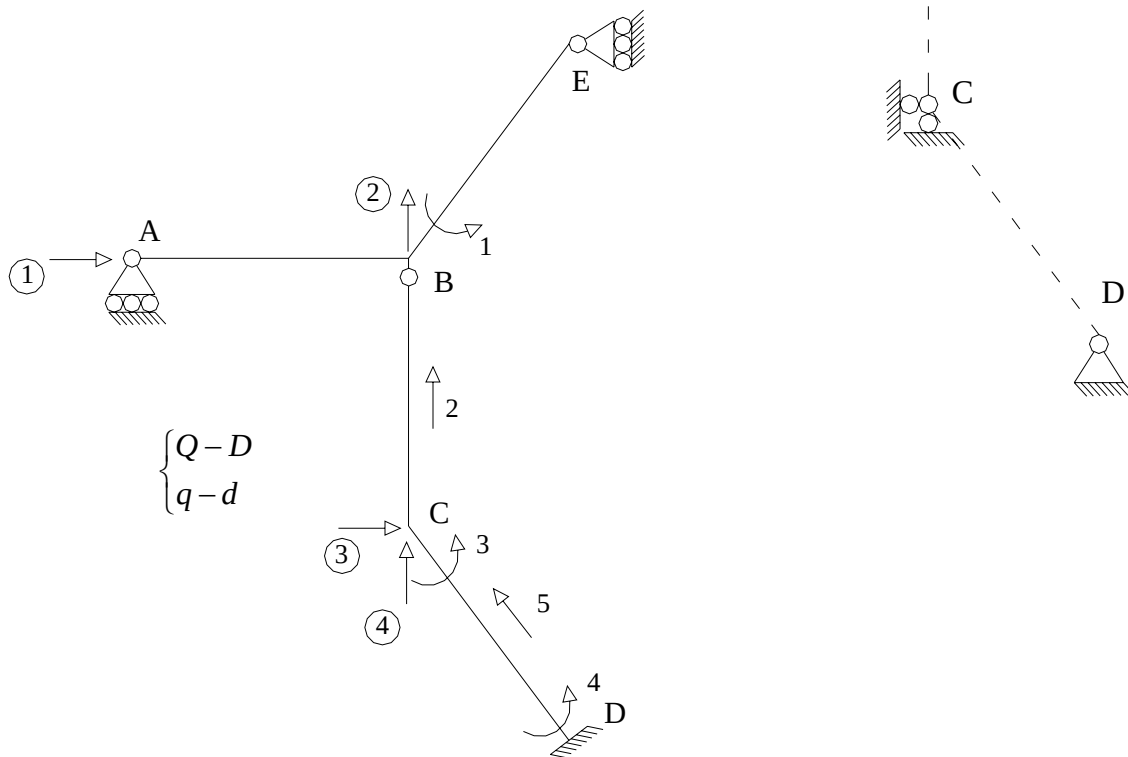
$$a) I_g = I_g^{S/R} - R;$$

$$I_g^{S/R} = 10; \quad R = 6$$

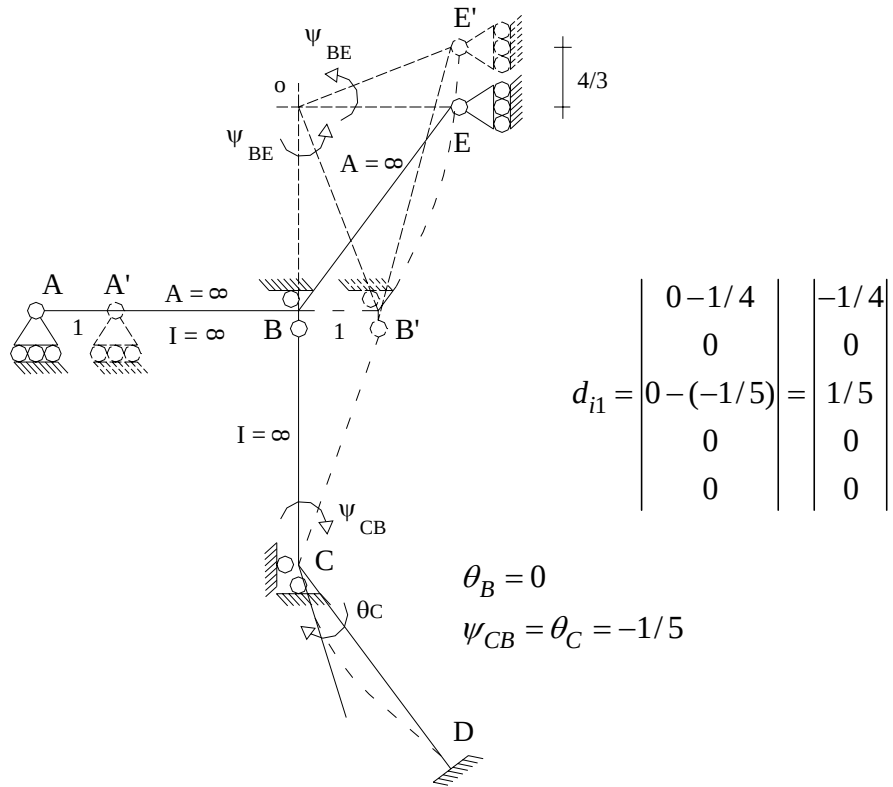
$$I_g = 4$$

$$b) I_g = G.D. + Rot^{IND}$$

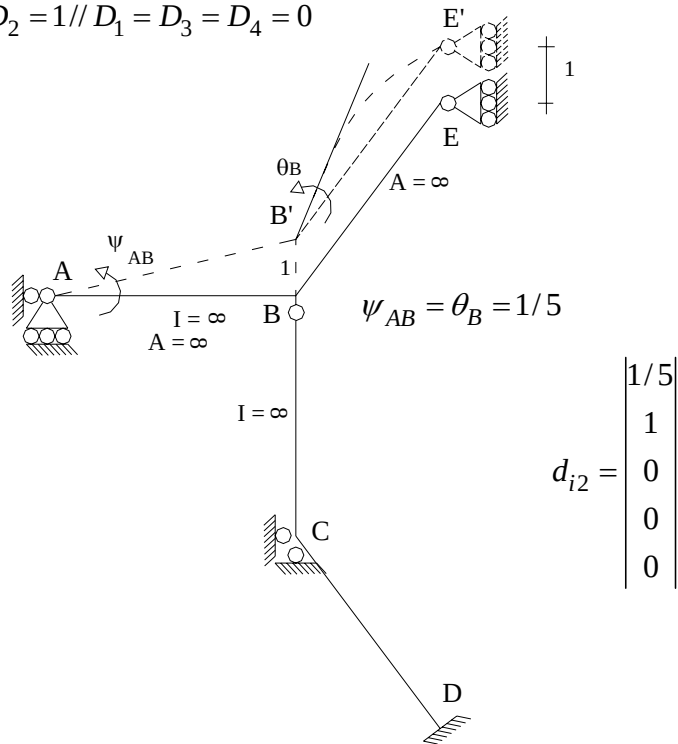
Imagen Cinemática

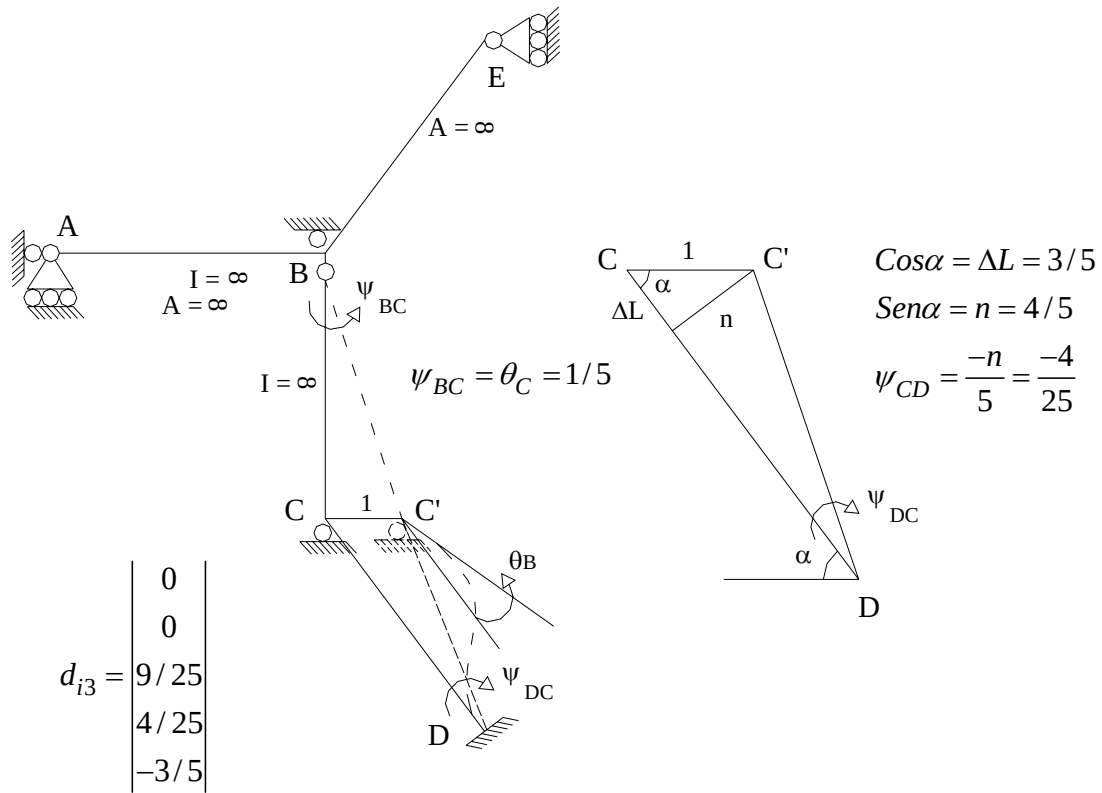


Modo 1 $D_1 = 1 // D_2 = D_3 = D_4 = 0$

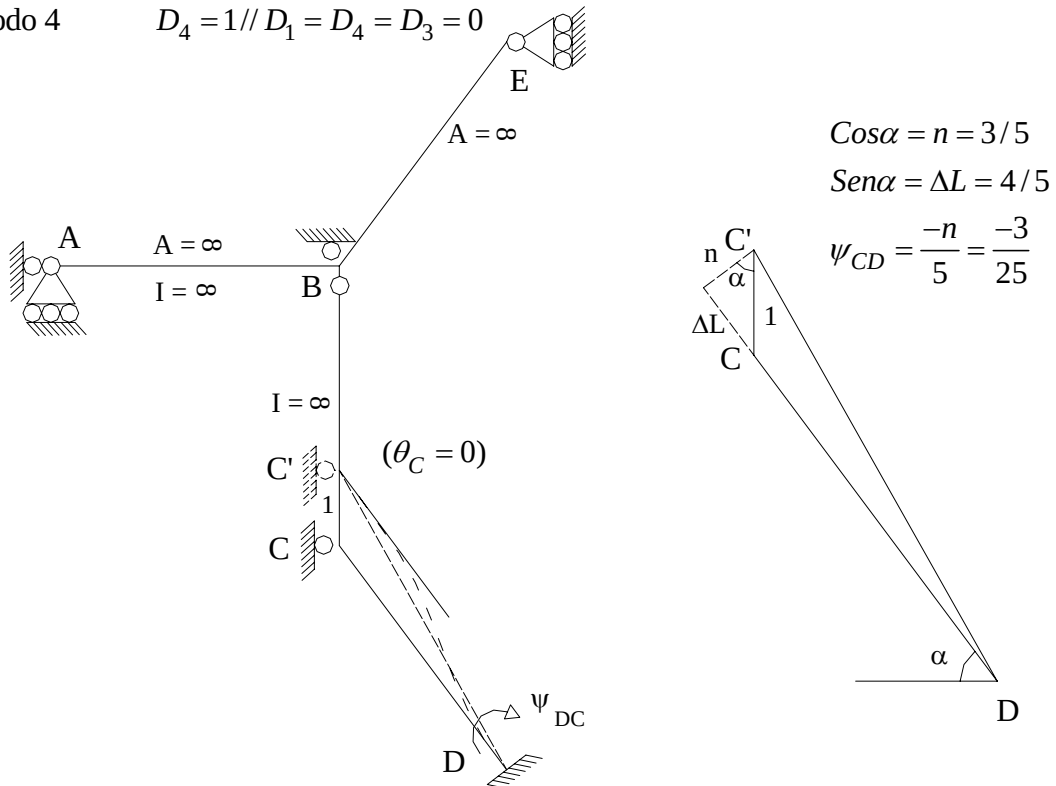


Modo 2 $D_2 = 1 // D_1 = D_3 = D_4 = 0$



$$D_3 = 1 // D_1 = D_2 = D_4 = 0$$


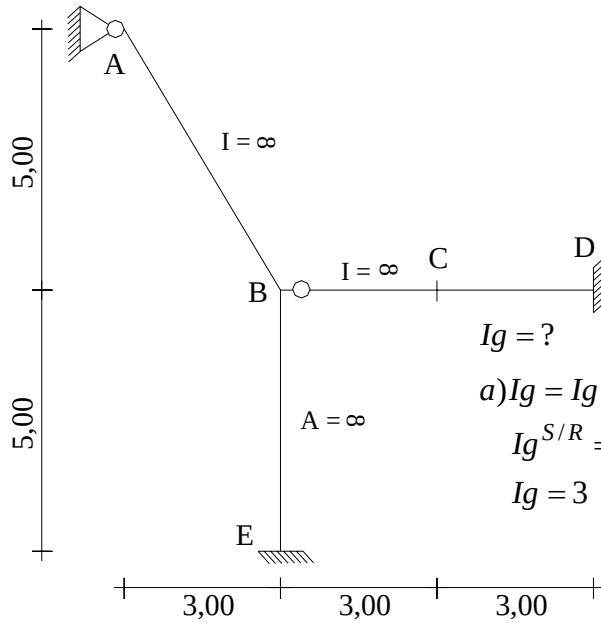
Modo 4

$$D_4=1//D_1=D_4=D_3=0$$


$$d_{i4} = \begin{vmatrix} 0 \\ -1 \\ 3/25 \\ 3/25 \\ 4/5 \end{vmatrix}$$

$$[A] = \begin{bmatrix} -1/4 & 1/5 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 1/5 & 0 & 9/25 & 3/25 \\ 0 & 0 & 4/25 & 3/25 \\ 0 & 0 & -3/5 & 4/5 \end{bmatrix}$$

22.-



$$I_g = ?$$

$$a) I_g = I_g^{S/R} - R;$$

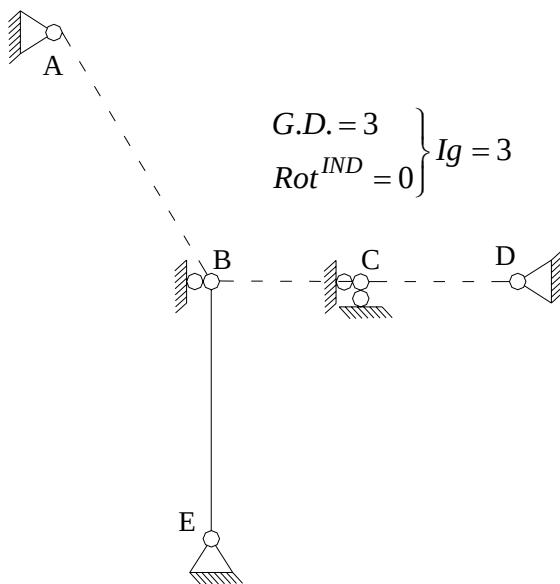
$$I_g^{S/R} = 8;$$

$$I_g = 3$$

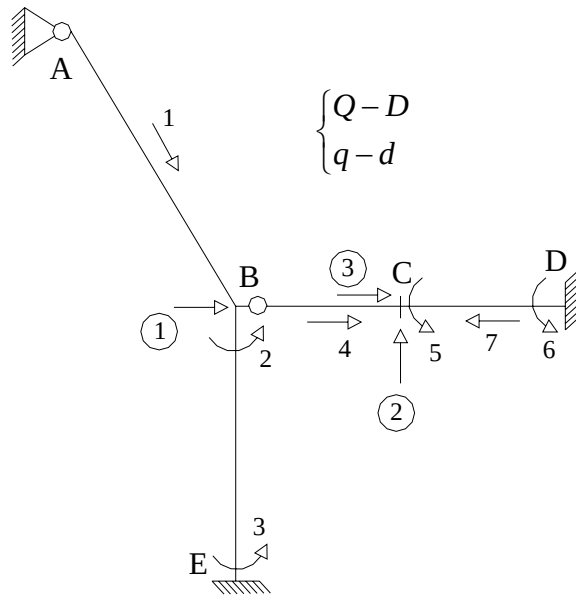
$$b) I_g = G.D. + Rot^{IND}$$

$$R = 5$$

Imagen Cinemática

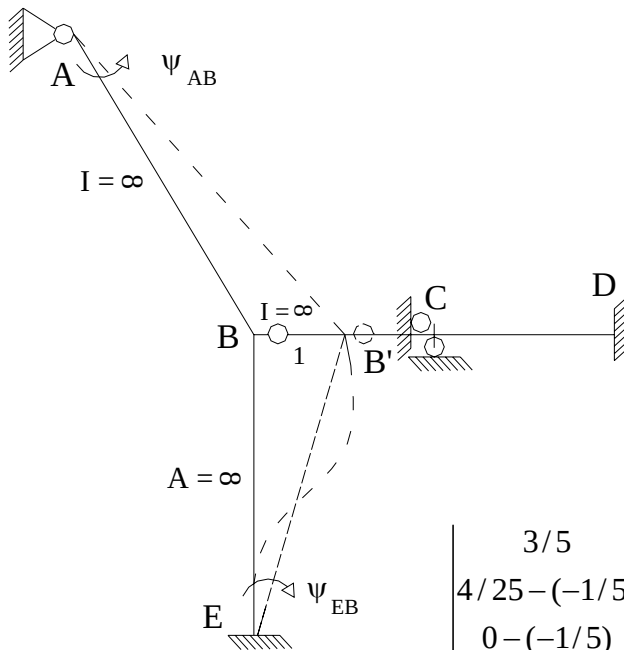


$$\left. \begin{array}{l} G.D. = 3 \\ Rot^{IND} = 0 \end{array} \right\} I_g = 3$$



$$\left\{ \begin{array}{l} Q-D \\ q-d \end{array} \right.$$

Modo 1 $D_1 = 1 // D_2 = D_3 = 0$

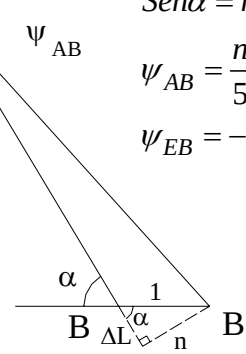


$$\cos \alpha = \Delta L = 3/5$$

$$\sin \alpha = n = 4/5$$

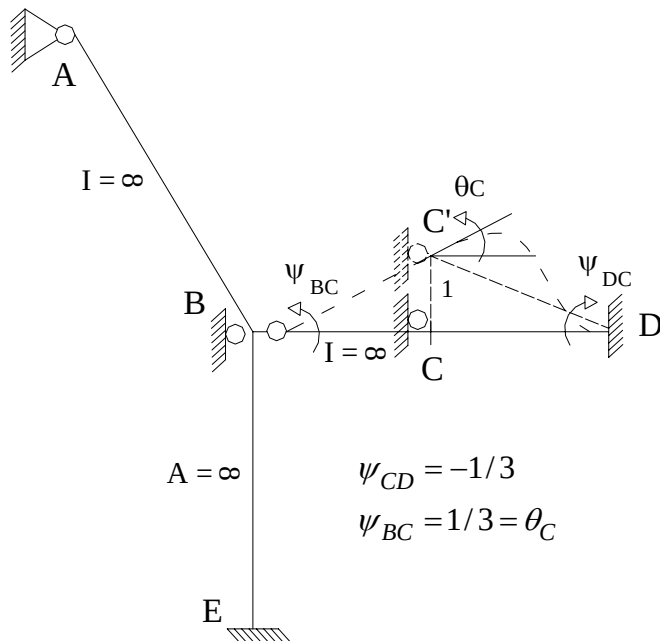
$$\psi_{AB} = \frac{n}{5} = \frac{4}{25}$$

$$\psi_{EB} = -1/5$$



$$d_{i1} = \begin{vmatrix} 3/5 \\ 4/25 - (-1/5) \\ 0 - (-1/5) \\ -1 \\ 0 \\ 0 \\ 0 \end{vmatrix} = \begin{vmatrix} 3/5 \\ 9/25 \\ 1/5 \\ -1 \\ 0 \\ 0 \\ 0 \end{vmatrix}$$

Modo 2 $D_2 = 1 // D_1 = D_3 = 0$

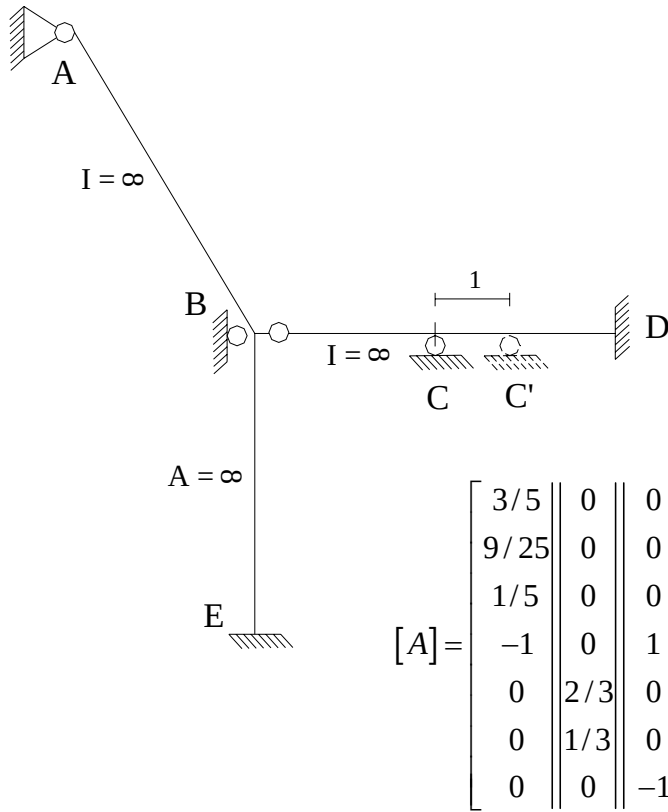


$$\psi_{CD} = -1/3$$

$$\psi_{BC} = 1/3 = \theta_C$$

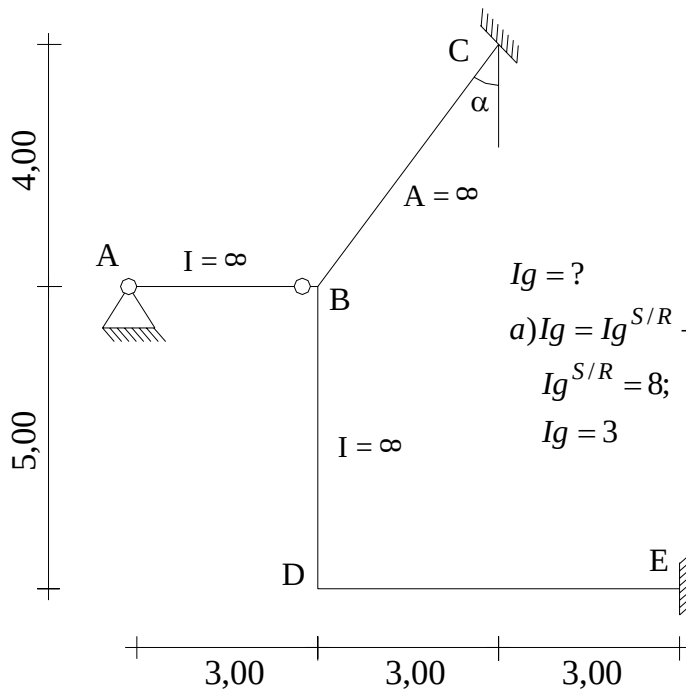
$$d_{i2} = \begin{vmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1/3 - (-1/3) \\ 0 - (-1/3) \\ 0 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 2/3 \\ 1/3 \\ 0 \end{vmatrix}$$

Modo 3 $D_3 = 1 // D_1 = D_2 = 0$



$$d_{i3} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$$

23.-



$Ig = ?$

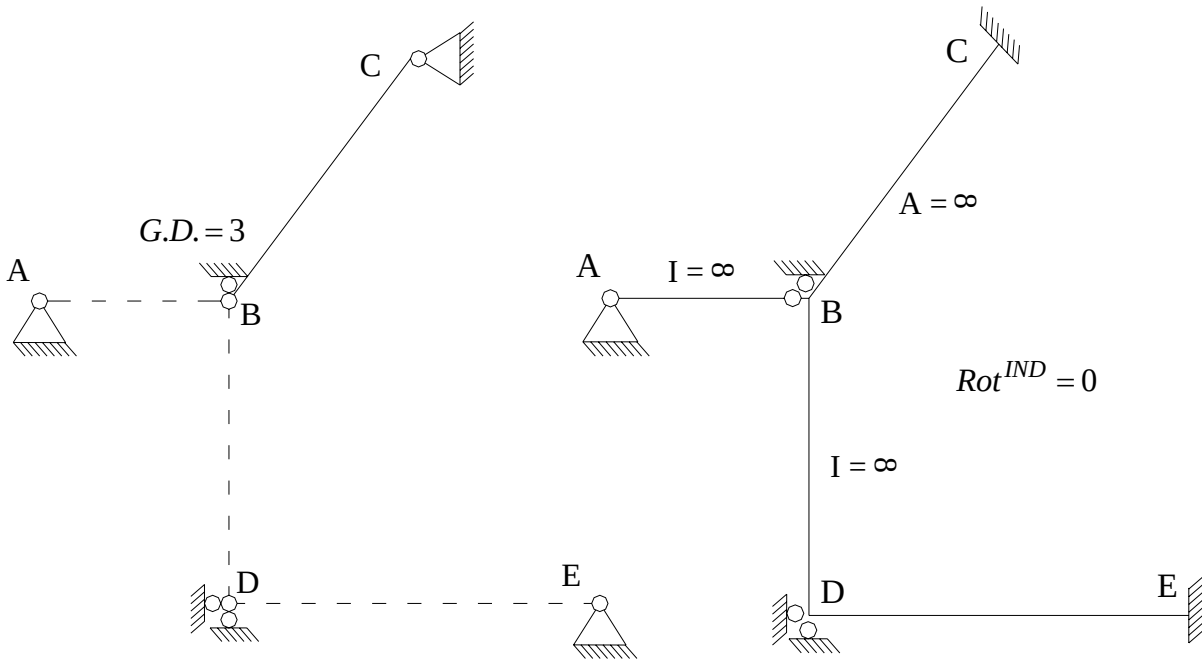
a) $Ig = Ig^{S/R} - R;$

$Ig^{S/R} = 8; \quad R = 5$

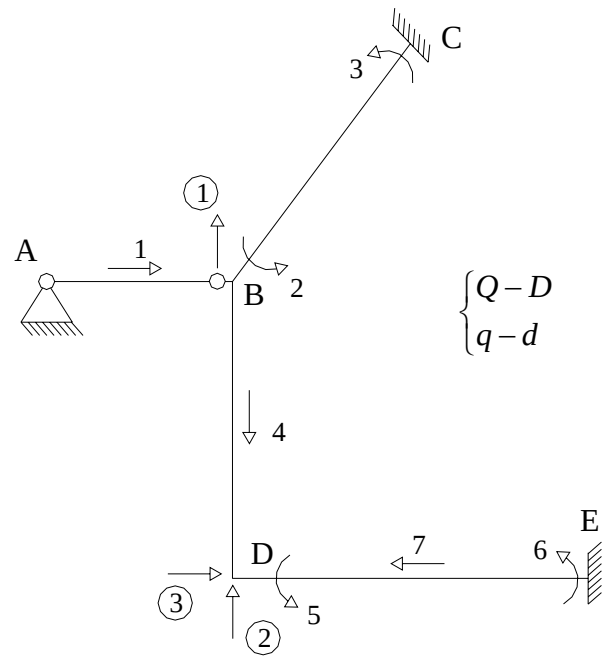
$Ig = 3$

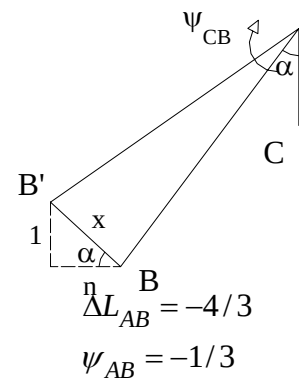
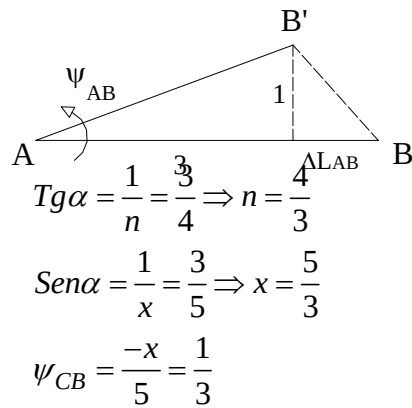
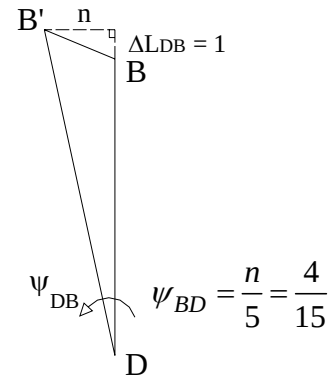
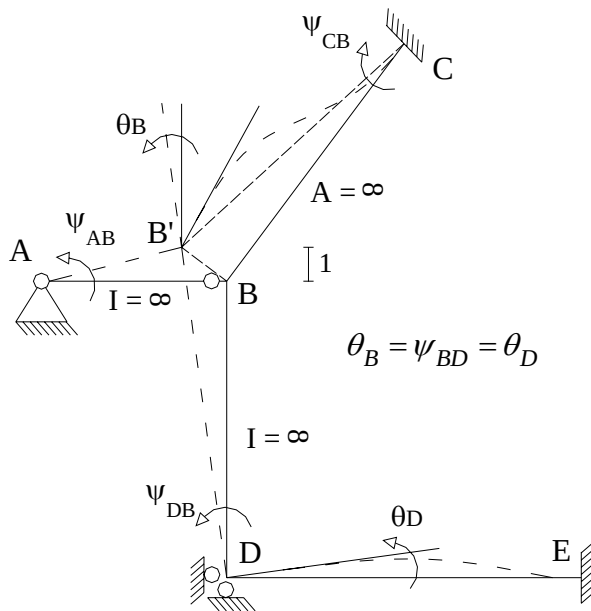
b) $Ig = G.D. + Rot^{IND}$

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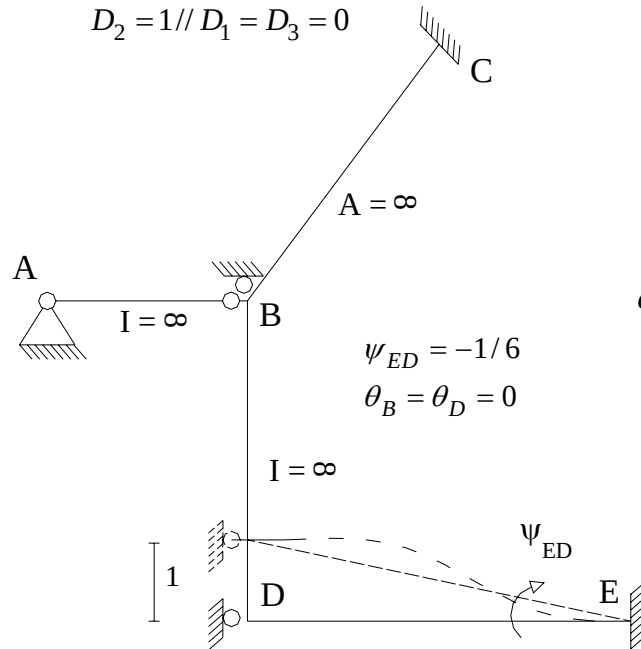
$$I_g = 3$$



$$D_1 = 1 // D_2 = D_3 = 0$$


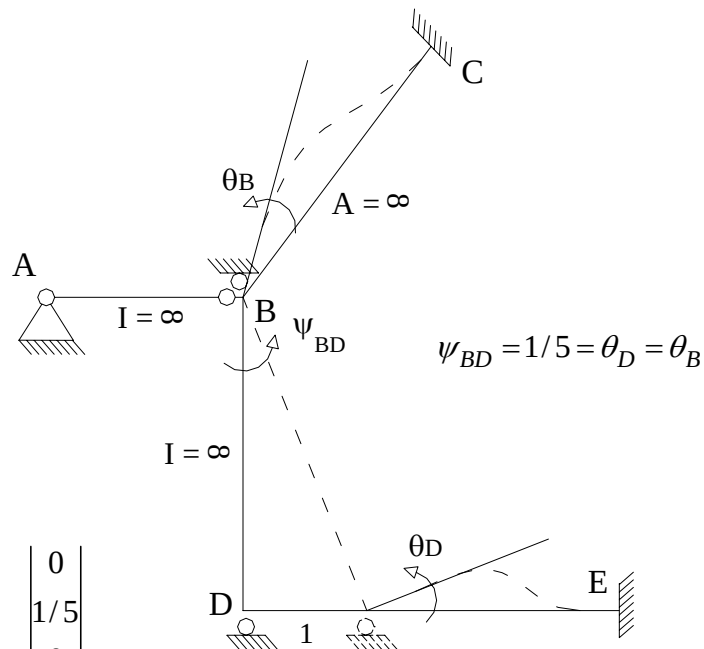
$$d_{i1} = \begin{vmatrix} -4/3 \\ 4/15 - (-1/3) \\ 0 - (-1/3) \\ 1 \\ 4/15 \\ 0 \\ 0 \end{vmatrix} = \begin{vmatrix} -4/3 \\ 9/15 \\ 1/3 \\ 1 \\ 4/15 \\ 0 \\ 0 \end{vmatrix}$$

Modo 2 $D_2 = 1 // D_1 = D_3 = 0$



$$d_{i2} = \begin{vmatrix} 0 \\ 0 \\ 0 \\ -1 \\ 0 - (-1/6) \\ 0 - (-1/6) \\ 0 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ 0 \\ -1 \\ 1/6 \\ 1/6 \\ 0 \end{vmatrix}$$

Modo 3 $D_3 = 1 // D_1 = D_2 = 0$



$$d_{i3} = \begin{vmatrix} 0 \\ 1/5 \\ 0 \\ 0 \\ 1/5 \\ 0 \\ -1 \end{vmatrix}$$

$$[A] = \begin{bmatrix} -4/3 & 0 & 0 \\ 9/15 & 0 & 1/5 \\ 1/3 & 0 & 0 \\ 1 & -1 & 0 \\ 4/15 & 1/6 & 1/5 \\ 0 & 1/6 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$